DISCO-TAB — Comprehensive Benchmark

all generators × 24 datasets × all metrics; cells are **mean ± 95% confidence interval** over random seeds (Student-t, up to 3 seeds); **bold** = best per row (by mean, where a direction is defined); – = missing run/metric; ± absent where only one seed is available.

Utility (downstream TSTR)

Table 1: TSTR F1-macro (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	**0.783 ± 0.184**	0.483 ± 0.237	0.135 ± 0.582	0.591 ± 0.252	0.406	0.635 ± 0.151	0.396 ± 0.174	0.600 ± 0.270	0.559 ± 0.371	0.649 ± 0.296	0.735 ± 0.205	0.394 ± 0.217
breast_cancer	0.930 ± 0.035	0.295 ± 0.304	0.934 ± 0.084	0.912 ± 0.041	0.924	0.934 ± 0.044	0.497 ± 0.461	0.937 ± 0.047	0.855 ± 0.082	0.940 ± 0.034	**0.947 ± 0.046**	0.513 ± 0.508
liver_disorders	0.509 ± 0.208	0.473 ± 0.224	0.527 ± 0.171	0.553 ± 0.098	0.592	0.542 ± 0.174	0.488 ± 0.263	0.490 ± 0.333	0.500 ± 0.106	**0.593 ± 0.140**	0.561 ± 0.271	0.380 ± 0.208
parkinsons	0.646 ± 0.517	0.480 ± 0.344	0.000 ± 0.000	0.630 ± 0.104	**0.928**	0.737 ± 0.215	0.424 ± 0.012	0.704 ± 0.131	0.695 ± 0.096	0.688 ± 0.183	0.759 ± 0.296	0.446 ± 0.139
obesity	**0.886 ± 0.011**	0.138 ± 0.161	0.490 ± 0.086	0.243 ± 0.045	0.795	0.655 ± 0.097	0.115 ± 0.033	0.607 ± 0.092	0.171 ± 0.091	0.757 ± 0.046	0.821 ± 0.023	0.102 ± 0.026
diabetes	0.843 ± 0.186	0.836 ± 0.042	**0.875 ± 0.003**	0.485 ± 0.014	–	0.797 ± 0.086	0.478 ± 0.000	0.727 ± 0.121	0.658 ± 0.070	0.853 ± 0.012	0.704 ± 0.071	0.377 ± 0.082
german_credit	0.516 ± 0.572	0.448 ± 0.046	0.137 ± 0.591	0.496 ± 0.118	–	0.594 ± 0.194	0.468 ± 0.203	0.501 ± 0.164	0.449 ± 0.145	0.584 ± 0.121	**0.638 ± 0.179**	0.459 ± 0.077
bank_marketing	**0.673 ± 0.205**	0.604 ± 0.039	0.673 ± 0.043	0.469 ± 0.001	–	0.608 ± 0.082	0.469 ± 0.000	0.498 ± 0.058	0.471 ± 0.005	0.558 ± 0.054	0.551 ± 0.100	0.236 ± 0.317
census_income	0.765 ± 0.039	0.727 ± 0.031	0.746 ± 0.014	0.459 ± 0.038	–	0.669 ± 0.126	0.433 ± 0.002	0.549 ± 0.126	0.615 ± 0.145	0.744 ± 0.011	**0.767 ± 0.005**	0.524 ± 0.264
surv_melanoma	0.536 ± 0.169	–	–	–	–	0.557 ± 0.099	0.442 ± 0.070	0.529 ± 0.128	0.508 ± 0.201	**0.573 ± 0.354**	0.570 ± 0.345	0.445 ± 0.588
surv_pbc	0.522 ± 0.515	–	–	–	–	0.590 ± 0.152	0.492 ± 0.182	0.646 ± 0.177	0.566 ± 0.220	0.634 ± 0.162	**0.679 ± 0.101**	0.428 ± 0.123
surv_pbc3	0.468 ± 0.398	–	–	–	–	0.632 ± 0.095	0.453 ± 0.000	0.469 ± 0.098	0.477 ± 0.101	0.553 ± 0.215	**0.707 ± 0.091**	0.497 ± 0.131
surv_retinopathy	0.518 ± 0.146	–	–	–	–	0.539 ± 0.211	0.480 ± 0.201	0.478 ± 0.147	0.513 ± 0.146	**0.586 ± 0.116**	0.493 ± 0.159	0.511 ± 0.042
surv_whas500	**0.739 ± 0.032**	–	–	–	–	0.699 ± 0.159	0.395 ± 0.161	0.619 ± 0.114	0.709 ± 0.046	0.671 ± 0.081	0.673 ± 0.189	0.324 ± 0.096
surv_gbsg2	0.580 ± 0.067	–	–	–	–	0.538 ± 0.086	0.474 ± 0.204	0.556 ± 0.082	**0.590 ± 0.196**	0.572 ± 0.067	0.547 ± 0.061	0.475 ± 0.310
surv_colon	**0.587 ± 0.120**	–	–	–	–	0.497 ± 0.250	0.426 ± 0.203	0.564 ± 0.075	0.526 ± 0.083	0.581 ± 0.037	0.539 ± 0.147	0.450 ± 0.097
surv_grace	**0.609 ± 0.052**	–	–	–	–	0.591 ± 0.100	0.456 ± 0.074	0.508 ± 0.058	0.543 ± 0.224	0.593 ± 0.072	0.521 ± 0.254	0.463 ± 0.033
surv_trace	0.676 ± 0.019	–	–	–	–	0.646 ± 0.163	0.445 ± 0.065	0.516 ± 0.148	0.609 ± 0.192	**0.704 ± 0.042**	0.684 ± 0.036	0.516 ± 0.045
surv_framingham	**0.581 ± 0.088**	–	–	–	–	0.546 ± 0.088	0.474 ± 0.112	0.486 ± 0.104	0.453 ± 0.122	0.532 ± 0.033	0.475 ± 0.112	0.448 ± 0.110
tab_blood_transfusion	0.351 ± 0.755	–	–	–	–	0.498 ± 0.141	0.510 ± 0.243	0.517 ± 0.078	0.456 ± 0.215	**0.580 ± 0.132**	0.528 ± 0.157	0.502 ± 0.198
tab_credit_g	0.643 ± 0.103	–	–	–	–	0.577 ± 0.077	0.435 ± 0.099	0.465 ± 0.128	0.448 ± 0.085	0.526 ± 0.095	**0.653 ± 0.076**	0.434 ± 0.200
tab_qsar_biodeg	0.796 ± 0.129	–	–	–	–	0.729 ± 0.081	0.428 ± 0.232	0.770 ± 0.071	0.612 ± 0.298	0.780 ± 0.119	**0.850 ± 0.139**	0.393 ± 0.338
tab_maternal_risk	**0.624 ± 0.257**	–	–	–	–	0.463 ± 0.240	0.235 ± 0.187	0.527 ± 0.099	0.395 ± 0.082	0.573 ± 0.106	0.214 ± 0.160	0.234 ± 0.260
tab_website_phishing	**0.614 ± 0.075**	–	–	–	–	0.555 ± 0.090	0.263 ± 0.217	0.492 ± 0.047	0.454 ± 0.013	0.562 ± 0.101	0.551 ± 0.100	0.286 ± 0.087

Table 2: TSTR F1-weighted (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	**0.799 ± 0.195**	0.559 ± 0.243	0.185 ± 0.796	0.679 ± 0.178	0.555	0.710 ± 0.122	0.500 ± 0.317	0.676 ± 0.173	0.655 ± 0.256	0.722 ± 0.219	0.768 ± 0.183	0.416 ± 0.438
breast_cancer	0.935 ± 0.033	0.284 ± 0.354	0.939 ± 0.079	0.920 ± 0.038	0.929	0.939 ± 0.042	0.489 ± 0.601	0.942 ± 0.045	0.869 ± 0.068	0.944 ± 0.033	**0.950 ± 0.044**	0.558 ± 0.428
liver_disorders	0.511 ± 0.205	0.472 ± 0.228	0.529 ± 0.169	0.553 ± 0.099	0.592	0.543 ± 0.175	0.488 ± 0.265	0.490 ± 0.333	0.501 ± 0.105	**0.593 ± 0.140**	0.562 ± 0.271	0.384 ± 0.206
parkinsons	0.732 ± 0.378	0.632 ± 0.164	0.000 ± 0.000	0.729 ± 0.116	**0.947**	0.801 ± 0.156	0.630 ± 0.018	0.784 ± 0.111	0.768 ± 0.100	0.756 ± 0.184	0.829 ± 0.201	0.584 ± 0.126
obesity	**0.888 ± 0.010**	0.138 ± 0.168	0.493 ± 0.088	0.250 ± 0.046	0.796	0.655 ± 0.108	0.120 ± 0.036	0.610 ± 0.098	0.176 ± 0.096	0.759 ± 0.050	0.827 ± 0.022	0.101 ± 0.023
diabetes	0.956 ± 0.045	0.952 ± 0.017	**0.962 ± 0.001**	0.876 ± 0.003	–	0.945 ± 0.021	0.874 ± 0.001	0.927 ± 0.029	0.901 ± 0.039	0.959 ± 0.003	0.923 ± 0.016	0.584 ± 0.144
german_credit	0.502 ± 0.722	0.585 ± 0.036	0.192 ± 0.827	0.624 ± 0.074	–	0.680 ± 0.112	0.601 ± 0.099	0.631 ± 0.105	0.587 ± 0.049	0.687 ± 0.082	**0.715 ± 0.119**	0.516 ± 0.219
bank_marketing	**0.865 ± 0.036**	0.846 ± 0.049	0.854 ± 0.030	0.828 ± 0.000	–	0.862 ± 0.019	0.828 ± 0.000	0.835 ± 0.015	0.829 ± 0.001	0.852 ± 0.014	0.848 ± 0.024	0.273 ± 0.618
census_income	0.830 ± 0.014	0.812 ± 0.018	0.818 ± 0.016	0.671 ± 0.019	–	0.782 ± 0.067	0.656 ± 0.002	0.718 ± 0.069	0.748 ± 0.076	0.824 ± 0.006	**0.838 ± 0.009**	0.672 ± 0.164
surv_melanoma	0.561 ± 0.205	–	–	–	–	0.676 ± 0.045	0.588 ± 0.059	0.645 ± 0.130	0.614 ± 0.026	**0.684 ± 0.262**	0.672 ± 0.247	0.454 ± 0.688
surv_pbc	0.507 ± 0.610	–	–	–	–	0.592 ± 0.167	0.528 ± 0.205	0.662 ± 0.191	0.586 ± 0.156	0.648 ± 0.161	**0.692 ± 0.107**	0.444 ± 0.124
surv_pbc3	0.541 ± 0.652	–	–	–	–	0.786 ± 0.079	0.751 ± 0.000	0.749 ± 0.037	0.748 ± 0.012	0.791 ± 0.087	**0.845 ± 0.042**	0.675 ± 0.082
surv_retinopathy	0.542 ± 0.133	–	–	–	–	0.565 ± 0.183	0.481 ± 0.265	0.507 ± 0.136	0.531 ± 0.148	**0.601 ± 0.135**	0.513 ± 0.132	0.517 ± 0.032
surv_whas500	**0.752 ± 0.024**	–	–	–	–	0.719 ± 0.127	0.428 ± 0.101	0.642 ± 0.126	0.721 ± 0.042	0.697 ± 0.071	0.696 ± 0.178	0.339 ± 0.098
surv_gbsg2	0.587 ± 0.063	–	–	–	–	0.542 ± 0.110	0.486 ± 0.213	0.564 ± 0.073	**0.596 ± 0.186**	0.582 ± 0.065	0.564 ± 0.061	0.474 ± 0.344
surv_colon	**0.586 ± 0.122**	–	–	–	–	0.495 ± 0.255	0.429 ± 0.200	0.565 ± 0.074	0.526 ± 0.084	0.582 ± 0.037	0.540 ± 0.149	0.451 ± 0.092
surv_grace	0.639 ± 0.066	–	–	–	–	**0.656 ± 0.050**	0.526 ± 0.150	0.591 ± 0.033	0.634 ± 0.153	0.652 ± 0.061	0.617 ± 0.157	0.503 ± 0.096
surv_trace	0.676 ± 0.018	–	–	–	–	0.645 ± 0.166	0.444 ± 0.069	0.516 ± 0.147	0.609 ± 0.191	**0.704 ± 0.042**	0.684 ± 0.036	0.516 ± 0.047
surv_framingham	0.622 ± 0.118	–	–	–	–	**0.635 ± 0.054**	0.590 ± 0.057	0.580 ± 0.101	0.580 ± 0.057	0.628 ± 0.030	0.595 ± 0.064	0.490 ± 0.119
tab_blood_transfusion	0.417 ± 0.898	–	–	–	–	0.607 ± 0.048	0.636 ± 0.128	0.615 ± 0.105	0.590 ± 0.192	**0.678 ± 0.075**	0.652 ± 0.094	0.551 ± 0.266
tab_credit_g	0.675 ± 0.121	–	–	–	–	0.657 ± 0.038	0.582 ± 0.025	0.606 ± 0.075	0.588 ± 0.063	0.645 ± 0.065	**0.725 ± 0.053**	0.501 ± 0.385
tab_qsar_biodeg	0.819 ± 0.106	–	–	–	–	0.767 ± 0.066	0.465 ± 0.314	0.799 ± 0.059	0.674 ± 0.210	0.809 ± 0.101	**0.870 ± 0.119**	0.361 ± 0.442
tab_maternal_risk	**0.660 ± 0.234**	–	–	–	–	0.510 ± 0.333	0.306 ± 0.153	0.592 ± 0.068	0.485 ± 0.087	0.637 ± 0.113	0.191 ± 0.165	0.251 ± 0.394
tab_website_phishing	**0.746 ± 0.068**	–	–	–	–	0.694 ± 0.052	0.346 ± 0.299	0.662 ± 0.035	0.614 ± 0.058	0.734 ± 0.045	0.727 ± 0.070	0.368 ± 0.100

Table 3: TSTR accuracy (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	**0.794 ± 0.204**	0.572 ± 0.305	0.228 ± 0.980	0.733 ± 0.110	0.683	0.750 ± 0.110	0.572 ± 0.514	0.711 ± 0.086	0.717 ± 0.143	0.761 ± 0.157	0.767 ± 0.190	0.472 ± 0.335
breast_cancer	0.936 ± 0.033	0.316 ± 0.278	0.939 ± 0.079	0.921 ± 0.038	0.930	0.939 ± 0.044	0.561 ± 0.437	0.942 ± 0.045	0.874 ± 0.055	0.944 ± 0.033	**0.950 ± 0.045**	0.579 ± 0.363
liver_disorders	0.546 ± 0.150	0.502 ± 0.181	0.551 ± 0.144	0.556 ± 0.091	**0.594**	0.551 ± 0.191	0.493 ± 0.252	0.493 ± 0.330	0.507 ± 0.095	**0.594 ± 0.144**	0.565 ± 0.272	0.473 ± 0.162
parkinsons	0.735 ± 0.368	0.675 ± 0.074	0.000 ± 0.000	0.744 ± 0.169	**0.949**	0.803 ± 0.147	0.735 ± 0.037	0.795 ± 0.127	0.769 ± 0.127	0.752 ± 0.224	0.846 ± 0.169	0.607 ± 0.224
obesity	**0.887 ± 0.012**	0.154 ± 0.127	0.499 ± 0.077	0.297 ± 0.056	0.801	0.655 ± 0.120	0.149 ± 0.042	0.627 ± 0.083	0.270 ± 0.101	0.766 ± 0.048	0.828 ± 0.019	0.115 ± 0.024
diabetes	0.961 ± 0.031	0.954 ± 0.022	**0.963 ± 0.000**	0.915 ± 0.001	–	0.953 ± 0.015	0.915 ± 0.001	0.941 ± 0.020	0.909 ± 0.095	0.963 ± 0.002	0.938 ± 0.010	0.479 ± 0.151
german_credit	0.542 ± 0.498	0.663 ± 0.147	0.233 ± 1.004	0.700 ± 0.033	–	0.712 ± 0.040	0.680 ± 0.045	0.715 ± 0.054	0.668 ± 0.094	**0.743 ± 0.050**	0.742 ± 0.068	0.545 ± 0.255
bank_marketing	0.867 ± 0.064	0.858 ± 0.094	0.845 ± 0.072	0.883 ± 0.000	–	0.889 ± 0.003	0.883 ± 0.000	0.884 ± 0.004	0.883 ± 0.000	**0.891 ± 0.004**	0.886 ± 0.003	0.266 ± 0.436
census_income	0.832 ± 0.004	0.823 ± 0.016	0.821 ± 0.022	0.766 ± 0.005	–	0.812 ± 0.032	0.758 ± 0.007	0.782 ± 0.034	0.784 ± 0.036	0.836 ± 0.004	**0.846 ± 0.015**	0.698 ± 0.183
surv_melanoma	0.545 ± 0.185	–	–	–	–	**0.707 ± 0.000**	0.626 ± 0.185	0.667 ± 0.195	0.634 ± 0.160	**0.707 ± 0.242**	0.683 ± 0.218	0.463 ± 0.655

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
surv_pbc	0.566 ± 0.373	–	–	–	–	0.593 ± 0.159	0.561 ± 0.277	0.667 ± 0.209	0.603 ± 0.079	0.651 ± 0.172	**0.693 ± 0.120**	0.439 ± 0.127
surv_pbc3	0.543 ± 0.663	–	–	–	–	0.781 ± 0.102	0.829 ± 0.000	0.810 ± 0.020	0.805 ± 0.102	0.848 ± 0.041	**0.857 ± 0.035**	0.652 ± 0.164
surv_retinopathy	0.554 ± 0.121	–	–	–	–	0.581 ± 0.146	0.514 ± 0.220	0.527 ± 0.146	0.532 ± 0.155	**0.608 ± 0.154**	0.518 ± 0.108	0.514 ± 0.034
surv_whas500	**0.753 ± 0.027**	–	–	–	–	0.728 ± 0.086	0.452 ± 0.080	0.649 ± 0.147	0.720 ± 0.046	0.713 ± 0.067	0.706 ± 0.178	0.333 ± 0.093
surv_gbsg2	0.589 ± 0.068	–	–	–	–	0.560 ± 0.153	0.493 ± 0.230	0.565 ± 0.065	**0.597 ± 0.180**	0.587 ± 0.065	0.587 ± 0.082	0.488 ± 0.280
surv_colon	**0.593 ± 0.098**	–	–	–	–	0.550 ± 0.089	0.492 ± 0.193	0.569 ± 0.071	0.535 ± 0.066	0.582 ± 0.037	0.542 ± 0.147	0.475 ± 0.056
surv_grace	0.630 ± 0.069	–	–	–	–	0.678 ± 0.029	0.545 ± 0.228	0.625 ± 0.062	**0.692 ± 0.072**	0.663 ± 0.059	0.682 ± 0.014	0.495 ± 0.097
surv_trace	0.679 ± 0.017	–	–	–	–	0.660 ± 0.119	0.501 ± 0.159	0.518 ± 0.148	0.611 ± 0.187	**0.704 ± 0.042**	0.684 ± 0.036	0.519 ± 0.040
surv_framingham	0.616 ± 0.140	–	–	–	–	0.674 ± 0.046	0.660 ± 0.046	0.613 ± 0.132	0.668 ± 0.046	0.674 ± 0.036	**0.676 ± 0.023**	0.472 ± 0.125
tab_blood_transfusion	0.427 ± 0.927	–	–	–	–	0.645 ± 0.163	0.682 ± 0.061	0.623 ± 0.155	0.629 ± 0.228	**0.701 ± 0.061**	0.698 ± 0.075	0.542 ± 0.304
tab_credit_g	0.663 ± 0.127	–	–	–	–	0.683 ± 0.133	0.678 ± 0.093	0.700 ± 0.012	0.668 ± 0.050	0.712 ± 0.040	**0.747 ± 0.047**	0.540 ± 0.388
tab_qsar_biodeg	0.820 ± 0.096	–	–	–	–	0.779 ± 0.058	0.485 ± 0.296	0.806 ± 0.059	0.709 ± 0.106	0.815 ± 0.093	**0.874 ± 0.111**	0.423 ± 0.276
tab_maternal_risk	0.656 ± 0.262	–	–	–	–	0.524 ± 0.417	0.381 ± 0.316	0.626 ± 0.082	0.557 ± 0.057	**0.678 ± 0.167**	0.278 ± 0.103	0.289 ± 0.356
tab_website_phishing	**0.761 ± 0.049**	–	–	–	–	0.708 ± 0.079	0.434 ± 0.223	0.683 ± 0.030	0.637 ± 0.081	0.749 ± 0.036	0.747 ± 0.065	0.336 ± 0.109

Statistical fidelity

Table 4: Jensen–Shannon div. (↓) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.054 ± 0.054	0.094 ± 0.020	0.200 ± 0.033	0.042 ± 0.002	0.194	0.070 ± 0.006	0.192 ± 0.132	0.265 ± 0.048	0.040 ± 0.004	0.041 ± 0.007	**0.034 ± 0.007**	0.189 ± 0.014
breast_cancer	0.062 ± 0.010	0.143 ± 0.007	0.111 ± 0.022	0.027 ± 0.011	0.294	0.072 ± 0.019	0.568 ± 0.357	0.466 ± 0.024	0.024 ± 0.014	0.016 ± 0.005	**0.013 ± 0.002**	0.293 ± 0.032
liver_disorders	0.068 ± 0.052	0.184 ± 0.038	0.129 ± 0.058	0.028 ± 0.005	0.028	0.113 ± 0.031	0.118 ± 0.095	0.450 ± 0.074	0.040 ± 0.004	**0.026 ± 0.002**	0.029 ± 0.001	0.316 ± 0.019
parkinsons	0.082 ± 0.044	0.175 ± 0.029	0.136 ± 0.013	0.056 ± 0.013	0.227	0.120 ± 0.015	0.143 ± 0.175	0.578 ± 0.075	0.049 ± 0.004	0.053 ± 0.016	**0.039 ± 0.013**	0.269 ± 0.016
obesity	0.022 ± 0.002	0.044 ± 0.017	0.090 ± 0.015	0.057 ± 0.011	0.044	0.047 ± 0.075	0.126 ± 0.168	0.158 ± 0.006	0.033 ± 0.011	0.030 ± 0.004	**0.010 ± 0.001**	0.200 ± 0.006
diabetes	0.024 ± 0.016	0.030 ± 0.003	0.035 ± 0.014	0.067 ± 0.007	–	0.062 ± 0.041	0.174 ± 0.103	0.094 ± 0.055	0.032 ± 0.020	0.039 ± 0.005	**0.007 ± 0.001**	0.225 ± 0.020
german_credit	0.207 ± 0.571	0.020 ± 0.008	0.171 ± 0.014	0.008 ± 0.003	–	0.019 ± 0.001	0.046 ± 0.025	0.067 ± 0.009	0.023 ± 0.002	0.007 ± 0.004	**0.005 ± 0.004**	0.140 ± 0.009
bank_marketing	0.016 ± 0.004	**0.009 ± 0.005**	0.040 ± 0.019	0.015 ± 0.006	–	0.040 ± 0.016	0.088 ± 0.110	0.083 ± 0.034	0.040 ± 0.017	0.014 ± 0.002	0.014 ± 0.001	0.275 ± 0.032
census_income	0.050 ± 0.039	**0.019 ± 0.005**	0.055 ± 0.022	0.049 ± 0.060	–	0.082 ± 0.067	0.182 ± 0.344	0.145 ± 0.108	0.085 ± 0.029	0.037 ± 0.007	0.052 ± 0.020	0.347 ± 0.009
surv_melanoma	0.064 ± 0.119	–	–	–	–	0.077 ± 0.079	0.187 ± 0.239	0.331 ± 0.072	0.033 ± 0.005	0.027 ± 0.016	**0.023 ± 0.005**	0.127 ± 0.033
surv_pbc	0.054 ± 0.086	–	–	–	–	0.021 ± 0.011	0.053 ± 0.012	0.093 ± 0.163	0.014 ± 0.009	0.005 ± 0.001	**0.004 ± 0.003**	0.070 ± 0.011
surv_pbc3	0.045 ± 0.023	–	–	–	–	0.051 ± 0.009	0.182 ± 0.169	0.303 ± 0.043	0.031 ± 0.002	0.023 ± 0.008	**0.016 ± 0.002**	0.179 ± 0.015
surv_retinopathy	0.126 ± 0.376	–	–	–	–	0.025 ± 0.022	0.043 ± 0.044	0.092 ± 0.007	0.020 ± 0.010	0.005 ± 0.004	**0.004 ± 0.003**	0.032 ± 0.006
surv_whas500	0.034 ± 0.030	–	–	–	–	0.030 ± 0.013	0.129 ± 0.116	0.244 ± 0.020	0.015 ± 0.005	0.010 ± 0.001	**0.008 ± 0.003**	0.129 ± 0.010
surv_gbsg2	0.047 ± 0.074	–	–	–	–	0.081 ± 0.110	0.131 ± 0.218	0.163 ± 0.058	0.016 ± 0.003	**0.011 ± 0.003**	0.033 ± 0.002	0.258 ± 0.017
surv_colon	0.012 ± 0.011	–	–	–	–	0.051 ± 0.064	0.104 ± 0.082	0.087 ± 0.019	0.008 ± 0.004	**0.003 ± 0.000**	0.005 ± 0.001	0.132 ± 0.017
surv_grace	0.023 ± 0.008	–	–	–	–	0.069 ± 0.042	0.225 ± 0.495	0.269 ± 0.055	0.015 ± 0.012	**0.005 ± 0.001**	0.007 ± 0.001	0.149 ± 0.019
surv_trace	0.010 ± 0.012	–	–	–	–	0.058 ± 0.041	0.100 ± 0.047	0.180 ± 0.130	0.020 ± 0.009	**0.005 ± 0.000**	0.006 ± 0.001	0.134 ± 0.003
surv_framingham	0.025 ± 0.005	–	–	–	–	0.060 ± 0.020	0.297 ± 0.057	0.226 ± 0.165	0.035 ± 0.011	0.009 ± 0.002	**0.007 ± 0.002**	0.205 ± 0.008
tab_blood_transfusion	0.123 ± 0.173	–	–	–	–	0.173 ± 0.049	0.228 ± 0.092	0.425 ± 0.105	0.043 ± 0.020	0.025 ± 0.014	**0.024 ± 0.002**	0.333 ± 0.025
tab_credit_g	0.185 ± 0.003	–	–	–	–	0.019 ± 0.016	0.067 ± 0.091	0.063 ± 0.006	0.021 ± 0.009	0.007 ± 0.001	**0.004 ± 0.000**	0.143 ± 0.004
tab_qsar_biodeg	0.071 ± 0.009	–	–	–	–	0.039 ± 0.016	0.272 ± 0.226	0.144 ± 0.022	0.029 ± 0.005	0.016 ± 0.003	**0.008 ± 0.001**	0.391 ± 0.011
tab_maternal_risk	**0.043 ± 0.035**	–	–	–	–	0.235 ± 0.132	0.362 ± 0.175	0.476 ± 0.047	0.140 ± 0.020	0.125 ± 0.022	0.125 ± 0.015	0.360 ± 0.014
tab_website_phishing	0.100 ± 0.394	–	–	–	–	0.005 ± 0.006	0.060 ± 0.052	0.003 ± 0.005	0.008 ± 0.005	**0.000 ± 0.000**	0.001 ± 0.001	0.028 ± 0.004

Table 5: KS statistic (↓) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.190 ± 0.134	0.201 ± 0.045	0.335 ± 0.035	**0.084 ± 0.022**	0.262	0.162 ± 0.021	0.356 ± 0.130	0.246 ± 0.036	0.101 ± 0.045	0.092 ± 0.008	0.089 ± 0.061	0.313 ± 0.032
breast_cancer	0.206 ± 0.055	0.329 ± 0.017	0.252 ± 0.085	0.103 ± 0.016	0.348	0.191 ± 0.029	0.736 ± 0.296	0.432 ± 0.062	0.088 ± 0.040	0.073 ± 0.035	**0.059 ± 0.016**	0.489 ± 0.028
liver_disorders	0.229 ± 0.141	0.415 ± 0.127	0.265 ± 0.099	0.074 ± 0.014	0.073	0.241 ± 0.021	0.278 ± 0.215	0.358 ± 0.098	0.123 ± 0.082	**0.071 ± 0.015**	0.080 ± 0.019	0.479 ± 0.063
parkinsons	0.184 ± 0.046	0.382 ± 0.051	0.268 ± 0.028	0.098 ± 0.016	0.293	0.253 ± 0.069	0.301 ± 0.247	0.454 ± 0.050	0.112 ± 0.017	0.106 ± 0.059	**0.074 ± 0.014**	0.454 ± 0.037
obesity	0.258 ± 0.011	0.239 ± 0.111	0.235 ± 0.033	0.216 ± 0.024	0.156	0.217 ± 0.063	0.354 ± 0.349	0.289 ± 0.023	0.161 ± 0.023	0.123 ± 0.022	**0.113 ± 0.003**	0.305 ± 0.017
diabetes	0.150 ± 0.075	0.121 ± 0.025	0.097 ± 0.020	0.120 ± 0.042	–	0.129 ± 0.030	0.262 ± 0.082	0.158 ± 0.058	0.130 ± 0.050	0.067 ± 0.010	**0.046 ± 0.001**	0.400 ± 0.028
german_credit	0.148 ± 0.144	0.125 ± 0.034	0.381 ± 0.075	0.046 ± 0.022	–	0.099 ± 0.049	0.249 ± 0.094	0.143 ± 0.012	0.095 ± 0.041	0.043 ± 0.006	**0.040 ± 0.003**	0.359 ± 0.017
bank_marketing	0.104 ± 0.024	0.126 ± 0.022	0.171 ± 0.025	0.212 ± 0.048	–	0.169 ± 0.144	0.414 ± 0.263	0.285 ± 0.027	0.140 ± 0.018	**0.061 ± 0.007**	0.110 ± 0.011	0.643 ± 0.015
census_income	0.445 ± 0.094	0.189 ± 0.135	0.227 ± 0.051	0.282 ± 0.273	–	0.272 ± 0.021	0.492 ± 0.639	0.211 ± 0.127	0.224 ± 0.167	**0.126 ± 0.009**	0.197 ± 0.007	0.607 ± 0.021
surv_melanoma	0.182 ± 0.297	–	–	–	–	0.156 ± 0.117	0.317 ± 0.285	0.239 ± 0.080	0.098 ± 0.016	**0.068 ± 0.015**	0.071 ± 0.033	0.236 ± 0.016
surv_pbc	0.116 ± 0.054	–	–	–	–	0.119 ± 0.058	0.189 ± 0.073	0.147 ± 0.295	0.103 ± 0.044	0.032 ± 0.038	**0.030 ± 0.019**	0.174 ± 0.008
surv_pbc3	0.143 ± 0.067	–	–	–	–	0.178 ± 0.041	0.347 ± 0.210	0.302 ± 0.049	0.103 ± 0.027	0.080 ± 0.013	**0.057 ± 0.018**	0.360 ± 0.006
surv_retinopathy	0.098 ± 0.077	–	–	–	–	0.107 ± 0.050	0.164 ± 0.162	0.138 ± 0.025	0.105 ± 0.053	0.039 ± 0.007	**0.032 ± 0.015**	0.174 ± 0.014
surv_whas500	0.115 ± 0.088	–	–	–	–	0.092 ± 0.041	0.230 ± 0.154	0.185 ± 0.060	0.065 ± 0.016	0.040 ± 0.012	**0.037 ± 0.004**	0.268 ± 0.017
surv_gbsg2	0.158 ± 0.207	–	–	–	–	0.286 ± 0.231	0.397 ± 0.292	0.321 ± 0.052	0.076 ± 0.012	0.095 ± 0.022	**0.054 ± 0.019**	0.570 ± 0.040
surv_colon	0.068 ± 0.031	–	–	–	–	0.161 ± 0.150	0.254 ± 0.205	0.102 ± 0.028	0.067 ± 0.027	**0.032 ± 0.004**	0.045 ± 0.003	0.320 ± 0.027
surv_grace	0.087 ± 0.040	–	–	–	–	0.185 ± 0.116	0.363 ± 0.481	0.160 ± 0.028	0.091 ± 0.063	**0.030 ± 0.013**	0.050 ± 0.009	0.241 ± 0.024
surv_trace	0.061 ± 0.066	–	–	–	–	0.110 ± 0.041	0.230 ± 0.058	0.190 ± 0.231	0.086 ± 0.036	0.025 ± 0.006	**0.024 ± 0.003**	0.285 ± 0.013
surv_framingham	0.079 ± 0.039	–	–	–	–	0.132 ± 0.066	0.436 ± 0.021	0.297 ± 0.357	0.122 ± 0.056	0.042 ± 0.009	**0.032 ± 0.001**	0.316 ± 0.018
tab_blood_transfusion	0.264 ± 0.336	–	–	–	–	0.336 ± 0.008	0.458 ± 0.259	0.280 ± 0.087	0.133 ± 0.064	**0.067 ± 0.038**	0.099 ± 0.009	0.537 ± 0.029
tab_credit_g	0.110 ± 0.034	–	–	–	–	0.113 ± 0.075	0.273 ± 0.324	0.137 ± 0.012	0.091 ± 0.024	0.056 ± 0.017	**0.042 ± 0.013**	0.365 ± 0.012
tab_qsar_biodeg	0.192 ± 0.065	–	–	–	–	0.131 ± 0.030	0.412 ± 0.171	0.206 ± 0.019	0.105 ± 0.005	0.068 ± 0.012	**0.061 ± 0.003**	0.573 ± 0.015
tab_maternal_risk	0.234 ± 0.041	–	–	–	–	0.291 ± 0.258	0.435 ± 0.263	0.313 ± 0.029	0.144 ± 0.039	**0.111 ± 0.026**	0.181 ± 0.021	0.454 ± 0.010
tab_website_phishing	–	–	–	–	–	–	–	–	–	–	–	–

Table 6: KL divergence (↓) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.396 ± 0.118	0.417 ± 0.141	5.073 ± 2.677	0.218 ± 0.021	0.889	0.739 ± 0.149	0.988 ± 1.095	6.855 ± 1.418	0.261 ± 0.126	**0.189 ± 0.060**	0.193 ± 0.014	0.548 ± 0.087
breast_cancer	0.499 ± 0.214	0.531 ± 0.050	1.804 ± 0.285	0.260 ± 0.236	1.136	0.522 ± 0.361	6.607 ± 11.311	11.996 ± 1.070	0.173 ± 0.047	0.146 ± 0.014	**0.071 ± 0.009**	0.801 ± 0.102
liver_disorders	0.432 ± 0.222	0.889 ± 0.385	3.393 ± 2.232	0.304 ± 0.132	0.261	1.205 ± 0.209	0.703 ± 1.668	11.831 ± 0.843	0.367 ± 0.232	**0.191 ± 0.103**	0.253 ± 0.051	0.887 ± 0.088
parkinsons	1.233 ± 0.905	1.241 ± 0.611	2.603 ± 0.351	0.430 ± 0.264	1.399	1.338 ± 0.547	0.973 ± 1.641	15.105 ± 2.132	0.508 ± 0.330	0.536 ± 0.018	**0.396 ± 0.294**	0.766 ± 0.056
obesity	0.109 ± 0.106	0.122 ± 0.045	1.345 ± 0.151	0.233 ± 0.141	0.124	0.360 ± 0.562	0.456 ± 0.765	4.246 ± 0.378	0.101 ± 0.038	0.089 ± 0.016	**0.026 ± 0.000**	0.568 ± 0.006
diabetes	0.104 ± 0.126	0.079 ± 0.006	0.360 ± 0.468	0.175 ± 0.041	–	0.380 ± 0.574	0.584 ± 0.449	2.691 ± 0.966	0.093 ± 0.045	0.088 ± 0.014	**0.015 ± 0.003**	0.570 ± 0.056
german_credit	2.010 ± 2.123	0.058 ± 0.035	3.982 ± 0.208	0.036 ± 0.017	–	0.142 ± 0.176	0.180 ± 0.244	1.708 ± 0.256	0.076 ± 0.052	0.030 ± 0.008	**0.015 ± 0.011**	0.368 ± 0.027
bank_marketing	0.180 ± 0.117	**0.028 ± 0.012**	0.615 ± 0.457	0.111 ± 0.106	–	0.235 ± 0.192	0.300 ± 0.530	2.197 ± 1.182	0.133 ± 0.087	0.039 ± 0.005	0.039 ± 0.003	0.819 ± 0.139
census_income	0.450 ± 0.102	**0.062 ± 0.018**	0.987 ± 0.673	0.155 ± 0.086	–	0.463 ± 0.390	1.749 ± 6.259	3.233 ± 3.534	0.257 ± 0.069	0.090 ± 0.017	0.132 ± 0.081	1.137 ± 0.033
surv_melanoma	0.355 ± 0.307	–	–	–	–	0.812 ± 1.141	1.628 ± 4.267	8.892 ± 1.341	0.200 ± 0.149	0.198 ± 0.144	**0.162 ± 0.109**	0.343 ± 0.092
surv_pbc	0.545 ± 1.516	–	–	–	–	0.183 ± 0.314	0.157 ± 0.086	1.650 ± 3.313	0.087 ± 0.091	0.031 ± 0.043	**0.011 ± 0.008**	0.175 ± 0.028
surv_pbc3	0.350 ± 0.042	–	–	–	–	0.488 ± 0.077	1.163 ± 2.106	7.724 ± 1.264	0.231 ± 0.188	0.162 ± 0.110	**0.092 ± 0.032**	0.502 ± 0.071
surv_retinopathy	2.306 ± 9.510	–	–	–	–	0.106 ± 0.085	0.178 ± 0.227	2.314 ± 0.104	0.059 ± 0.027	0.016 ± 0.013	**0.011 ± 0.008**	0.092 ± 0.022
surv_whas500	0.258 ± 0.113	–	–	–	–	0.241 ± 0.108	0.495 ± 0.674	5.838 ± 0.587	0.106 ± 0.104	0.064 ± 0.024	**0.048 ± 0.029**	0.339 ± 0.018
surv_gbsg2	0.311 ± 0.195	–	–	–	–	0.586 ± 0.792	0.609 ± 1.635	4.183 ± 1.379	0.138 ± 0.071	**0.094 ± 0.033**	0.153 ± 0.043	0.780 ± 0.034
surv_colon	0.108 ± 0.095	–	–	–	–	0.273 ± 0.301	0.616 ± 0.901	2.297 ± 1.226	0.031 ± 0.010	**0.018 ± 0.016**	0.023 ± 0.005	0.350 ± 0.047
surv_grace	0.289 ± 0.166	–	–	–	–	0.862 ± 1.509	2.834 ± 10.957	6.324 ± 1.459	0.085 ± 0.060	**0.032 ± 0.015**	0.034 ± 0.026	0.384 ± 0.060
surv_trace	0.093 ± 0.018	–	–	–	–	0.304 ± 0.308	0.298 ± 0.127	3.033 ± 1.847	0.059 ± 0.026	**0.015 ± 0.007**	0.017 ± 0.004	0.331 ± 0.009
surv_framingham	0.312 ± 0.015	–	–	–	–	0.407 ± 0.393	1.058 ± 0.287	3.481 ± 6.399	0.114 ± 0.039	0.029 ± 0.010	**0.021 ± 0.008**	0.525 ± 0.027
tab_blood_transfusion	1.269 ± 2.769	–	–	–	–	2.008 ± 0.824	1.265 ± 1.770	11.252 ± 1.432	0.251 ± 0.073	0.167 ± 0.107	**0.164 ± 0.087**	0.906 ± 0.074
tab_credit_g	3.869 ± 0.059	–	–	–	–	0.098 ± 0.075	0.366 ± 0.862	1.565 ± 0.130	0.064 ± 0.038	0.023 ± 0.004	**0.015 ± 0.006**	0.377 ± 0.013
tab_qsar_biodeg	0.776 ± 0.298	–	–	–	–	0.296 ± 0.175	3.983 ± 4.591	3.813 ± 0.579	0.180 ± 0.084	0.109 ± 0.014	**0.051 ± 0.012**	1.125 ± 0.028
tab_maternal_risk	0.715 ± 0.658	–	–	–	–	1.545 ± 1.718	3.575 ± 4.765	12.353 ± 0.949	0.412 ± 0.222	**0.335 ± 0.050**	0.452 ± 0.173	1.023 ± 0.059
tab_website_phishing	0.854 ± 3.562	–	–	–	–	0.013 ± 0.017	0.194 ± 0.199	0.009 ± 0.014	0.022 ± 0.011	**0.001 ± 0.000**	0.002 ± 0.002	0.074 ± 0.012

Table 7: Hellinger (↓) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.183 ± 0.099	0.208 ± 0.010	0.419 ± 0.047	0.137 ± 0.008	0.326	0.195 ± 0.025	0.319 ± 0.127	0.376 ± 0.039	0.147 ± 0.013	0.137 ± 0.016	**0.134 ± 0.032**	0.322 ± 0.018
breast_cancer	0.209 ± 0.015	0.313 ± 0.013	0.297 ± 0.028	0.138 ± 0.015	0.490	0.219 ± 0.031	0.668 ± 0.263	0.636 ± 0.019	0.132 ± 0.033	0.111 ± 0.015	**0.099 ± 0.006**	0.477 ± 0.027
liver_disorders	0.219 ± 0.094	0.360 ± 0.045	0.331 ± 0.090	0.147 ± 0.013	0.147	0.290 ± 0.047	0.281 ± 0.104	0.633 ± 0.042	0.176 ± 0.011	**0.144 ± 0.014**	0.150 ± 0.013	0.501 ± 0.018
parkinsons	0.238 ± 0.079	0.357 ± 0.033	0.333 ± 0.018	0.208 ± 0.028	0.427	0.301 ± 0.022	0.320 ± 0.186	0.718 ± 0.051	0.195 ± 0.010	0.204 ± 0.030	**0.178 ± 0.033**	0.453 ± 0.017
obesity	0.098 ± 0.018	0.149 ± 0.022	0.245 ± 0.024	0.133 ± 0.011	0.154	0.143 ± 0.126	0.241 ± 0.170	0.277 ± 0.014	0.128 ± 0.028	0.103 ± 0.011	**0.059 ± 0.003**	0.367 ± 0.007
diabetes	0.118 ± 0.043	0.135 ± 0.011	0.142 ± 0.028	0.148 ± 0.005	–	0.161 ± 0.056	0.302 ± 0.127	0.208 ± 0.060	0.133 ± 0.038	0.116 ± 0.006	**0.051 ± 0.005**	0.402 ± 0.020
german_credit	0.309 ± 0.604	0.087 ± 0.010	0.368 ± 0.015	0.043 ± 0.005	–	0.082 ± 0.009	0.126 ± 0.043	0.138 ± 0.011	0.112 ± 0.016	0.040 ± 0.008	**0.039 ± 0.007**	0.300 ± 0.010
bank_marketing	0.102 ± 0.019	0.069 ± 0.021	0.163 ± 0.041	0.065 ± 0.010	–	0.112 ± 0.029	0.200 ± 0.145	0.178 ± 0.048	0.140 ± 0.036	0.066 ± 0.002	**0.065 ± 0.011**	0.416 ± 0.020
census_income	0.182 ± 0.065	**0.109 ± 0.020**	0.186 ± 0.037	0.111 ± 0.069	–	0.194 ± 0.067	0.308 ± 0.328	0.274 ± 0.092	0.213 ± 0.050	0.124 ± 0.009	0.126 ± 0.017	0.491 ± 0.007
surv_melanoma	0.189 ± 0.222	–	–	–	–	0.200 ± 0.101	0.328 ± 0.231	0.436 ± 0.036	0.144 ± 0.019	0.120 ± 0.028	**0.110 ± 0.017**	0.269 ± 0.044
surv_pbc	0.164 ± 0.146	–	–	–	–	0.098 ± 0.019	0.157 ± 0.078	0.153 ± 0.175	0.090 ± 0.024	0.036 ± 0.011	**0.035 ± 0.020**	0.184 ± 0.017
surv_pbc3	0.165 ± 0.040	–	–	–	–	0.170 ± 0.025	0.311 ± 0.156	0.417 ± 0.041	0.142 ± 0.005	0.110 ± 0.020	**0.093 ± 0.003**	0.332 ± 0.016
surv_retinopathy	0.212 ± 0.343	–	–	–	–	0.105 ± 0.051	0.140 ± 0.067	0.144 ± 0.010	0.094 ± 0.005	0.042 ± 0.015	**0.037 ± 0.010**	0.115 ± 0.013
surv_whas500	0.131 ± 0.069	–	–	–	–	0.113 ± 0.023	0.233 ± 0.122	0.332 ± 0.045	0.086 ± 0.015	0.065 ± 0.005	**0.060 ± 0.012**	0.273 ± 0.006
surv_gbsg2	0.162 ± 0.127	–	–	–	–	0.206 ± 0.180	0.254 ± 0.261	0.294 ± 0.049	0.100 ± 0.024	**0.077 ± 0.012**	0.116 ± 0.014	0.396 ± 0.015
surv_colon	0.081 ± 0.046	–	–	–	–	0.145 ± 0.107	0.187 ± 0.107	0.162 ± 0.024	0.062 ± 0.017	**0.031 ± 0.005**	0.045 ± 0.005	0.262 ± 0.017
surv_grace	0.119 ± 0.038	–	–	–	–	0.193 ± 0.069	0.341 ± 0.442	0.383 ± 0.049	0.094 ± 0.032	**0.050 ± 0.004**	0.062 ± 0.007	0.271 ± 0.008
surv_trace	0.069 ± 0.065	–	–	–	–	0.148 ± 0.045	0.218 ± 0.079	0.275 ± 0.061	0.100 ± 0.021	**0.036 ± 0.005**	0.040 ± 0.006	0.282 ± 0.004
surv_framingham	0.133 ± 0.003	–	–	–	–	0.190 ± 0.036	0.432 ± 0.041	0.382 ± 0.152	0.150 ± 0.025	0.065 ± 0.008	**0.058 ± 0.003**	0.359 ± 0.003
tab_blood_transfusion	0.289 ± 0.247	–	–	–	–	0.364 ± 0.044	0.399 ± 0.075	0.609 ± 0.078	0.178 ± 0.040	0.138 ± 0.038	**0.132 ± 0.014**	0.499 ± 0.029
tab_credit_g	0.287 ± 0.015	–	–	–	–	0.086 ± 0.043	0.165 ± 0.131	0.135 ± 0.002	0.099 ± 0.030	0.041 ± 0.003	**0.037 ± 0.006**	0.304 ± 0.004
tab_qsar_biodeg	0.208 ± 0.006	–	–	–	–	0.144 ± 0.027	0.401 ± 0.195	0.290 ± 0.020	0.133 ± 0.013	0.095 ± 0.008	**0.069 ± 0.002**	0.550 ± 0.008
tab_maternal_risk	**0.175 ± 0.057**	–	–	–	–	0.397 ± 0.119	0.513 ± 0.162	0.618 ± 0.026	0.304 ± 0.025	0.288 ± 0.014	0.287 ± 0.019	0.538 ± 0.010
tab_website_phishing	0.167 ± 0.437	–	–	–	–	0.050 ± 0.023	0.184 ± 0.070	0.039 ± 0.033	0.066 ± 0.009	**0.015 ± 0.004**	0.020 ± 0.008	0.121 ± 0.010

Table 8: Pearson corr. preservation (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.927 ± 0.008	0.904 ± 0.030	0.926 ± 0.013	0.930 ± 0.012	0.822	0.912 ± 0.018	0.844 ± 0.042	0.918 ± 0.012	0.908 ± 0.011	**0.939 ± 0.019**	0.907 ± 0.037	0.900 ± 0.005
breast_cancer	0.868 ± 0.033	0.606 ± 0.006	0.799 ± 0.009	**0.948 ± 0.018**	0.773	0.887 ± 0.039	0.566 ± 0.047	0.753 ± 0.028	0.694 ± 0.048	0.899 ± 0.008	0.945 ± 0.028	0.607 ± 0.012
liver_disorders	0.917 ± 0.103	0.744 ± 0.034	0.831 ± 0.125	0.948 ± 0.005	0.912	0.862 ± 0.092	0.640 ± 0.144	0.734 ± 0.068	0.814 ± 0.021	**0.949 ± 0.010**	0.899 ± 0.063	0.743 ± 0.034
parkinsons	0.845 ± 0.085	0.487 ± 0.042	0.653 ± 0.012	0.936 ± 0.010	0.739	0.814 ± 0.054	0.481 ± 0.057	0.606 ± 0.043	0.610 ± 0.052	0.863 ± 0.021	**0.950 ± 0.006**	0.486 ± 0.030
obesity	0.945 ± 0.024	0.871 ± 0.007	0.895 ± 0.014	0.941 ± 0.014	0.847	0.953 ± 0.088	0.856 ± 0.053	0.923 ± 0.004	0.892 ± 0.022	0.964 ± 0.021	**0.974 ± 0.008**	0.874 ± 0.010
diabetes	0.928 ± 0.102	0.898 ± 0.033	0.934 ± 0.037	0.916 ± 0.011	–	0.968 ± 0.008	0.882 ± 0.039	0.905 ± 0.015	0.918 ± 0.032	**0.982 ± 0.005**	0.956 ± 0.005	0.868 ± 0.013
german_credit	0.926 ± 0.061	0.896 ± 0.015	0.885 ± 0.052	0.933 ± 0.020	–	0.933 ± 0.023	0.879 ± 0.022	0.927 ± 0.029	0.905 ± 0.004	0.953 ± 0.012	**0.961 ± 0.022**	0.897 ± 0.008
bank_marketing	0.952 ± 0.025	0.943 ± 0.028	0.896 ± 0.062	0.966 ± 0.022	–	0.969 ± 0.016	0.922 ± 0.008	0.969 ± 0.031	0.947 ± 0.006	**0.979 ± 0.014**	0.965 ± 0.011	0.940 ± 0.002
census_income	0.951 ± 0.029	0.953 ± 0.018	0.942 ± 0.004	0.980 ± 0.005	–	0.970 ± 0.013	0.914 ± 0.030	0.961 ± 0.014	0.946 ± 0.021	0.980 ± 0.007	**0.982 ± 0.003**	0.941 ± 0.005
surv_melanoma	0.910 ± 0.054	–	–	–	–	0.855 ± 0.119	0.794 ± 0.028	0.848 ± 0.022	0.865 ± 0.109	**0.935 ± 0.015**	0.924 ± 0.027	0.827 ± 0.065
surv_pbc	0.877 ± 0.148	–	–	–	–	0.924 ± 0.023	0.862 ± 0.055	0.887 ± 0.034	0.897 ± 0.063	0.938 ± 0.050	**0.943 ± 0.037**	0.861 ± 0.029
surv_pbc3	0.912 ± 0.031	–	–	–	–	0.916 ± 0.030	0.857 ± 0.023	0.889 ± 0.007	0.882 ± 0.014	0.939 ± 0.011	**0.942 ± 0.010**	0.863 ± 0.021
surv_retinopathy	0.958 ± 0.041	–	–	–	–	0.868 ± 0.017	0.784 ± 0.037	0.852 ± 0.077	0.813 ± 0.027	0.899 ± 0.040	**0.964 ± 0.079**	0.779 ± 0.034
surv_whas500	0.918 ± 0.025	–	–	–	–	0.931 ± 0.011	0.870 ± 0.023	0.917 ± 0.003	0.914 ± 0.010	0.942 ± 0.003	**0.944 ± 0.011**	0.897 ± 0.009
surv_gbsg2	0.902 ± 0.088	–	–	–	–	0.882 ± 0.116	0.803 ± 0.059	0.884 ± 0.081	0.887 ± 0.048	**0.949 ± 0.054**	0.921 ± 0.036	0.848 ± 0.018
surv_colon	0.957 ± 0.026	–	–	–	–	0.915 ± 0.062	0.911 ± 0.004	0.947 ± 0.005	0.931 ± 0.009	**0.967 ± 0.018**	0.940 ± 0.006	0.917 ± 0.013
surv_grace	0.951 ± 0.031	–	–	–	–	0.942 ± 0.031	0.883 ± 0.074	0.918 ± 0.014	0.950 ± 0.036	**0.978 ± 0.009**	0.923 ± 0.038	0.919 ± 0.041
surv_trace	0.965 ± 0.018	–	–	–	–	0.931 ± 0.058	0.853 ± 0.021	0.877 ± 0.046	0.924 ± 0.057	0.977 ± 0.017	**0.978 ± 0.014**	0.876 ± 0.009
surv_framingham	0.969 ± 0.012	–	–	–	–	0.964 ± 0.004	0.792 ± 0.016	0.887 ± 0.017	0.883 ± 0.025	**0.980 ± 0.005**	0.921 ± 0.015	0.833 ± 0.006
tab_blood_transfusion	0.683 ± 0.613	–	–	–	–	0.880 ± 0.110	0.576 ± 0.145	0.680 ± 0.163	0.731 ± 0.142	**0.934 ± 0.039**	0.864 ± 0.076	0.549 ± 0.092
tab_credit_g	0.959 ± 0.033	–	–	–	–	0.930 ± 0.033	0.889 ± 0.033	0.930 ± 0.002	0.903 ± 0.016	0.954 ± 0.025	**0.966 ± 0.006**	0.888 ± 0.009

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
tab_qsar_biodeg	0.791 ± 0.045	–	–	–	–	0.906 ± 0.010	0.809 ± 0.025	0.873 ± 0.002	0.837 ± 0.014	0.933 ± 0.009	$\mathbf{0.966 \pm 0.006}$	0.829 ± 0.003
tab_maternal_risk	0.916 ± 0.059	–	–	–	–	0.906 ± 0.081	0.751 ± 0.036	0.882 ± 0.039	0.822 ± 0.033	$\mathbf{0.957 \pm 0.009}$	0.910 ± 0.048	0.750 ± 0.035
tab_website_phishing	–	–	–	–	–	–	–	–	–	–	–	–

Table 9: JS numeric (↓) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.054 ± 0.054	0.094 ± 0.020	0.200 ± 0.033	0.042 ± 0.002	0.194	0.070 ± 0.006	0.192 ± 0.132	0.265 ± 0.048	0.040 ± 0.004	0.041 ± 0.007	$\mathbf{0.034 \pm 0.007}$	0.189 ± 0.014
breast_cancer	0.062 ± 0.010	0.143 ± 0.007	0.111 ± 0.022	0.027 ± 0.011	0.294	0.072 ± 0.019	0.568 ± 0.357	0.466 ± 0.024	0.024 ± 0.014	0.016 ± 0.005	$\mathbf{0.013 \pm 0.002}$	0.293 ± 0.032
liver_disorders	0.068 ± 0.052	0.184 ± 0.038	0.129 ± 0.058	0.028 ± 0.005	0.028	0.113 ± 0.031	0.118 ± 0.095	0.450 ± 0.074	0.040 ± 0.004	$\mathbf{0.026 \pm 0.002}$	0.029 ± 0.001	0.316 ± 0.019
parkinsons	0.082 ± 0.044	0.175 ± 0.029	0.136 ± 0.013	0.056 ± 0.013	0.227	0.120 ± 0.015	0.143 ± 0.175	0.578 ± 0.075	0.049 ± 0.004	0.053 ± 0.016	$\mathbf{0.039 \pm 0.013}$	0.269 ± 0.016
obesity	0.040 ± 0.003	0.081 ± 0.036	0.136 ± 0.030	0.114 ± 0.022	0.072	0.093 ± 0.144	0.237 ± 0.334	0.313 ± 0.013	0.052 ± 0.014	0.059 ± 0.008	$\mathbf{0.019 \pm 0.002}$	0.206 ± 0.008
diabetes	0.028 ± 0.022	0.033 ± 0.003	0.030 ± 0.011	0.089 ± 0.009	–	0.082 ± 0.054	0.216 ± 0.114	0.125 ± 0.072	0.036 ± 0.022	0.052 ± 0.007	$\mathbf{0.009 \pm 0.002}$	0.248 ± 0.024
german_credit	0.050 ± 0.076	0.047 ± 0.021	0.202 ± 0.060	0.023 ± 0.009	–	0.044 ± 0.004	0.111 ± 0.054	0.181 ± 0.026	0.031 ± 0.014	0.019 ± 0.010	$\mathbf{0.013 \pm 0.011}$	0.160 ± 0.013
bank_marketing	0.019 ± 0.006	0.010 ± 0.006	0.043 ± 0.030	0.035 ± 0.013	–	0.027 ± 0.013	0.098 ± 0.163	0.123 ± 0.003	0.025 ± 0.015	0.008 ± 0.002	$\mathbf{0.004 \pm 0.002}$	0.454 ± 0.076
census_income	0.052 ± 0.049	0.020 ± 0.008	0.071 ± 0.036	0.115 ± 0.139	–	0.069 ± 0.034	0.188 ± 0.498	0.139 ± 0.098	0.046 ± 0.023	0.033 ± 0.004	$\mathbf{0.019 \pm 0.002}$	0.445 ± 0.025
surv_melanoma	0.064 ± 0.119	–	–	–	–	0.077 ± 0.079	0.187 ± 0.239	0.331 ± 0.072	0.033 ± 0.005	0.027 ± 0.016	$\mathbf{0.023 \pm 0.005}$	0.127 ± 0.033
surv_pbc	0.039 ± 0.074	–	–	–	–	0.025 ± 0.014	0.062 ± 0.012	0.111 ± 0.198	0.016 ± 0.012	0.005 ± 0.001	$\mathbf{0.005 \pm 0.003}$	0.058 ± 0.005
surv_pbc3	0.042 ± 0.028	–	–	–	–	0.055 ± 0.010	0.197 ± 0.181	0.330 ± 0.047	0.030 ± 0.003	0.024 ± 0.009	$\mathbf{0.017 \pm 0.002}$	0.194 ± 0.016
surv_retinopathy	0.018 ± 0.007	–	–	–	–	0.036 ± 0.023	0.065 ± 0.082	0.159 ± 0.013	0.031 ± 0.020	0.009 ± 0.008	$\mathbf{0.007 \pm 0.005}$	0.054 ± 0.012
surv_whas500	0.034 ± 0.030	–	–	–	–	0.030 ± 0.013	0.129 ± 0.116	0.244 ± 0.020	0.015 ± 0.005	0.010 ± 0.001	$\mathbf{0.008 \pm 0.003}$	0.129 ± 0.010
surv_gbsg2	0.038 ± 0.057	–	–	–	–	0.109 ± 0.127	0.198 ± 0.306	0.260 ± 0.092	0.022 ± 0.004	0.017 ± 0.005	$\mathbf{0.014 \pm 0.006}$	0.392 ± 0.027
surv_colon	0.011 ± 0.007	–	–	–	–	0.056 ± 0.077	0.128 ± 0.102	0.110 ± 0.024	0.009 ± 0.006	$\mathbf{0.004 \pm 0.000}$	0.006 ± 0.000	0.145 ± 0.019
surv_grace	0.023 ± 0.008	–	–	–	–	0.069 ± 0.042	0.225 ± 0.495	0.269 ± 0.055	0.015 ± 0.012	$\mathbf{0.005 \pm 0.001}$	0.007 ± 0.001	0.149 ± 0.019
surv_trace	0.010 ± 0.012	–	–	–	–	0.058 ± 0.041	0.100 ± 0.047	0.180 ± 0.130	0.020 ± 0.009	$\mathbf{0.005 \pm 0.000}$	0.006 ± 0.001	0.134 ± 0.003
surv_framingham	0.025 ± 0.005	–	–	–	–	0.060 ± 0.020	0.297 ± 0.057	0.226 ± 0.165	0.035 ± 0.011	0.009 ± 0.002	$\mathbf{0.007 \pm 0.002}$	0.205 ± 0.008
tab_blood_transfusion	0.123 ± 0.173	–	–	–	–	0.173 ± 0.049	0.228 ± 0.092	0.425 ± 0.105	0.043 ± 0.020	0.025 ± 0.014	$\mathbf{0.024 \pm 0.002}$	0.333 ± 0.025
tab_credit_g	0.026 ± 0.012	–	–	–	–	0.043 ± 0.037	0.145 ± 0.226	0.170 ± 0.018	0.029 ± 0.008	0.019 ± 0.002	$\mathbf{0.010 \pm 0.002}$	0.166 ± 0.014
tab_qsar_biodeg	0.071 ± 0.009	–	–	–	–	0.039 ± 0.016	0.272 ± 0.226	0.144 ± 0.022	0.029 ± 0.005	0.016 ± 0.003	$\mathbf{0.008 \pm 0.001}$	0.391 ± 0.011
tab_maternal_risk	$\mathbf{0.043 \pm 0.035}$	–	–	–	–	0.235 ± 0.132	0.362 ± 0.175	0.476 ± 0.047	0.140 ± 0.020	0.125 ± 0.022	0.125 ± 0.015	0.360 ± 0.014
tab_website_phishing	–	–	–	–	–	–	–	–	–	–	–	–

Table 10: JS categorical (↓) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	–	–	–	–	–	–	–	–	–	–	–	–
breast_cancer	–	–	–	–	–	–	–	–	–	–	–	–
liver_disorders	–	–	–	–	–	–	–	–	–	–	–	–
parkinsons	–	–	–	–	–	–	–	–	–	–	–	–
obesity	0.004 ± 0.003	0.006 ± 0.003	0.044 ± 0.001	$\mathbf{0.000 \pm 0.000}$	0.015	0.002 ± 0.005	0.015 ± 0.018	0.002 ± 0.003	0.014 ± 0.018	0.001 ± 0.001	0.000 ± 0.001	0.195 ± 0.012
diabetes	0.010 ± 0.015	0.020 ± 0.013	0.049 ± 0.025	$\mathbf{0.000 \pm 0.000}$	–	0.003 ± 0.005	0.047 ± 0.104	0.002 ± 0.005	0.022 ± 0.028	0.000 ± 0.000	0.000 ± 0.000	0.155 ± 0.009
german_credit	0.290 ± 0.844	0.005 ± 0.001	0.153 ± 0.023	0.001 ± 0.000	–	0.005 ± 0.000	0.011 ± 0.010	0.006 ± 0.002	0.019 ± 0.005	$\mathbf{0.001 \pm 0.000}$	0.001 ± 0.000	0.129 ± 0.007
bank_marketing	0.014 ± 0.005	0.008 ± 0.004	0.038 ± 0.012	$\mathbf{0.000 \pm 0.000}$	–	0.050 ± 0.027	0.079 ± 0.070	0.052 ± 0.061	0.051 ± 0.040	0.018 ± 0.002	0.023 ± 0.000	0.135 ± 0.003
census_income	0.049 ± 0.032	0.018 ± 0.014	0.043 ± 0.014	$\mathbf{0.000 \pm 0.000}$	–	0.092 ± 0.109	0.177 ± 0.239	0.150 ± 0.195	0.114 ± 0.036	0.040 ± 0.009	0.076 ± 0.034	0.274 ± 0.003
surv_melanoma	–	–	–	–	–	–	–	–	–	–	–	–
surv_pbc	0.129 ± 0.460	–	–	–	–	0.004 ± 0.006	0.012 ± 0.019	0.007 ± 0.016	0.005 ± 0.011	$\mathbf{0.000 \pm 0.001}$	0.001 ± 0.002	0.129 ± 0.048
surv_pbc3	0.077 ± 0.039	–	–	–	–	0.008 ± 0.016	0.023 ± 0.037	0.008 ± 0.013	0.050 ± 0.013	0.005 ± 0.003	$\mathbf{0.003 \pm 0.002}$	0.007 ± 0.007
surv_retinopathy	0.270 ± 0.868	–	–	–	–	0.011 ± 0.025	0.013 ± 0.017	0.001 ± 0.005	0.006 ± 0.006	0.001 ± 0.001	$\mathbf{0.001 \pm 0.001}$	0.003 ± 0.003
surv_whas500	–	–	–	–	–	–	–	–	–	–	–	–
surv_gbsg2	0.061 ± 0.202	–	–	–	–	0.033 ± 0.083	0.018 ± 0.070	0.001 ± 0.001	0.005 ± 0.013	$\mathbf{0.000 \pm 0.001}$	0.066 ± 0.004	0.035 ± 0.016
surv_colon	0.012 ± 0.022	–	–	–	–	0.032 ± 0.034	0.018 ± 0.019	0.006 ± 0.006	0.007 ± 0.010	$\mathbf{0.001 \pm 0.001}$	0.002 ± 0.002	0.087 ± 0.010
surv_grace	–	–	–	–	–	–	–	–	–	–	–	–
surv_trace	–	–	–	–	–	–	–	–	–	–	–	–
surv_framingham	–	–	–	–	–	–	–	–	–	–	–	–
tab_blood_transfusion	–	–	–	–	–	–	–	–	–	–	–	–
tab_credit_g	0.271 ± 0.003	–	–	–	–	0.006 ± 0.007	0.025 ± 0.044	0.005 ± 0.001	0.017 ± 0.016	$\mathbf{0.001 \pm 0.000}$	0.001 ± 0.001	0.131 ± 0.001
tab_qsar_biodeg	–	–	–	–	–	–	–	–	–	–	–	–
tab_maternal_risk	–	–	–	–	–	–	–	–	–	–	–	–
tab_website_phishing	0.100 ± 0.394	–	–	–	–	0.005 ± 0.006	0.060 ± 0.052	0.003 ± 0.005	0.008 ± 0.005	$\mathbf{0.000 \pm 0.000}$	0.001 ± 0.001	0.028 ± 0.004

Privacy / disclosure

Table 11: DCR median (↑=less memorization) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.731 ± 0.064	0.869 ± 0.158	0.553 ± 0.234	0.736 ± 0.007	**1.971**	0.703 ± 0.090	1.043 ± 0.312	0.617 ± 0.114	0.715 ± 0.050	0.737 ± 0.022	0.722 ± 0.072	1.873 ± 0.111
breast_cancer	0.550 ± 0.162	1.245 ± 0.120	0.476 ± 0.047	0.555 ± 0.037	**2.769**	0.672 ± 0.083	1.726 ± 0.372	0.433 ± 0.045	0.850 ± 0.176	0.607 ± 0.027	0.488 ± 0.083	2.494 ± 0.208
liver_disorders	0.412 ± 0.158	0.682 ± 0.169	0.240 ± 0.013	0.382 ± 0.048	0.307	0.316 ± 0.036	0.570 ± 0.144	0.379 ± 0.087	0.376 ± 0.124	0.355 ± 0.033	0.359 ± 0.060	$\mathbf{1.677 \pm 0.319}$
parkinsons	0.483 ± 0.222	1.033 ± 0.097	0.510 ± 0.052	0.538 ± 0.063	0.928	0.625 ± 0.124	1.019 ± 0.193	0.457 ± 0.028	0.742 ± 0.102	0.570 ± 0.060	0.437 ± 0.022	$\mathbf{1.923 \pm 0.271}$
obesity	0.277 ± 0.035	0.437 ± 0.042	0.247 ± 0.025	0.403 ± 0.032	0.325	0.342 ± 0.088	0.471 ± 0.041	0.349 ± 0.023	0.426 ± 0.019	0.305 ± 0.007	0.269 ± 0.015	$\mathbf{0.706 \pm 0.031}$
diabetes	0.053 ± 0.042	0.082 ± 0.021	0.038 ± 0.009	0.116 ± 0.021	–	0.065 ± 0.008	0.118 ± 0.169	0.068 ± 0.041	0.081 ± 0.021	0.079 ± 0.001	0.060 ± 0.003	$\mathbf{0.545 \pm 0.125}$
german_credit	0.432 ± 0.113	0.456 ± 0.004	0.308 ± 0.040	0.426 ± 0.008	–	0.418 ± 0.023	0.431 ± 0.001	0.402 ± 0.022	0.446 ± 0.024	0.415 ± 0.004	0.399 ± 0.007	$\mathbf{0.642 \pm 0.028}$
bank_marketing	0.160 ± 0.082	0.254 ± 0.020	0.096 ± 0.006	0.269 ± 0.015	–	0.250 ± 0.039	0.331 ± 0.128	0.226 ± 0.033	0.289 ± 0.037	0.247 ± 0.008	0.243 ± 0.009	$\mathbf{2.937 \pm 5.289}$

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
census_income	0.087 ± 0.058	0.195 ± 0.012	0.044 ± 0.007	0.254 ± 0.137	–	0.188 ± 0.056	0.247 ± 0.050	0.189 ± 0.077	0.222 ± 0.039	0.195 ± 0.005	0.196 ± 0.001	$\mathbf{0.683 \pm 0.097}$
surv_melanoma	0.364 ± 0.093	–	–	–	–	0.375 ± 0.047	0.748 ± 0.091	0.348 ± 0.097	0.419 ± 0.084	0.376 ± 0.051	0.379 ± 0.035	$\mathbf{0.891 \pm 0.076}$
surv_pbc	0.247 ± 0.707	–	–	–	–	0.075 ± 0.031	0.158 ± 0.152	0.242 ± 0.773	0.084 ± 0.007	0.092 ± 0.021	0.087 ± 0.011	$\mathbf{0.571 \pm 0.057}$
surv_pbc3	0.568 ± 0.107	–	–	–	–	0.600 ± 0.044	0.809 ± 0.259	0.464 ± 0.025	0.575 ± 0.060	0.606 ± 0.042	0.598 ± 0.033	$\mathbf{1.414 \pm 0.109}$
surv_retinopathy	0.342 ± 0.315	–	–	–	–	0.300 ± 0.012	0.454 ± 0.134	0.321 ± 0.021	0.373 ± 0.161	0.285 ± 0.047	0.230 ± 0.050	$\mathbf{0.542 \pm 0.036}$
surv_whas500	0.643 ± 0.102	–	–	–	–	0.678 ± 0.070	1.125 ± 0.123	0.571 ± 0.116	0.702 ± 0.018	0.701 ± 0.039	0.692 ± 0.003	$\mathbf{1.822 \pm 0.304}$
surv_gbsg2	0.311 ± 0.142	–	–	–	–	0.324 ± 0.187	0.534 ± 0.309	0.261 ± 0.049	0.334 ± 0.046	0.306 ± 0.024	0.371 ± 0.052	$\mathbf{1.615 \pm 0.146}$
surv_colon	0.185 ± 0.081	–	–	–	–	0.271 ± 0.292	0.430 ± 0.059	0.142 ± 0.079	0.206 ± 0.107	0.215 ± 0.049	0.202 ± 0.019	$\mathbf{1.057 \pm 0.203}$
surv_grace	0.228 ± 0.035	–	–	–	–	0.207 ± 0.032	0.676 ± 0.890	0.191 ± 0.019	0.209 ± 0.014	0.231 ± 0.022	0.236 ± 0.010	$\mathbf{0.975 \pm 0.106}$
surv_trace	0.081 ± 0.036	–	–	–	–	0.072 ± 0.025	0.230 ± 0.356	0.238 ± 0.562	0.085 ± 0.017	0.083 ± 0.009	0.079 ± 0.002	$\mathbf{0.768 \pm 0.023}$
surv_framingham	0.306 ± 0.017	–	–	–	–	0.330 ± 0.051	0.724 ± 0.431	0.466 ± 0.464	0.342 ± 0.036	0.324 ± 0.009	0.329 ± 0.006	$\mathbf{1.356 \pm 0.084}$
tab_blood_transfusion	0.260 ± 0.312	–	–	–	–	0.172 ± 0.069	0.313 ± 0.135	0.244 ± 0.035	0.230 ± 0.058	0.196 ± 0.001	0.160 ± 0.032	$\mathbf{2.031 \pm 0.130}$
tab_credit_g	0.422 ± 0.022	–	–	–	–	0.405 ± 0.033	0.453 ± 0.097	0.395 ± 0.013	0.432 ± 0.007	0.416 ± 0.001	0.395 ± 0.011	$\mathbf{0.642 \pm 0.025}$
tab_qsar_biodeg	0.474 ± 0.102	–	–	–	–	0.591 ± 0.164	1.033 ± 0.515	0.482 ± 0.014	0.829 ± 0.365	0.473 ± 0.013	0.376 ± 0.007	$\mathbf{4.355 \pm 0.251}$
tab_maternal_risk	0.260 ± 0.125	–	–	–	–	0.353 ± 0.079	0.486 ± 0.099	0.307 ± 0.084	0.440 ± 0.081	0.359 ± 0.016	0.373 ± 0.056	$\mathbf{1.381 \pm 0.189}$
tab_website_phishing	0.322 ± 0.168	–	–	–	–	0.283 ± 0.000	$\mathbf{0.400 \pm 0.000}$	0.283 ± 0.000	0.322 ± 0.168	0.283 ± 0.000	0.283 ± 0.000	$\mathbf{0.400 \pm 0.000}$

Table 12: DCR real–real reference — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.801 ± 0.024	0.801 ± 0.024	0.801 ± 0.024	0.801 ± 0.024	0.796	0.801 ± 0.024	0.801 ± 0.024	0.801 ± 0.024	0.801 ± 0.024	0.801 ± 0.024	0.801 ± 0.024	0.801 ± 0.024
breast_cancer	0.465 ± 0.057	0.465 ± 0.057	0.465 ± 0.057	0.465 ± 0.057	0.468	0.465 ± 0.057	0.465 ± 0.057	0.465 ± 0.057	0.465 ± 0.057	0.465 ± 0.057	0.465 ± 0.057	0.465 ± 0.057
liver_disorders	0.380 ± 0.131	0.380 ± 0.131	0.380 ± 0.131	0.380 ± 0.131	0.342	0.380 ± 0.131	0.380 ± 0.131	0.380 ± 0.131	0.380 ± 0.131	0.380 ± 0.131	0.380 ± 0.131	0.380 ± 0.131
parkinsons	0.524 ± 0.058	0.524 ± 0.058	0.524 ± 0.058	0.524 ± 0.058	0.538	0.524 ± 0.058	0.524 ± 0.058	0.524 ± 0.058	0.524 ± 0.058	0.524 ± 0.058	0.524 ± 0.058	0.524 ± 0.058
obesity	0.279 ± 0.028	0.279 ± 0.028	0.279 ± 0.028	0.279 ± 0.028	0.284	0.276 ± 0.129	0.279 ± 0.028	0.279 ± 0.028	0.279 ± 0.028	0.279 ± 0.028	0.279 ± 0.028	0.279 ± 0.028
diabetes	0.071 ± 0.000	0.071 ± 0.000	0.071 ± 0.000	0.071 ± 0.000	–	0.071 ± 0.000	0.071 ± 0.000	0.071 ± 0.000	0.071 ± 0.000	0.071 ± 0.000	0.071 ± 0.000	0.071 ± 0.000
german_credit	0.419 ± 0.015	0.419 ± 0.015	0.419 ± 0.015	0.419 ± 0.015	–	0.419 ± 0.015	0.419 ± 0.015	0.419 ± 0.015	0.419 ± 0.015	0.419 ± 0.015	0.419 ± 0.015	0.419 ± 0.015
bank_marketing	0.209 ± 0.002	0.209 ± 0.002	0.209 ± 0.002	0.209 ± 0.002	–	0.209 ± 0.002	0.209 ± 0.002	0.209 ± 0.002	0.209 ± 0.002	0.209 ± 0.002	0.209 ± 0.002	0.209 ± 0.002
census_income	0.136 ± 0.001	0.136 ± 0.001	0.136 ± 0.001	0.136 ± 0.001	–	0.136 ± 0.001	0.136 ± 0.001	0.136 ± 0.001	0.136 ± 0.001	0.136 ± 0.001	0.136 ± 0.001	0.136 ± 0.001
surv_melanoma	0.524 ± 0.041	–	–	–	–	0.524 ± 0.041	0.524 ± 0.041	0.524 ± 0.041	0.524 ± 0.041	0.524 ± 0.041	0.524 ± 0.041	0.524 ± 0.041
surv_pbc	0.196 ± 0.163	–	–	–	–	0.196 ± 0.163	0.196 ± 0.163	0.196 ± 0.163	0.196 ± 0.163	0.196 ± 0.163	0.196 ± 0.163	0.196 ± 0.163
surv_pbc3	0.626 ± 0.053	–	–	–	–	0.626 ± 0.053	0.626 ± 0.053	0.626 ± 0.053	0.626 ± 0.053	0.626 ± 0.053	0.626 ± 0.053	0.626 ± 0.053
surv_retinopathy	0.318 ± 0.016	–	–	–	–	0.318 ± 0.016	0.318 ± 0.016	0.318 ± 0.016	0.318 ± 0.016	0.318 ± 0.016	0.318 ± 0.016	0.318 ± 0.016
surv_whas500	0.745 ± 0.023	–	–	–	–	0.745 ± 0.023	0.745 ± 0.023	0.745 ± 0.023	0.745 ± 0.023	0.745 ± 0.023	0.745 ± 0.023	0.745 ± 0.023
surv_gbsg2	0.323 ± 0.105	–	–	–	–	0.323 ± 0.105	0.323 ± 0.105	0.323 ± 0.105	0.323 ± 0.105	0.323 ± 0.105	0.323 ± 0.105	0.323 ± 0.105
surv_colon	0.337 ± 0.202	–	–	–	–	0.337 ± 0.202	0.337 ± 0.202	0.337 ± 0.202	0.337 ± 0.202	0.337 ± 0.202	0.337 ± 0.202	0.337 ± 0.202
surv_grace	0.290 ± 0.029	–	–	–	–	0.290 ± 0.029	0.290 ± 0.029	0.290 ± 0.029	0.290 ± 0.029	0.290 ± 0.029	0.290 ± 0.029	0.290 ± 0.029
surv_trace	0.109 ± 0.006	–	–	–	–	0.109 ± 0.006	0.109 ± 0.006	0.109 ± 0.006	0.109 ± 0.006	0.109 ± 0.006	0.109 ± 0.006	0.109 ± 0.006
surv_framingham	0.352 ± 0.009	–	–	–	–	0.352 ± 0.009	0.352 ± 0.009	0.352 ± 0.009	0.352 ± 0.009	0.352 ± 0.009	0.352 ± 0.009	0.352 ± 0.009
tab_blood_transfusion	0.201 ± 0.026	–	–	–	–	0.201 ± 0.026	0.201 ± 0.026	0.201 ± 0.026	0.201 ± 0.026	0.201 ± 0.026	0.201 ± 0.026	0.201 ± 0.026
tab_credit_g	0.410 ± 0.019	–	–	–	–	0.410 ± 0.019	0.410 ± 0.019	0.410 ± 0.019	0.410 ± 0.019	0.410 ± 0.019	0.410 ± 0.019	0.410 ± 0.019
tab_qsar_biodeg	0.436 ± 0.030	–	–	–	–	0.436 ± 0.030	0.436 ± 0.030	0.436 ± 0.030	0.436 ± 0.030	0.436 ± 0.030	0.436 ± 0.030	0.436 ± 0.030
tab_maternal_risk	0.357 ± 0.174	–	–	–	–	0.357 ± 0.174	0.357 ± 0.174	0.357 ± 0.174	0.357 ± 0.174	0.357 ± 0.174	0.357 ± 0.174	0.357 ± 0.174
tab_website_phishing	0.400 ± 0.000	–	–	–	–	0.400 ± 0.000	0.400 ± 0.000	0.400 ± 0.000	0.400 ± 0.000	0.400 ± 0.000	0.400 ± 0.000	0.400 ± 0.000

Table 13: DCR encoded (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	2.153 ± 0.553	2.316 ± 0.299	3.165 ± 0.177	1.999 ± 0.143	2.022	2.025 ± 0.662	2.880 ± 0.566	1.582 ± 0.178	1.918 ± 0.340	1.946 ± 0.085	2.079 ± 0.405	$\mathbf{3.810 \pm 0.402}$
breast_cancer	3.280 ± 0.411	5.484 ± 0.782	3.199 ± 0.489	2.429 ± 0.233	2.308	3.052 ± 0.345	6.717 ± 3.045	2.951 ± 0.615	3.495 ± 0.493	2.793 ± 0.561	2.273 ± 0.426	$\mathbf{12.169 \pm 0.265}$
liver_disorders	0.641 ± 0.058	0.989 ± 0.150	0.570 ± 0.047	0.560 ± 0.121	0.470	0.676 ± 0.243	0.797 ± 0.059	0.937 ± 0.602	0.594 ± 0.033	0.551 ± 0.035	0.577 ± 0.075	$\mathbf{1.627 \pm 0.322}$
parkinsons	2.372 ± 0.729	3.982 ± 0.120	2.595 ± 0.205	1.978 ± 0.149	1.756	2.676 ± 0.238	3.538 ± 0.665	2.529 ± 0.171	2.767 ± 0.446	2.100 ± 0.217	1.513 ± 0.310	$\mathbf{8.250 \pm 1.029}$
obesity	1.166 ± 0.222	1.909 ± 0.085	1.959 ± 0.111	1.860 ± 0.310	1.273	1.516 ± 0.066	2.083 ± 0.472	1.459 ± 0.129	1.770 ± 0.063	1.264 ± 0.182	1.023 ± 0.019	$\mathbf{3.013 \pm 0.145}$
diabetes	0.498 ± 0.059	0.558 ± 0.027	0.546 ± 0.076	0.660 ± 0.105	–	0.517 ± 0.052	0.822 ± 0.185	0.524 ± 0.106	0.568 ± 0.060	0.501 ± 0.007	0.484 ± 0.011	$\mathbf{1.604 \pm 0.012}$
german_credit	2.974 ± 0.742	3.107 ± 0.154	3.516 ± 0.132	3.007 ± 0.024	–	2.914 ± 0.062	3.187 ± 0.138	2.880 ± 0.062	3.105 ± 0.140	2.916 ± 0.123	2.866 ± 0.062	$\mathbf{4.129 \pm 0.139}$
bank_marketing	1.652 ± 0.047	1.951 ± 0.121	1.893 ± 0.243	2.006 ± 0.020	–	1.994 ± 0.168	2.382 ± 0.464	1.959 ± 0.405	2.111 ± 0.111	1.841 ± 0.051	1.855 ± 0.009	$\mathbf{5.609 \pm 1.075}$
census_income	1.640 ± 0.098	2.048 ± 0.035	1.781 ± 0.231	2.226 ± 0.404	–	2.049 ± 0.320	2.686 ± 0.876	2.221 ± 0.626	2.212 ± 0.155	1.902 ± 0.051	1.786 ± 0.285	$\mathbf{4.370 \pm 0.304}$
surv_melanoma	0.587 ± 0.084	–	–	–	–	0.597 ± 0.283	$\mathbf{1.072 \pm 0.963}$	0.865 ± 0.360	0.525 ± 0.056	0.521 ± 0.081	0.514 ± 0.062	0.904 ± 0.178
surv_pbc	$\mathbf{0.630 \pm 1.356}$	–	–	–	–	0.075 ± 0.041	0.171 ± 0.108	0.574 ± 1.595	0.068 ± 0.032	0.060 ± 0.023	0.064 ± 0.011	0.193 ± 0.108
surv_pbc3	2.000 ± 0.207	–	–	–	–	2.127 ± 0.176	2.730 ± 0.463	1.971 ± 0.114	2.029 ± 0.322	1.983 ± 0.100	1.981 ± 0.055	$\mathbf{3.829 \pm 0.484}$
surv_retinopathy	0.628 ± 1.706	–	–	–	–	0.352 ± 0.184	0.820 ± 0.071	0.507 ± 0.241	0.617 ± 0.261	0.253 ± 0.121	0.132 ± 0.003	$\mathbf{0.852 \pm 0.202}$
surv_whas500	2.270 ± 0.345	–	–	–	–	2.260 ± 0.094	3.447 ± 0.792	2.218 ± 0.077	2.387 ± 0.362	2.251 ± 0.158	2.356 ± 0.238	$\mathbf{4.183 \pm 0.777}$
surv_gbsg2	0.789 ± 0.403	–	–	–	–	0.868 ± 0.408	1.240 ± 0.741	0.742 ± 0.116	0.692 ± 0.103	0.663 ± 0.073	0.964 ± 0.210	$\mathbf{2.470 \pm 0.172}$
surv_colon	0.341 ± 0.356	–	–	–	–	0.513 ± 0.233	1.220 ± 1.010	0.433 ± 0.005	0.341 ± 0.016	0.334 ± 0.067	0.368 ± 0.088	$\mathbf{1.772 \pm 0.360}$
surv_grace	0.354 ± 0.049	–	–	–	–	0.395 ± 0.162	$\mathbf{1.192 \pm 2.913}$	0.603 ± 0.047	0.321 ± 0.026	0.329 ± 0.025	0.332 ± 0.026	0.666 ± 0.091
surv_trace	0.089 ± 0.027	–	–	–	–	0.177 ± 0.106	0.238 ± 0.079	$\mathbf{0.737 \pm 1.768}$	0.108 ± 0.013	0.090 ± 0.002	0.101 ± 0.013	0.294 ± 0.008
surv_framingham	0.726 ± 0.014	–	–	–	–	0.780 ± 0.126	1.368 ± 0.122	1.252 ± 1.082	0.786 ± 0.089	0.739 ± 0.008	0.754 ± 0.042	$\mathbf{1.567 \pm 0.089}$
tab_blood_transfusion	0.554 ± 0.418	–	–	–	–	0.353 ± 0.125	0.569 ± 0.578	0.542 ± 0.148	0.276 ± 0.073	0.245 ± 0.012	0.225 ± 0.047	$\mathbf{0.975 \pm 0.200}$
tab_credit_g	3.156 ± 0.053	–	–	–	–	2.866 ± 0.088	3.297 ± 0.324	2.866 ± 0.170	3.038 ± 0.155	2.870 ± 0.104	2.859 ± 0.126	$\mathbf{4.188 \pm 0.121}$
tab_qsar_biodeg	3.690 ± 0.252	–	–	–	–	3.096 ± 0.058	6.437 ± 2.976	3.072 ± 0.274	3.949 ± 0.307	2.546 ± 0.177	1.938 ± 0.077	$\mathbf{19.399 \pm 7.340}$
tab_maternal_risk	0.452 ± 0.309	–	–	–	–	0.767 ± 0.248	1.263 ± 0.936	0.714 ± 0.079	0.651 ± 0.015	0.549 ± 0.196	0.569 ± 0.105	$\mathbf{1.582 \pm 0.269}$
tab_website_phishing	1.520 ± 0.456	–	–	–	–	1.414 ± 0.000	$\mathbf{1.609 \pm 0.840}$	1.414 ± 0.000	1.414 ± 0.000	1.414 ± 0.000	1.414 ± 0.000	1.414 ± 0.000

Table 14: NNDR (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.892 ± 0.046	0.918 ± 0.017	**0.983 ± 0.016**	0.880 ± 0.024	0.912	0.906 ± 0.089	0.930 ± 0.014	0.909 ± 0.067	0.891 ± 0.037	0.897 ± 0.050	0.922 ± 0.039	0.927 ± 0.061
breast_cancer	**0.980 ± 0.004**	0.958 ± 0.022	0.965 ± 0.003	0.931 ± 0.012	0.929	0.949 ± 0.033	0.877 ± 0.241	0.975 ± 0.010	0.956 ± 0.004	0.953 ± 0.006	0.950 ± 0.001	0.972 ± 0.025
liver_disorders	0.922 ± 0.075	0.877 ± 0.049	0.916 ± 0.077	0.834 ± 0.034	0.867	0.897 ± 0.053	0.860 ± 0.027	**1.000 ± 0.000**	0.856 ± 0.105	0.851 ± 0.052	0.876 ± 0.016	0.864 ± 0.059
parkinsons	0.951 ± 0.033	0.956 ± 0.040	0.953 ± 0.019	0.907 ± 0.063	0.828	0.935 ± 0.012	0.944 ± 0.036	**0.992 ± 0.016**	0.935 ± 0.010	0.916 ± 0.057	0.922 ± 0.012	0.969 ± 0.022
obesity	0.902 ± 0.023	0.929 ± 0.014	**0.963 ± 0.010**	0.925 ± 0.020	0.903	0.910 ± 0.019	0.935 ± 0.023	0.910 ± 0.014	0.923 ± 0.002	0.893 ± 0.014	0.911 ± 0.020	0.956 ± 0.013
diabetes	0.831 ± 0.019	0.834 ± 0.010	0.897 ± 0.027	0.847 ± 0.012	–	0.846 ± 0.026	0.871 ± 0.034	0.848 ± 0.056	0.831 ± 0.013	0.832 ± 0.003	0.821 ± 0.011	**0.905 ± 0.021**
german_credit	0.953 ± 0.003	0.960 ± 0.001	**0.986 ± 0.006**	0.957 ± 0.017	–	0.951 ± 0.015	0.956 ± 0.007	0.949 ± 0.010	0.953 ± 0.017	0.959 ± 0.024	0.973 ± 0.004	0.962 ± 0.005
bank_marketing	0.927 ± 0.008	0.938 ± 0.008	**0.966 ± 0.010**	0.945 ± 0.006	–	0.943 ± 0.003	0.955 ± 0.010	0.949 ± 0.018	0.945 ± 0.008	0.933 ± 0.003	0.939 ± 0.006	0.892 ± 0.164
census_income	0.931 ± 0.008	0.944 ± 0.007	**0.970 ± 0.009**	0.948 ± 0.005	–	0.947 ± 0.005	0.962 ± 0.010	0.966 ± 0.025	0.952 ± 0.004	0.936 ± 0.001	0.932 ± 0.024	0.957 ± 0.028
surv_melanoma	0.857 ± 0.153	–	–	–	–	0.830 ± 0.053	0.832 ± 0.049	**1.000 ± 0.000**	0.805 ± 0.071	0.777 ± 0.025	0.851 ± 0.118	0.759 ± 0.029
surv_pbc	0.712 ± 0.379	–	–	–	–	0.542 ± 0.145	0.638 ± 0.298	**0.817 ± 0.409**	0.538 ± 0.026	0.497 ± 0.035	0.542 ± 0.143	0.493 ± 0.096
surv_pbc3	0.917 ± 0.035	–	–	–	–	0.911 ± 0.033	0.918 ± 0.024	**0.940 ± 0.030**	0.916 ± 0.034	0.900 ± 0.029	0.926 ± 0.056	0.932 ± 0.029
surv_retinopathy	0.725 ± 0.617	–	–	–	–	0.688 ± 0.059	0.810 ± 0.093	**1.000 ± 0.000**	0.680 ± 0.274	0.588 ± 0.195	0.490 ± 0.177	0.772 ± 0.036
surv_whas500	0.933 ± 0.027	–	–	–	–	0.897 ± 0.072	0.926 ± 0.026	0.929 ± 0.013	0.907 ± 0.056	0.906 ± 0.032	**0.938 ± 0.046**	0.910 ± 0.073
surv_gbsg2	0.871 ± 0.085	–	–	–	–	0.887 ± 0.038	0.888 ± 0.088	**0.973 ± 0.055**	0.850 ± 0.019	0.851 ± 0.028	0.870 ± 0.053	0.847 ± 0.087
surv_colon	0.734 ± 0.207	–	–	–	–	0.791 ± 0.091	0.841 ± 0.085	**1.000 ± 0.000**	0.703 ± 0.028	0.691 ± 0.097	0.739 ± 0.018	0.884 ± 0.055
surv_grace	0.814 ± 0.058	–	–	–	–	0.824 ± 0.102	0.815 ± 0.068	**1.000 ± 0.000**	0.787 ± 0.066	0.787 ± 0.068	0.808 ± 0.033	0.789 ± 0.083
surv_trace	0.720 ± 0.084	–	–	–	–	0.786 ± 0.114	0.727 ± 0.024	**0.980 ± 0.086**	0.734 ± 0.056	0.679 ± 0.083	0.718 ± 0.039	0.687 ± 0.056
surv_framingham	0.895 ± 0.013	–	–	–	–	0.898 ± 0.021	0.898 ± 0.028	**0.919 ± 0.037**	0.895 ± 0.004	0.890 ± 0.005	0.888 ± 0.024	0.883 ± 0.014
tab_blood_transfusion	0.946 ± 0.081	–	–	–	–	0.870 ± 0.109	0.846 ± 0.037	**1.000 ± 0.000**	0.823 ± 0.019	0.822 ± 0.012	0.824 ± 0.037	0.805 ± 0.133
tab_credit_g	0.964 ± 0.005	–	–	–	–	0.948 ± 0.020	0.960 ± 0.010	0.948 ± 0.008	0.954 ± 0.009	0.948 ± 0.018	**0.976 ± 0.001**	0.963 ± 0.021
tab_qsar_biodeg	**0.970 ± 0.009**	–	–	–	–	0.949 ± 0.009	0.970 ± 0.028	0.949 ± 0.006	0.961 ± 0.011	0.930 ± 0.007	0.922 ± 0.010	0.915 ± 0.284
tab_maternal_risk	0.964 ± 0.146	–	–	–	–	0.876 ± 0.082	0.889 ± 0.054	**1.000 ± 0.000**	0.861 ± 0.043	0.823 ± 0.069	0.851 ± 0.032	0.898 ± 0.031
tab_website_phishing	**1.000 ± 0.000**	–	–	–	–	**1.000 ± 0.000**	**1.000 ± 0.000**	0.902 ± 0.420	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**

Table 15: NNAA (0.5 ideal) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.604 ± 0.174	0.738 ± 0.052	0.967 ± 0.038	0.508 ± 0.131	0.655	0.600 ± 0.050	0.875 ± 0.151	0.629 ± 0.086	0.536 ± 0.023	0.455 ± 0.056	0.472 ± 0.109	0.927 ± 0.014
breast_cancer	0.975 ± 0.000	0.958 ± 0.029	0.944 ± 0.017	0.639 ± 0.058	0.713	0.882 ± 0.055	0.996 ± 0.007	0.932 ± 0.030	0.753 ± 0.032	0.564 ± 0.030	0.311 ± 0.048	0.951 ± 0.003
liver_disorders	0.361 ± 0.076	0.850 ± 0.037	0.673 ± 0.145	0.505 ± 0.080	0.458	0.711 ± 0.060	0.772 ± 0.160	0.905 ± 0.077	0.562 ± 0.070	0.470 ± 0.039	0.511 ± 0.054	0.891 ± 0.043
parkinsons	0.903 ± 0.072	0.953 ± 0.017	0.934 ± 0.059	0.637 ± 0.053	0.638	0.907 ± 0.029	0.886 ± 0.122	0.936 ± 0.036	0.740 ± 0.072	0.619 ± 0.063	0.386 ± 0.024	0.893 ± 0.056
obesity	0.591 ± 0.013	0.844 ± 0.027	0.935 ± 0.010	0.864 ± 0.088	0.726	0.775 ± 0.079	0.900 ± 0.109	0.761 ± 0.014	0.819 ± 0.062	0.614 ± 0.020	0.452 ± 0.015	0.950 ± 0.009
diabetes	0.585 ± 0.036	0.628 ± 0.015	0.682 ± 0.042	0.691 ± 0.060	–	0.630 ± 0.065	0.828 ± 0.124	0.624 ± 0.155	0.636 ± 0.059	0.541 ± 0.010	0.537 ± 0.036	0.920 ± 0.013
german_credit	0.790 ± 0.393	0.703 ± 0.112	0.948 ± 0.003	0.614 ± 0.019	–	0.578 ± 0.048	0.775 ± 0.076	0.550 ± 0.011	0.686 ± 0.039	0.463 ± 0.010	0.303 ± 0.029	0.910 ± 0.014
bank_marketing	0.612 ± 0.041	0.703 ± 0.034	0.806 ± 0.097	0.750 ± 0.033	–	0.775 ± 0.084	0.903 ± 0.087	0.784 ± 0.220	0.791 ± 0.040	0.626 ± 0.023	0.693 ± 0.024	0.995 ± 0.005
census_income	0.663 ± 0.068	0.734 ± 0.034	0.789 ± 0.078	0.840 ± 0.163	–	0.828 ± 0.179	0.944 ± 0.119	0.908 ± 0.165	0.868 ± 0.029	0.658 ± 0.044	0.723 ± 0.171	0.990 ± 0.003
surv_melanoma	0.536 ± 0.098	–	–	–	–	0.641 ± 0.203	0.797 ± 0.225	0.884 ± 0.062	0.534 ± 0.160	0.439 ± 0.098	0.527 ± 0.033	0.763 ± 0.024
surv_pbc	0.679 ± 0.631	–	–	–	–	0.562 ± 0.055	0.673 ± 0.064	0.802 ± 0.468	0.533 ± 0.055	0.477 ± 0.030	0.440 ± 0.010	0.701 ± 0.036
surv_pbc3	0.556 ± 0.102	–	–	–	–	0.643 ± 0.049	0.880 ± 0.203	0.719 ± 0.035	0.624 ± 0.062	0.483 ± 0.064	0.290 ± 0.065	0.912 ± 0.005
surv_retinopathy	0.571 ± 0.897	–	–	–	–	0.550 ± 0.178	0.799 ± 0.084	0.691 ± 0.076	0.661 ± 0.089	0.405 ± 0.026	0.327 ± 0.057	0.811 ± 0.057
surv_whas500	0.674 ± 0.034	–	–	–	–	0.586 ± 0.020	0.898 ± 0.169	0.654 ± 0.125	0.594 ± 0.080	0.482 ± 0.045	0.337 ± 0.065	0.912 ± 0.017
surv_gbsg2	0.634 ± 0.319	–	–	–	–	0.750 ± 0.244	0.852 ± 0.164	0.749 ± 0.073	0.593 ± 0.071	0.496 ± 0.063	0.701 ± 0.034	0.959 ± 0.009
surv_colon	0.449 ± 0.206	–	–	–	–	0.699 ± 0.188	0.872 ± 0.220	0.762 ± 0.054	0.568 ± 0.083	0.412 ± 0.071	0.547 ± 0.040	0.875 ± 0.030
surv_grace	0.504 ± 0.022	–	–	–	–	0.672 ± 0.141	0.821 ± 0.399	0.919 ± 0.027	0.536 ± 0.031	0.480 ± 0.056	0.516 ± 0.015	0.831 ± 0.029
surv_trace	0.536 ± 0.063	–	–	–	–	0.761 ± 0.060	0.830 ± 0.033	0.946 ± 0.048	0.645 ± 0.028	0.529 ± 0.060	0.539 ± 0.024	0.830 ± 0.003
surv_framingham	0.556 ± 0.039	–	–	–	–	0.644 ± 0.165	0.936 ± 0.015	0.839 ± 0.402	0.636 ± 0.043	0.490 ± 0.011	0.546 ± 0.023	0.924 ± 0.013
tab_blood_transfusion	0.862 ± 0.269	–	–	–	–	0.814 ± 0.057	0.912 ± 0.133	0.975 ± 0.007	0.718 ± 0.076	0.574 ± 0.035	0.623 ± 0.026	0.956 ± 0.009
tab_credit_g	0.911 ± 0.015	–	–	–	–	0.594 ± 0.060	0.828 ± 0.226	0.540 ± 0.036	0.691 ± 0.028	0.461 ± 0.056	0.297 ± 0.019	0.916 ± 0.010
tab_qsar_biodeg	0.980 ± 0.015	–	–	–	–	0.890 ± 0.067	0.997 ± 0.002	0.897 ± 0.022	0.893 ± 0.024	0.639 ± 0.007	0.435 ± 0.016	0.995 ± 0.006
tab_maternal_risk	0.668 ± 0.083	–	–	–	–	0.860 ± 0.144	0.929 ± 0.089	0.890 ± 0.007	0.759 ± 0.016	0.636 ± 0.007	0.765 ± 0.039	0.911 ± 0.033
tab_website_phishing	0.368 ± 0.919	–	–	–	–	0.140 ± 0.047	0.520 ± 0.162	0.138 ± 0.026	0.174 ± 0.046	0.071 ± 0.016	0.130 ± 0.007	0.409 ± 0.049

Table 16: MIA AUROC (0.5 safe) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.499 ± 0.105	0.470 ± 0.077	0.477 ± 0.120	0.493 ± 0.061	0.486	0.515 ± 0.148	0.491 ± 0.105	0.458 ± 0.046	0.481 ± 0.049	0.539 ± 0.041	0.621 ± 0.106	0.492 ± 0.061
breast_cancer	0.516 ± 0.100	0.500 ± 0.056	0.497 ± 0.059	0.497 ± 0.070	0.491	0.511 ± 0.140	0.521 ± 0.132	0.499 ± 0.101	0.496 ± 0.095	0.523 ± 0.106	0.749 ± 0.045	0.498 ± 0.064
liver_disorders	0.632 ± 0.045	0.511 ± 0.085	0.500 ± 0.052	0.498 ± 0.078	0.503	0.525 ± 0.087	0.494 ± 0.049	0.514 ± 0.055	0.500 ± 0.063	0.516 ± 0.046	0.546 ± 0.067	0.508 ± 0.130
parkinsons	0.516 ± 0.214	0.501 ± 0.058	0.483 ± 0.110	0.486 ± 0.071	0.510	0.499 ± 0.039	0.487 ± 0.073	0.536 ± 0.118	0.502 ± 0.076	0.527 ± 0.074	0.695 ± 0.006	0.536 ± 0.037
obesity	0.559 ± 0.026	0.494 ± 0.001	0.503 ± 0.007	0.488 ± 0.036	0.527	0.515 ± 0.067	0.492 ± 0.051	0.512 ± 0.024	0.498 ± 0.035	0.540 ± 0.015	0.621 ± 0.010	0.499 ± 0.019
diabetes	0.504 ± 0.016	0.508 ± 0.027	0.504 ± 0.014	0.497 ± 0.022	–	0.497 ± 0.010	0.498 ± 0.006	0.500 ± 0.012	0.490 ± 0.017	0.504 ± 0.033	0.504 ± 0.013	0.501 ± 0.022
german_credit	0.513 ± 0.056	0.520 ± 0.038	0.506 ± 0.061	0.522 ± 0.031	–	0.530 ± 0.026	0.497 ± 0.021	0.552 ± 0.028	0.511 ± 0.041	0.574 ± 0.026	0.776 ± 0.013	0.524 ± 0.044
bank_marketing	0.505 ± 0.021	0.503 ± 0.034	0.506 ± 0.022	0.504 ± 0.026	–	0.498 ± 0.006	0.502 ± 0.008	0.501 ± 0.006	0.503 ± 0.024	0.496 ± 0.004	0.504 ± 0.022	0.498 ± 0.017
census_income	0.503 ± 0.015	0.493 ± 0.015	0.501 ± 0.014	0.507 ± 0.020	–	0.507 ± 0.020	0.497 ± 0.013	0.507 ± 0.010	0.496 ± 0.028	0.508 ± 0.019	0.507 ± 0.027	0.500 ± 0.021
surv_melanoma	0.647 ± 0.090	–	–	–	–	0.538 ± 0.059	0.513 ± 0.057	0.563 ± 0.117	0.561 ± 0.084	0.585 ± 0.076	0.559 ± 0.059	0.486 ± 0.157
surv_pbc	0.571 ± 0.112	–	–	–	–	0.522 ± 0.028	0.511 ± 0.101	0.525 ± 0.083	0.518 ± 0.061	0.545 ± 0.038	0.581 ± 0.077	0.508 ± 0.061
surv_pbc3	0.552 ± 0.054	–	–	–	–	0.555 ± 0.113	0.496 ± 0.065	0.533 ± 0.082	0.507 ± 0.094	0.547 ± 0.058	0.790 ± 0.034	0.501 ± 0.106
surv_retinopathy	0.601 ± 0.171	–	–	–	–	0.500 ± 0.056	0.491 ± 0.088	0.535 ± 0.078	0.496 ± 0.079	0.520 ± 0.054	0.532 ± 0.114	0.499 ± 0.022
surv_whas500	0.516 ± 0.088	–	–	–	–	0.519 ± 0.034	0.507 ± 0.038	0.525 ± 0.030	0.499 ± 0.086	0.524 ± 0.056	0.742 ± 0.053	0.484 ± 0.057
surv_gbsg2	0.527 ± 0.083	–	–	–	–	0.501 ± 0.067	0.497 ± 0.078	0.495 ± 0.094	0.481 ± 0.064	0.522 ± 0.027	0.503 ± 0.057	0.481 ± 0.053
surv_colon	0.520 ± 0.058	–	–	–	–	0.489 ± 0.075	0.482 ± 0.039	0.478 ± 0.028	0.468 ± 0.011	0.513 ± 0.058	0.489 ± 0.009	0.516 ± 0.076
surv_grace	0.571 ± 0.026	–	–	–	–	0.495 ± 0.064	0.514 ± 0.069	0.502 ± 0.057	0.489 ± 0.012	0.549 ± 0.037	0.530 ± 0.002	0.493 ± 0.006
surv_trace	0.578 ± 0.006	–	–	–	–	0.498 ± 0.059	0.497 ± 0.010	0.499 ± 0.006	0.508 ± 0.028	0.525 ± 0.025	0.571 ± 0.032	0.497 ± 0.042
surv_framingham	0.512 ± 0.025	–	–	–	–	0.509 ± 0.014	0.505 ± 0.030	0.508 ± 0.024	0.509 ± 0.051	0.527 ± 0.017	0.512 ± 0.048	0.507 ± 0.057
tab_blood_transfusion	0.533 ± 0.094	–	–	–	–	0.505 ± 0.006	0.517 ± 0.058	0.512 ± 0.104	0.526 ± 0.041	0.545 ± 0.045	0.524 ± 0.027	0.499 ± 0.102
tab_credit_g	0.516 ± 0.025	–	–	–	–	0.494 ± 0.050	0.488 ± 0.038	0.533 ± 0.059	0.500 ± 0.067	0.548 ± 0.063	0.774 ± 0.015	0.523 ± 0.061

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
tab_qsar_biodeg	0.503 ± 0.026	–	–	–	–	0.511 ± 0.015	0.485 ± 0.048	0.507 ± 0.021	0.497 ± 0.051	0.527 ± 0.047	0.722 ± 0.037	0.484 ± 0.025
tab_maternal_risk	0.647 ± 0.020	–	–	–	–	0.501 ± 0.142	0.493 ± 0.096	0.502 ± 0.088	0.506 ± 0.112	0.525 ± 0.180	0.513 ± 0.110	0.500 ± 0.046
tab_website_phishing	0.530 ± 0.028	–	–	–	–	0.537 ± 0.073	0.498 ± 0.112	0.513 ± 0.041	0.520 ± 0.030	0.589 ± 0.015	0.511 ± 0.031	0.507 ± 0.019

Table 17: Duplicate rate (↓) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**
breast_cancer	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**
liver_disorders	0.307 ± 0.675	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	0.002 ± 0.005	**0.000 ± 0.000**
parkinsons	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**
obesity	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	0.000 ± 0.002	**0.000 ± 0.000**	**0.000 ± 0.000**
diabetes	**0.000 ± 0.000**	0.000 ± 0.000	0.001 ± 0.001	**0.000 ± 0.000**	–	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**
german_credit	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	–	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**
bank_marketing	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	–	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**
census_income	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	–	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**
surv_melanoma	0.006 ± 0.026	–	–	–	–	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**
surv_pbc	**0.000 ± 0.000**	–	–	–	–	**0.000 ± 0.000**	0.003 ± 0.006	0.017 ± 0.075	0.004 ± 0.010	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**
surv_pbc3	**0.000 ± 0.000**	–	–	–	–	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**
surv_retinopathy	0.263 ± 0.584	–	–	–	–	0.119 ± 0.110	0.024 ± 0.022	0.101 ± 0.047	0.056 ± 0.055	0.127 ± 0.043	0.204 ± 0.039	**0.008 ± 0.010**
surv_whas500	**0.000 ± 0.000**	–	–	–	–	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**
surv_gbsg2	0.001 ± 0.003	–	–	–	–	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**
surv_colon	**0.000 ± 0.000**	–	–	–	–	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	0.053 ± 0.027	**0.000 ± 0.000**	**0.000 ± 0.000**
surv_grace	0.052 ± 0.115	–	–	–	–	0.003 ± 0.003	0.001 ± 0.003	0.000 ± 0.002	0.003 ± 0.005	0.003 ± 0.004	0.010 ± 0.018	**0.000 ± 0.000**
surv_trace	**0.000 ± 0.000**	–	–	–	–	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**
surv_framingham	**0.000 ± 0.000**	–	–	–	–	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**
tab_blood_transfusion	**0.000 ± 0.000**	–	–	–	–	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	0.009 ± 0.006	0.005 ± 0.003	**0.000 ± 0.000**
tab_credit_g	**0.000 ± 0.000**	–	–	–	–	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**
tab_qsar_biodeg	**0.000 ± 0.000**	–	–	–	–	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**
tab_maternal_risk	**0.000 ± 0.000**	–	–	–	–	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**
tab_website_phishing	0.402 ± 0.866	–	–	–	–	0.343 ± 0.046	0.157 ± 0.079	0.319 ± 0.044	0.298 ± 0.129	0.431 ± 0.016	0.318 ± 0.024	**0.055 ± 0.005**

Diversity / coverage (PRDC)

Table 18: Precision (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.965 ± 0.074	0.812 ± 0.120	**1.000 ± 0.000**	0.943 ± 0.043	0.477	0.967 ± 0.021	0.619 ± 0.360	0.996 ± 0.018	0.940 ± 0.032	0.969 ± 0.012	0.986 ± 0.042	0.457 ± 0.146
breast_cancer	0.575 ± 0.156	0.004 ± 0.016	0.221 ± 0.149	0.853 ± 0.090	0.433	0.350 ± 0.035	0.001 ± 0.006	0.541 ± 0.033	0.035 ± 0.068	0.453 ± 0.024	**0.969 ± 0.036**	0.054 ± 0.121
liver_disorders	0.995 ± 0.010	0.571 ± 0.315	**0.999 ± 0.005**	0.981 ± 0.005	0.978	0.982 ± 0.024	0.704 ± 0.095	0.998 ± 0.005	0.937 ± 0.050	0.982 ± 0.024	0.969 ± 0.036	0.594 ± 0.172
parkinsons	0.872 ± 0.126	0.004 ± 0.018	0.553 ± 0.233	0.848 ± 0.112	0.814	0.372 ± 0.115	0.013 ± 0.016	0.842 ± 0.072	0.158 ± 0.113	0.741 ± 0.064	**0.991 ± 0.024**	0.107 ± 0.096
obesity	**0.937 ± 0.007**	0.484 ± 0.015	0.874 ± 0.047	0.587 ± 0.042	0.688	0.684 ± 0.331	0.374 ± 0.191	0.702 ± 0.062	0.526 ± 0.095	0.813 ± 0.027	0.900 ± 0.025	0.098 ± 0.120
diabetes	0.982 ± 0.011	0.940 ± 0.035	**0.995 ± 0.005**	0.874 ± 0.099	–	0.965 ± 0.032	0.757 ± 0.675	0.917 ± 0.214	0.916 ± 0.041	0.977 ± 0.009	0.960 ± 0.009	0.514 ± 0.068
german_credit	0.924 ± 0.199	0.734 ± 0.078	**1.000 ± 0.002**	0.822 ± 0.036	–	0.888 ± 0.016	0.700 ± 0.094	0.926 ± 0.022	0.671 ± 0.019	0.917 ± 0.016	0.990 ± 0.011	0.150 ± 0.053
bank_marketing	0.971 ± 0.023	0.720 ± 0.082	**0.990 ± 0.009**	0.664 ± 0.165	–	0.743 ± 0.089	0.392 ± 0.130	0.761 ± 0.254	0.636 ± 0.049	0.785 ± 0.035	0.776 ± 0.022	0.323 ± 0.658
census_income	0.967 ± 0.025	0.543 ± 0.086	**0.983 ± 0.012**	0.405 ± 0.417	–	0.480 ± 0.355	0.186 ± 0.298	0.377 ± 0.516	0.321 ± 0.141	0.539 ± 0.085	0.552 ± 0.019	0.017 ± 0.054
surv_melanoma	0.990 ± 0.009	–	–	–	–	**0.994 ± 0.026**	0.789 ± 0.091	0.986 ± 0.038	0.953 ± 0.068	0.986 ± 0.009	0.986 ± 0.023	0.831 ± 0.057
surv_pbc	0.771 ± 0.755	–	–	–	–	0.991 ± 0.012	0.914 ± 0.168	0.888 ± 0.467	0.993 ± 0.021	0.992 ± 0.010	**0.997 ± 0.006**	0.841 ± 0.066
surv_pbc3	0.982 ± 0.036	–	–	–	–	0.944 ± 0.094	0.456 ± 0.253	**0.995 ± 0.014**	0.906 ± 0.037	0.953 ± 0.050	0.994 ± 0.019	0.225 ± 0.181
surv_retinopathy	0.704 ± 0.774	–	–	–	–	0.815 ± 0.167	0.515 ± 0.182	0.737 ± 0.030	0.639 ± 0.156	0.926 ± 0.013	**0.999 ± 0.005**	0.442 ± 0.086
surv_whas500	0.987 ± 0.021	–	–	–	–	0.960 ± 0.031	0.341 ± 0.392	**1.000 ± 0.000**	0.938 ± 0.054	0.962 ± 0.018	0.991 ± 0.014	0.725 ± 0.314
surv_gbsg2	0.892 ± 0.368	–	–	–	–	0.937 ± 0.053	0.648 ± 0.300	0.916 ± 0.051	0.871 ± 0.072	**0.951 ± 0.020**	0.777 ± 0.008	0.540 ± 0.203
surv_colon	**0.994 ± 0.000**	–	–	–	–	0.976 ± 0.022	0.712 ± 0.280	0.961 ± 0.013	0.928 ± 0.083	0.978 ± 0.023	0.986 ± 0.009	0.593 ± 0.073
surv_grace	0.994 ± 0.016	–	–	–	–	0.989 ± 0.025	0.560 ± 1.175	**1.000 ± 0.002**	0.988 ± 0.010	0.991 ± 0.007	0.980 ± 0.014	0.665 ± 0.016
surv_trace	**0.999 ± 0.002**	–	–	–	–	0.995 ± 0.001	0.755 ± 0.208	0.899 ± 0.405	0.982 ± 0.004	0.991 ± 0.002	0.990 ± 0.005	0.641 ± 0.036
surv_framingham	**0.980 ± 0.013**	–	–	–	–	0.938 ± 0.080	0.301 ± 0.513	0.652 ± 1.006	0.904 ± 0.071	0.958 ± 0.012	0.936 ± 0.024	0.272 ± 0.078
tab_blood_transfusion	0.688 ± 0.618	–	–	–	–	0.829 ± 0.126	0.553 ± 0.401	0.581 ± 0.036	0.658 ± 0.207	0.879 ± 0.078	**0.972 ± 0.015**	0.347 ± 0.024
tab_credit_g	0.791 ± 0.094	–	–	–	–	0.912 ± 0.048	0.673 ± 0.327	0.933 ± 0.027	0.734 ± 0.032	0.910 ± 0.043	**0.992 ± 0.006**	0.122 ± 0.031
tab_qsar_biodeg	0.149 ± 0.037	–	–	–	–	0.251 ± 0.076	0.006 ± 0.010	0.348 ± 0.041	0.073 ± 0.031	0.536 ± 0.025	**0.961 ± 0.014**	0.040 ± 0.014
tab_maternal_risk	**0.992 ± 0.018**	–	–	–	–	0.757 ± 0.314	0.476 ± 0.331	0.838 ± 0.111	0.778 ± 0.134	0.905 ± 0.077	0.895 ± 0.077	0.778 ± 0.042
tab_website_phishing	0.990 ± 0.042	–	–	–	–	0.997 ± 0.007	0.969 ± 0.064	0.997 ± 0.007	0.992 ± 0.013	**0.999 ± 0.005**	0.997 ± 0.005	0.882 ± 0.071

Table 19: Recall (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.921 ± 0.089	0.886 ± 0.037	0.031 ± 0.069	0.918 ± 0.042	**0.979**	0.879 ± 0.055	0.907 ± 0.094	0.669 ± 0.154	0.923 ± 0.049	0.967 ± 0.027	0.971 ± 0.018	0.975 ± 0.091
breast_cancer	0.103 ± 0.102	0.651 ± 0.274	0.286 ± 0.066	0.895 ± 0.068	0.960	0.738 ± 0.152	0.588 ± 0.549	0.330 ± 0.109	0.913 ± 0.023	0.938 ± 0.033	**0.964 ± 0.019**	0.525 ± 0.210
liver_disorders	0.918 ± 0.026	0.891 ± 0.199	0.563 ± 0.307	0.972 ± 0.005	0.971	0.762 ± 0.185	0.960 ± 0.080	0.199 ± 0.293	0.969 ± 0.045	0.978 ± 0.031	0.986 ± 0.009	**1.000 ± 0.000**
parkinsons	0.372 ± 0.073	0.667 ± 0.016	0.434 ± 0.199	0.981 ± 0.000	**0.987**	0.684 ± 0.189	0.838 ± 0.151	0.363 ± 0.161	0.944 ± 0.115	0.979 ± 0.018	0.972 ± 0.051	0.447 ± 0.160
obesity	0.923 ± 0.026	0.915 ± 0.020	0.351 ± 0.032	0.854 ± 0.197	0.933	0.907 ± 0.011	0.810 ± 0.145	0.881 ± 0.026	0.907 ± 0.059	0.945 ± 0.010	**0.954 ± 0.009**	0.949 ± 0.054
diabetes	0.945 ± 0.063	0.963 ± 0.025	0.704 ± 0.143	0.973 ± 0.022	–	0.940 ± 0.070	0.910 ± 0.152	0.946 ± 0.056	0.964 ± 0.023	0.978 ± 0.006	0.984 ± 0.012	**0.988 ± 0.019**
german_credit	0.473 ± 1.310	0.907 ± 0.079	0.015 ± 0.002	0.932 ± 0.022	–	0.924 ± 0.035	0.917 ± 0.097	0.875 ± 0.005	0.940 ± 0.078	0.966 ± 0.008	0.958 ± 0.014	**0.988 ± 0.025**
bank_marketing	0.790 ± 0.146	0.936 ± 0.016	0.298 ± 0.184	0.814 ± 0.161	–	0.840 ± 0.125	0.871 ± 0.072	0.775 ± 0.157	0.892 ± 0.103	**0.966 ± 0.017**	0.949 ± 0.009	0.949 ± 0.184

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
census_income	0.712 ± 0.170	0.911 ± 0.096	0.304 ± 0.172	0.883 ± 0.062	–	0.787 ± 0.036	0.631 ± 1.049	0.458 ± 0.739	0.840 ± 0.097	$\mathbf{0.960 \pm 0.017}$	0.903 ± 0.083	0.917 ± 0.265
surv_melanoma	0.947 ± 0.038	–	–	–	–	0.872 ± 0.215	0.929 ± 0.229	0.522 ± 0.273	0.986 ± 0.009	0.988 ± 0.015	0.986 ± 0.023	$\mathbf{1.000 \pm 0.000}$
surv_pbc	0.782 ± 0.546	–	–	–	–	0.968 ± 0.043	0.968 ± 0.056	0.647 ± 0.709	0.972 ± 0.026	0.995 ± 0.006	0.992 ± 0.010	$\mathbf{1.000 \pm 0.000}$
surv_pbc3	0.878 ± 0.073	–	–	–	–	0.842 ± 0.098	0.858 ± 0.298	0.455 ± 0.099	0.914 ± 0.041	0.962 ± 0.021	0.956 ± 0.031	$\mathbf{0.986 \pm 0.032}$
surv_retinopathy	0.668 ± 1.305	–	–	–	–	0.929 ± 0.022	0.965 ± 0.065	0.982 ± 0.042	0.976 ± 0.055	0.995 ± 0.013	$\mathbf{0.998 \pm 0.005}$	0.997 ± 0.015
surv_whas500	0.684 ± 0.124	–	–	–	–	0.885 ± 0.110	0.906 ± 0.135	0.593 ± 0.238	0.925 ± 0.045	0.965 ± 0.018	0.957 ± 0.041	$\mathbf{1.000 \pm 0.000}$
surv_gbsg2	0.888 ± 0.133	–	–	–	–	0.853 ± 0.142	0.904 ± 0.327	0.763 ± 0.053	0.964 ± 0.022	0.974 ± 0.003	0.895 ± 0.053	$\mathbf{0.985 \pm 0.032}$
surv_colon	0.948 ± 0.063	–	–	–	–	0.911 ± 0.139	0.788 ± 0.419	0.743 ± 0.028	0.978 ± 0.019	0.987 ± 0.021	0.980 ± 0.013	$\mathbf{0.995 \pm 0.017}$
surv_grace	0.945 ± 0.041	–	–	–	–	0.811 ± 0.330	0.893 ± 0.419	0.299 ± 0.225	0.982 ± 0.006	0.994 ± 0.002	0.993 ± 0.002	$\mathbf{0.998 \pm 0.002}$
surv_trace	0.950 ± 0.064	–	–	–	–	0.838 ± 0.191	0.977 ± 0.041	0.316 ± 0.097	0.993 ± 0.005	0.994 ± 0.009	0.996 ± 0.004	$\mathbf{1.000 \pm 0.001}$
surv_framingham	0.905 ± 0.016	–	–	–	–	0.904 ± 0.095	0.869 ± 0.253	0.663 ± 0.764	0.951 ± 0.026	0.981 ± 0.015	0.973 ± 0.009	$\mathbf{0.998 \pm 0.006}$
tab_blood_transfusion	0.613 ± 0.912	–	–	–	–	0.825 ± 0.166	0.928 ± 0.171	0.244 ± 0.098	0.980 ± 0.045	$\mathbf{0.990 \pm 0.009}$	0.986 ± 0.031	0.978 ± 0.055
tab_credit_g	0.188 ± 0.098	–	–	–	–	0.870 ± 0.093	0.806 ± 0.609	0.876 ± 0.072	0.914 ± 0.028	0.967 ± 0.012	0.954 ± 0.025	$\mathbf{0.976 \pm 0.017}$
tab_qsar_biodeg	0.278 ± 0.104	–	–	–	–	0.801 ± 0.106	0.119 ± 0.281	0.640 ± 0.014	0.772 ± 0.120	0.954 ± 0.008	$\mathbf{0.973 \pm 0.025}$	0.472 ± 1.128
tab_maternal_risk	0.912 ± 0.041	–	–	–	–	0.809 ± 0.260	0.807 ± 0.332	0.700 ± 0.072	0.953 ± 0.072	$\mathbf{0.990 \pm 0.020}$	0.970 ± 0.021	0.972 ± 0.095
tab_website_phishing	0.834 ± 0.687	–	–	–	–	$\mathbf{1.000 \pm 0.000}$	0.957 ± 0.168	$\mathbf{1.000 \pm 0.000}$	0.999 ± 0.005	$\mathbf{1.000 \pm 0.000}$	$\mathbf{1.000 \pm 0.000}$	$\mathbf{1.000 \pm 0.000}$

Table 20: Density (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.866 ± 0.521	0.489 ± 0.259	1.616 ± 0.212	0.904 ± 0.147	0.574	1.036 ± 0.204	0.250 ± 0.195	$\mathbf{1.623 \pm 0.132}$	0.942 ± 0.128	0.954 ± 0.055	1.059 ± 0.228	0.139 ± 0.040
breast_cancer	0.311 ± 0.185	0.001 ± 0.003	0.078 ± 0.068	0.590 ± 0.106	0.507	0.135 ± 0.012	0.000 ± 0.001	0.556 ± 0.164	0.008 ± 0.016	0.192 ± 0.003	$\mathbf{0.963 \pm 0.203}$	0.011 ± 0.025
liver_disorders	1.139 ± 0.220	0.268 ± 0.231	$\mathbf{1.375 \pm 0.251}$	0.946 ± 0.069	1.162	1.090 ± 0.271	0.370 ± 0.012	1.100 ± 0.161	0.874 ± 0.274	1.019 ± 0.030	0.956 ± 0.084	0.258 ± 0.097
parkinsons	0.960 ± 0.632	0.001 ± 0.006	0.276 ± 0.178	0.575 ± 0.217	0.772	0.166 ± 0.143	0.003 ± 0.005	0.992 ± 0.094	0.054 ± 0.058	0.466 ± 0.026	$\mathbf{1.035 \pm 0.045}$	0.027 ± 0.032
obesity	$\mathbf{0.904 \pm 0.034}$	0.225 ± 0.040	0.808 ± 0.126	0.286 ± 0.046	0.565	0.430 ± 0.399	0.153 ± 0.082	0.458 ± 0.060	0.252 ± 0.096	0.626 ± 0.053	0.854 ± 0.025	0.023 ± 0.027
diabetes	1.021 ± 0.093	0.870 ± 0.156	$\mathbf{1.140 \pm 0.043}$	0.641 ± 0.120	–	0.941 ± 0.063	0.619 ± 0.783	0.836 ± 0.416	0.815 ± 0.133	0.928 ± 0.063	0.928 ± 0.038	0.213 ± 0.077
german_credit	0.742 ± 0.648	0.439 ± 0.132	$\mathbf{2.157 \pm 0.320}$	0.558 ± 0.014	–	0.784 ± 0.114	0.393 ± 0.095	0.911 ± 0.210	0.376 ± 0.051	0.838 ± 0.070	1.030 ± 0.050	0.039 ± 0.016
bank_marketing	1.104 ± 0.343	0.449 ± 0.112	$\mathbf{1.565 \pm 0.044}$	0.347 ± 0.157	–	0.447 ± 0.163	0.147 ± 0.076	0.501 ± 0.476	0.319 ± 0.041	0.526 ± 0.058	0.551 ± 0.026	0.076 ± 0.150
census_income	1.189 ± 0.481	0.328 ± 0.169	$\mathbf{1.667 \pm 0.184}$	0.156 ± 0.170	–	0.281 ± 0.381	0.055 ± 0.102	0.187 ± 0.351	0.132 ± 0.077	0.321 ± 0.074	0.434 ± 0.191	0.004 ± 0.011
surv_melanoma	1.107 ± 0.292	–	–	–	–	1.102 ± 0.087	0.512 ± 0.121	$\mathbf{1.292 \pm 0.174}$	0.923 ± 0.179	1.027 ± 0.055	0.963 ± 0.061	0.504 ± 0.131
surv_pbc	0.777 ± 0.939	–	–	–	–	1.007 ± 0.066	0.779 ± 0.305	0.867 ± 1.303	1.016 ± 0.063	$\mathbf{1.030 \pm 0.061}$	1.027 ± 0.106	0.592 ± 0.136
surv_pbc3	1.111 ± 0.614	–	–	–	–	0.881 ± 0.056	0.187 ± 0.100	$\mathbf{1.736 \pm 0.377}$	0.853 ± 0.235	0.907 ± 0.108	0.977 ± 0.039	0.066 ± 0.052
surv_retinopathy	0.706 ± 1.299	–	–	–	–	0.770 ± 0.443	0.341 ± 0.138	0.680 ± 0.047	0.508 ± 0.255	0.936 ± 0.071	$\mathbf{1.053 \pm 0.071}$	0.281 ± 0.122
surv_whas500	1.457 ± 0.428	–	–	–	–	1.035 ± 0.207	0.152 ± 0.237	$\mathbf{1.693 \pm 0.512}$	0.889 ± 0.054	0.973 ± 0.048	1.039 ± 0.066	0.357 ± 0.281
surv_gbsg2	0.870 ± 0.692	–	–	–	–	0.827 ± 0.514	0.304 ± 0.173	$\mathbf{0.923 \pm 0.108}$	0.728 ± 0.042	0.905 ± 0.096	0.592 ± 0.032	0.168 ± 0.090
surv_colon	$\mathbf{1.096 \pm 0.079}$	–	–	–	–	0.960 ± 0.116	0.372 ± 0.570	1.081 ± 0.013	0.893 ± 0.279	1.042 ± 0.022	0.982 ± 0.011	0.285 ± 0.047
surv_grace	1.056 ± 0.122	–	–	–	–	1.012 ± 0.083	0.376 ± 0.806	$\mathbf{1.108 \pm 0.088}$	1.013 ± 0.043	1.021 ± 0.032	0.938 ± 0.036	0.360 ± 0.071
surv_trace	$\mathbf{1.091 \pm 0.054}$	–	–	–	–	1.082 ± 0.115	0.525 ± 0.228	0.898 ± 0.838	1.023 ± 0.058	1.041 ± 0.026	1.059 ± 0.027	0.397 ± 0.022
surv_framingham	$\mathbf{1.035 \pm 0.123}$	–	–	–	–	0.823 ± 0.291	0.141 ± 0.272	0.498 ± 1.113	0.782 ± 0.154	0.945 ± 0.061	0.868 ± 0.012	0.098 ± 0.038
tab_blood_transfusion	0.537 ± 0.787	–	–	–	–	0.758 ± 0.137	0.338 ± 0.613	0.556 ± 0.052	0.574 ± 0.228	0.738 ± 0.035	$\mathbf{0.903 \pm 0.076}$	0.119 ± 0.043
tab_credit_g	0.436 ± 0.148	–	–	–	–	0.916 ± 0.278	0.390 ± 0.481	0.948 ± 0.050	0.457 ± 0.079	0.814 ± 0.062	$\mathbf{1.093 \pm 0.042}$	0.029 ± 0.006
tab_qsar_biodeg	0.043 ± 0.017	–	–	–	–	0.091 ± 0.030	0.001 ± 0.003	0.134 ± 0.031	0.021 ± 0.016	0.290 ± 0.027	$\mathbf{0.896 \pm 0.039}$	0.008 ± 0.003
tab_maternal_risk	$\mathbf{1.084 \pm 0.118}$	–	–	–	–	0.603 ± 0.727	0.238 ± 0.222	0.994 ± 0.262	0.591 ± 0.157	0.842 ± 0.105	0.757 ± 0.140	0.218 ± 0.034
tab_website_phishing	2.197 ± 2.674	–	–	–	–	2.272 ± 0.372	1.588 ± 0.531	2.224 ± 0.125	2.064 ± 0.492	$\mathbf{2.375 \pm 0.007}$	2.274 ± 0.068	0.979 ± 0.018

Table 21: Coverage (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.841 ± 0.320	0.681 ± 0.111	0.202 ± 0.097	0.968 ± 0.026	0.933	0.925 ± 0.027	0.400 ± 0.510	0.940 ± 0.026	0.961 ± 0.033	$\mathbf{0.979 \pm 0.021}$	0.947 ± 0.088	0.096 ± 0.000
breast_cancer	0.217 ± 0.025	0.001 ± 0.003	0.144 ± 0.082	0.789 ± 0.055	0.778	0.304 ± 0.087	0.001 ± 0.003	0.305 ± 0.036	0.017 ± 0.025	0.511 ± 0.057	$\mathbf{0.978 \pm 0.062}$	0.008 ± 0.003
liver_disorders	0.899 ± 0.039	0.349 ± 0.086	0.686 ± 0.218	0.961 ± 0.010	**0.989**	0.798 ± 0.090	0.566 ± 0.207	0.460 ± 0.289	0.929 ± 0.105	0.984 ± 0.029	0.966 ± 0.054	0.165 ± 0.034
parkinsons	0.556 ± 0.096	0.004 ± 0.018	0.267 ± 0.079	0.812 ± 0.144	0.872	0.318 ± 0.199	0.015 ± 0.024	0.476 ± 0.133	0.171 ± 0.187	0.825 ± 0.040	$\mathbf{0.981 \pm 0.042}$	0.021 ± 0.009
obesity	0.882 ± 0.070	0.304 ± 0.070	0.283 ± 0.014	0.295 ± 0.169	0.725	0.540 ± 0.233	0.221 ± 0.115	0.561 ± 0.031	0.354 ± 0.048	0.866 ± 0.028	$\mathbf{0.941 \pm 0.008}$	0.028 ± 0.004
diabetes	0.907 ± 0.047	0.852 ± 0.024	0.763 ± 0.123	0.726 ± 0.154	–	0.902 ± 0.061	0.529 ± 0.218	0.861 ± 0.189	0.833 ± 0.081	0.952 ± 0.028	$\mathbf{0.953 \pm 0.031}$	0.164 ± 0.022
german_credit	0.783 ± 0.627	0.674 ± 0.127	0.310 ± 0.043	0.826 ± 0.006	–	0.905 ± 0.039	0.522 ± 0.128	0.909 ± 0.050	0.670 ± 0.065	0.953 ± 0.011	$\mathbf{0.984 \pm 0.011}$	0.076 ± 0.018
bank_marketing	$\mathbf{0.898 \pm 0.058}$	0.693 ± 0.099	0.712 ± 0.146	0.581 ± 0.015	–	0.616 ± 0.163	0.237 ± 0.197	0.632 ± 0.369	0.502 ± 0.087	0.805 ± 0.048	0.788 ± 0.035	0.006 ± 0.008
census_income	$\mathbf{0.841 \pm 0.081}$	0.527 ± 0.060	0.703 ± 0.154	0.288 ± 0.317	–	0.474 ± 0.425	0.100 ± 0.186	0.327 ± 0.512	0.306 ± 0.164	0.684 ± 0.103	0.767 ± 0.350	0.003 ± 0.000
surv_melanoma	0.911 ± 0.197	–	–	–	–	0.850 ± 0.229	0.516 ± 0.441	0.673 ± 0.166	0.931 ± 0.168	$\mathbf{0.972 \pm 0.032}$	0.963 ± 0.045	0.557 ± 0.155
surv_pbc	0.477 ± 0.977	–	–	–	–	0.896 ± 0.095	0.778 ± 0.075	0.554 ± 0.781	0.940 ± 0.020	0.989 ± 0.023	$\mathbf{0.993 \pm 0.015}$	0.708 ± 0.120
surv_pbc3	0.919 ± 0.105	–	–	–	–	0.872 ± 0.029	0.331 ± 0.294	0.888 ± 0.005	0.889 ± 0.079	0.974 ± 0.031	$\mathbf{0.994 \pm 0.005}$	0.082 ± 0.031
surv_retinopathy	0.702 ± 1.147	–	–	–	–	0.867 ± 0.250	0.591 ± 0.117	0.917 ± 0.020	0.773 ± 0.148	$\mathbf{0.984 \pm 0.048}$	0.980 ± 0.051	0.612 ± 0.228
surv_whas500	0.915 ± 0.052	–	–	–	–	0.936 ± 0.016	0.281 ± 0.290	0.914 ± 0.068	0.894 ± 0.007	0.976 ± 0.017	$\mathbf{0.983 \pm 0.010}$	0.113 ± 0.044
surv_gbsg2	0.814 ± 0.291	–	–	–	–	0.691 ± 0.396	0.411 ± 0.316	0.804 ± 0.046	0.881 ± 0.121	$\mathbf{0.952 \pm 0.026}$	0.651 ± 0.011	0.069 ± 0.016
surv_colon	0.958 ± 0.170	–	–	–	–	0.794 ± 0.301	0.399 ± 0.498	0.817 ± 0.036	0.935 ± 0.025	$\mathbf{0.983 \pm 0.019}$	0.938 ± 0.052	0.255 ± 0.073
surv_grace	0.940 ± 0.040	–	–	–	–	0.739 ± 0.269	0.464 ± 0.996	0.481 ± 0.041	0.942 ± 0.035	$\mathbf{0.981 \pm 0.006}$	0.952 ± 0.034	0.447 ± 0.081
surv_trace	0.933 ± 0.081	–	–	–	–	0.673 ± 0.156	0.519 ± 0.165	0.297 ± 0.427	0.890 ± 0.048	$\mathbf{0.979 \pm 0.025}$	0.971 ± 0.012	0.463 ± 0.053
surv_framingham	0.926 ± 0.034	–	–	–	–	0.844 ± 0.185	0.202 ± 0.072	0.405 ± 0.946	0.828 ± 0.036	$\mathbf{0.962 \pm 0.014}$	0.933 ± 0.023	0.138 ± 0.027
tab_blood_transfusion	0.376 ± 0.375	–	–	–	–	0.592 ± 0.222	0.282 ± 0.426	0.251 ± 0.157	0.785 ± 0.141	$\mathbf{0.933 \pm 0.053}$	0.915 ± 0.029	0.104 ± 0.022
tab_credit_g	0.623 ± 0.075	–	–	–	–	0.909 ± 0.044	0.422 ± 0.297	0.917 ± 0.032	0.735 ± 0.117	0.947 ± 0.027	$\mathbf{0.988 \pm 0.009}$	0.062 ± 0.010
tab_qsar_biodeg	0.085 ± 0.044	–	–	–	–	0.212 ± 0.065	0.003 ± 0.005	0.248 ± 0.042	0.048 ± 0.032	0.651 ± 0.046	$\mathbf{0.973 \pm 0.014}$	0.001 ± 0.000
tab_maternal_risk	0.899 ± 0.021	–	–	–	–	0.597 ± 0.484	0.270 ± 0.409	0.661 ± 0.068	0.715 ± 0.113	$\mathbf{0.918 \pm 0.067}$	0.770 ± 0.068	0.173 ± 0.046
tab_website_phishing	0.883 ± 0.440	–	–	–	–	0.988 ± 0.005	0.752 ± 0.176	0.982 ± 0.020	0.983 ± 0.013	$\mathbf{0.999 \pm 0.002}$	0.990 ± 0.011	0.838 ± 0.031

Detection / distribution

Table 22: C2ST AUROC (↓, 0.5 best) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.969 ± 0.039	0.956 ± 0.039	0.999 ± 0.003	**0.902 ± 0.033**	0.958	0.956 ± 0.014	0.976 ± 0.012	0.999 ± 0.003	0.939 ± 0.014	0.921 ± 0.048	0.932 ± 0.013	0.987 ± 0.003
breast_cancer	1.000 ± 0.000	1.000 ± 0.000	0.993 ± 0.005	0.817 ± 0.265	0.828	0.989 ± 0.014	0.998 ± 0.001	1.000 ± 0.000	0.988 ± 0.008	0.860 ± 0.004	**0.462 ± 0.065**	0.998 ± 0.002
liver_disorders	0.662 ± 0.068	0.960 ± 0.054	0.822 ± 0.062	0.593 ± 0.034	**0.461**	0.879 ± 0.055	0.914 ± 0.088	1.000 ± 0.000	0.648 ± 0.074	0.477 ± 0.005	0.560 ± 0.024	0.972 ± 0.002
parkinsons	0.999 ± 0.005	0.995 ± 0.016	0.984 ± 0.012	0.732 ± 0.150	0.728	0.976 ± 0.021	0.993 ± 0.006	0.993 ± 0.016	0.951 ± 0.043	0.793 ± 0.066	**0.483 ± 0.053**	0.983 ± 0.010
obesity	0.998 ± 0.002	0.993 ± 0.002	0.999 ± 0.001	0.991 ± 0.008	**0.839**	0.994 ± 0.003	0.997 ± 0.008	1.000 ± 0.000	0.993 ± 0.002	0.889 ± 0.002	0.977 ± 0.002	1.000 ± 0.000
diabetes	1.000 ± 0.001	0.951 ± 0.022	**0.927 ± 0.011**	0.984 ± 0.002	–	0.999 ± 0.001	0.997 ± 0.002	1.000 ± 0.000	0.997 ± 0.002	0.989 ± 0.004	0.996 ± 0.004	0.997 ± 0.002
german_credit	1.000 ± 0.000	0.937 ± 0.035	0.999 ± 0.000	0.877 ± 0.013	–	0.892 ± 0.078	0.977 ± 0.013	1.000 ± 0.001	0.924 ± 0.011	0.843 ± 0.020	**0.775 ± 0.028**	0.996 ± 0.002
bank_marketing	0.888 ± 0.048	0.910 ± 0.025	0.964 ± 0.023	0.985 ± 0.026	–	0.969 ± 0.062	0.998 ± 0.005	1.000 ± 0.000	0.968 ± 0.020	**0.848 ± 0.021**	0.966 ± 0.006	1.000 ± 0.000
census_income	1.000 ± 0.000	0.965 ± 0.020	0.980 ± 0.014	0.990 ± 0.034	–	0.995 ± 0.004	0.998 ± 0.002	1.000 ± 0.001	0.993 ± 0.001	**0.942 ± 0.009**	0.995 ± 0.008	1.000 ± 0.000
surv_melanoma	0.853 ± 0.047	–	–	–	–	0.841 ± 0.123	0.922 ± 0.147	0.999 ± 0.002	0.717 ± 0.075	**0.682 ± 0.048**	0.714 ± 0.107	0.890 ± 0.077
surv_pbc	0.729 ± 0.575	–	–	–	–	0.633 ± 0.068	0.790 ± 0.093	0.875 ± 0.458	0.536 ± 0.115	**0.476 ± 0.077**	0.485 ± 0.067	0.832 ± 0.045
surv_pbc3	0.913 ± 0.005	–	–	–	–	0.912 ± 0.017	0.970 ± 0.064	1.000 ± 0.000	0.849 ± 0.014	0.761 ± 0.057	**0.758 ± 0.013**	0.986 ± 0.001
surv_retinopathy	0.744 ± 0.580	–	–	–	–	0.795 ± 0.202	0.948 ± 0.023	0.981 ± 0.014	0.881 ± 0.065	0.653 ± 0.066	**0.469 ± 0.122**	0.944 ± 0.012
surv_whas500	1.000 ± 0.000	–	–	–	–	0.785 ± 0.049	0.971 ± 0.066	1.000 ± 0.000	0.675 ± 0.032	0.568 ± 0.023	**0.503 ± 0.058**	0.985 ± 0.004
surv_gbsg2	0.857 ± 0.256	–	–	–	–	0.914 ± 0.151	0.959 ± 0.066	1.000 ± 0.000	0.767 ± 0.045	**0.636 ± 0.042**	0.878 ± 0.016	0.994 ± 0.002
surv_colon	**0.600 ± 0.062**	–	–	–	–	0.994 ± 0.003	0.997 ± 0.004	1.000 ± 0.000	0.995 ± 0.004	0.910 ± 0.040	0.989 ± 0.003	0.989 ± 0.005
surv_grace	0.671 ± 0.038	–	–	–	–	0.867 ± 0.080	0.927 ± 0.167	1.000 ± 0.000	0.734 ± 0.040	**0.654 ± 0.030**	0.699 ± 0.033	0.942 ± 0.018
surv_trace	0.999 ± 0.002	–	–	–	–	0.995 ± 0.003	0.992 ± 0.002	0.999 ± 0.003	0.993 ± 0.003	**0.902 ± 0.024**	0.994 ± 0.004	0.985 ± 0.004
surv_framingham	0.804 ± 0.036	–	–	–	–	0.910 ± 0.081	0.988 ± 0.003	0.996 ± 0.012	0.864 ± 0.031	**0.785 ± 0.017**	0.792 ± 0.007	0.988 ± 0.004
tab_blood_transfusion	0.992 ± 0.027	–	–	–	–	0.986 ± 0.010	0.989 ± 0.014	1.000 ± 0.000	0.979 ± 0.016	**0.898 ± 0.017**	0.971 ± 0.009	0.992 ± 0.005
tab_credit_g	1.000 ± 0.000	–	–	–	–	0.894 ± 0.044	0.977 ± 0.057	1.000 ± 0.001	0.920 ± 0.014	0.847 ± 0.019	**0.784 ± 0.007**	0.997 ± 0.001
tab_qsar_biodeg	1.000 ± 0.002	–	–	–	–	0.999 ± 0.001	0.999 ± 0.003	1.000 ± 0.000	0.999 ± 0.001	**0.967 ± 0.011**	0.987 ± 0.002	1.000 ± 0.000
tab_maternal_risk	0.999 ± 0.002	–	–	–	–	0.993 ± 0.004	0.995 ± 0.008	0.999 ± 0.004	0.983 ± 0.002	0.974 ± 0.015	**0.973 ± 0.023**	0.995 ± 0.003
tab_website_phishing	0.771 ± 0.501	–	–	–	–	0.635 ± 0.052	0.927 ± 0.034	0.664 ± 0.047	0.707 ± 0.081	**0.495 ± 0.038**	0.584 ± 0.034	0.904 ± 0.029

Table 23: MMD^2 (↓) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.050 ± 0.074	0.099 ± 0.049	0.392 ± 0.088	**0.000 ± 0.000**	0.188	0.033 ± 0.010	0.242 ± 0.210	0.027 ± 0.007	0.007 ± 0.011	0.002 ± 0.004	0.009 ± 0.023	0.399 ± 0.025
breast_cancer	0.104 ± 0.055	0.218 ± 0.041	0.110 ± 0.068	0.009 ± 0.003	0.227	0.058 ± 0.013	0.806 ± 0.460	0.101 ± 0.028	0.039 ± 0.057	0.004 ± 0.005	**0.001 ± 0.002**	0.573 ± 0.050
liver_disorders	0.051 ± 0.056	0.276 ± 0.126	0.118 ± 0.148	0.001 ± 0.002	**0.000**	0.066 ± 0.048	0.132 ± 0.156	0.066 ± 0.030	0.021 ± 0.034	0.001 ± 0.002	0.003 ± 0.002	0.480 ± 0.041
parkinsons	0.083 ± 0.117	0.202 ± 0.086	0.140 ± 0.050	**0.000 ± 0.000**	0.121	0.096 ± 0.059	0.145 ± 0.265	0.119 ± 0.032	0.022 ± 0.016	0.004 ± 0.010	**0.000 ± 0.000**	0.488 ± 0.007
obesity	0.005 ± 0.006	0.037 ± 0.027	0.081 ± 0.022	0.031 ± 0.042	0.014	0.008 ± 0.020	0.109 ± 0.200	0.006 ± 0.002	0.019 ± 0.010	0.004 ± 0.003	**0.000 ± 0.000**	0.202 ± 0.030
diabetes	0.024 ± 0.072	0.019 ± 0.007	0.017 ± 0.019	0.019 ± 0.030	–	0.007 ± 0.007	0.157 ± 0.159	0.025 ± 0.086	0.019 ± 0.019	**0.000 ± 0.000**	0.001 ± 0.001	0.381 ± 0.085
german_credit	0.046 ± 0.027	0.024 ± 0.030	0.299 ± 0.013	0.001 ± 0.003	–	0.008 ± 0.009	0.049 ± 0.039	0.007 ± 0.004	0.015 ± 0.010	0.000 ± 0.000	**0.000 ± 0.000**	0.238 ± 0.022
bank_marketing	0.012 ± 0.012	0.010 ± 0.006	0.033 ± 0.014	0.012 ± 0.009	–	0.021 ± 0.015	0.090 ± 0.130	0.019 ± 0.017	0.027 ± 0.018	**0.005 ± 0.003**	0.007 ± 0.001	0.873 ± 0.104
census_income	0.039 ± 0.047	0.012 ± 0.010	0.039 ± 0.040	0.123 ± 0.294	–	0.048 ± 0.057	0.224 ± 0.647	0.065 ± 0.111	0.044 ± 0.032	**0.009 ± 0.004**	0.026 ± 0.006	0.676 ± 0.035
surv_melanoma	0.063 ± 0.192	–	–	–	–	0.036 ± 0.064	0.223 ± 0.252	0.031 ± 0.043	0.008 ± 0.013	**0.000 ± 0.000**	0.006 ± 0.010	0.179 ± 0.051
surv_pbc	0.038 ± 0.054	–	–	–	–	0.029 ± 0.032	0.106 ± 0.069	0.109 ± 0.386	0.024 ± 0.023	0.000 ± 0.001	**0.000 ± 0.000**	0.153 ± 0.025
surv_pbc3	0.026 ± 0.021	–	–	–	–	0.040 ± 0.015	0.225 ± 0.202	0.049 ± 0.023	0.016 ± 0.009	0.001 ± 0.002	**0.000 ± 0.000**	0.340 ± 0.021
surv_retinopathy	0.042 ± 0.129	–	–	–	–	0.018 ± 0.035	0.052 ± 0.049	0.007 ± 0.006	0.028 ± 0.019	**0.000 ± 0.000**	**0.000 ± 0.000**	0.045 ± 0.019
surv_whas500	0.027 ± 0.036	–	–	–	–	0.017 ± 0.021	0.259 ± 0.275	0.034 ± 0.043	0.007 ± 0.003	0.000 ± 0.000	**0.000 ± 0.000**	0.394 ± 0.016
surv_gbsg2	0.038 ± 0.070	–	–	–	–	0.087 ± 0.190	0.200 ± 0.235	0.020 ± 0.008	0.006 ± 0.008	**0.004 ± 0.001**	0.039 ± 0.004	0.661 ± 0.073
surv_colon	0.007 ± 0.012	–	–	–	–	0.052 ± 0.097	0.187 ± 0.228	0.009 ± 0.010	0.007 ± 0.005	**0.000 ± 0.000**	0.003 ± 0.002	0.386 ± 0.055
surv_grace	0.013 ± 0.029	–	–	–	–	0.059 ± 0.072	0.422 ± 1.087	0.011 ± 0.001	0.011 ± 0.020	**0.000 ± 0.000**	0.002 ± 0.002	0.271 ± 0.073
surv_trace	0.008 ± 0.028	–	–	–	–	0.017 ± 0.020	0.213 ± 0.094	0.140 ± 0.430	0.013 ± 0.010	0.000 ± 0.001	**0.000 ± 0.000**	0.232 ± 0.022
surv_framingham	0.007 ± 0.002	–	–	–	–	0.027 ± 0.036	0.380 ± 0.213	0.188 ± 0.399	0.019 ± 0.018	**0.001 ± 0.002**	0.003 ± 0.003	0.365 ± 0.077
tab_blood_transfusion	0.108 ± 0.355	–	–	–	–	0.089 ± 0.049	0.292 ± 0.265	0.026 ± 0.017	0.014 ± 0.017	**0.000 ± 0.000**	0.012 ± 0.008	0.632 ± 0.039
tab_credit_g	0.085 ± 0.016	–	–	–	–	0.010 ± 0.017	0.079 ± 0.127	0.006 ± 0.003	0.014 ± 0.007	0.001 ± 0.002	**0.000 ± 0.000**	0.244 ± 0.031
tab_qsar_biodeg	0.067 ± 0.073	–	–	–	–	0.034 ± 0.027	0.568 ± 0.387	0.025 ± 0.016	0.036 ± 0.056	0.004 ± 0.003	**0.002 ± 0.001**	0.680 ± 0.030
tab_maternal_risk	0.007 ± 0.010	–	–	–	–	0.066 ± 0.165	0.316 ± 0.525	0.019 ± 0.023	0.017 ± 0.019	**0.000 ± 0.001**	0.026 ± 0.010	0.360 ± 0.064
tab_website_phishing	0.043 ± 0.155	–	–	–	–	0.005 ± 0.009	0.083 ± 0.069	0.003 ± 0.005	0.009 ± 0.008	**0.000 ± 0.000**	0.000 ± 0.000	0.034 ± 0.006

Table 24: Fréchet-PCA (↓) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	2.197 ± 1.768	4.871 ± 3.551	10.777 ± 1.890	**0.309 ± 0.172**	36.258	1.474 ± 0.435	11.078 ± 7.624	1.620 ± 0.978	0.651 ± 0.427	0.331 ± 0.176	0.664 ± 0.694	36.526 ± 6.670
breast_cancer	24.230 ± 35.796	29.409 ± 5.766	13.849 ± 4.611	0.790 ± 0.154	262.006	5.478 ± 2.510	107.847 ± 59.046	12.971 ± 1.348	11.090 ± 12.826	1.360 ± 0.375	**0.471 ± 0.211**	216.031 ± 50.704
liver_disorders	0.799 ± 0.631	5.592 ± 4.179	2.911 ± 0.827	0.102 ± 0.040	0.163	1.003 ± 1.006	3.041 ± 2.091	1.369 ± 0.855	0.689 ± 0.564	**0.071 ± 0.009**	0.171 ± 0.140	26.046 ± 2.381
parkinsons	6.899 ± 3.221	20.827 ± 6.418	12.444 ± 2.154	0.514 ± 0.286	77.241	7.191 ± 3.318	19.322 ± 15.333	12.159 ± 1.849	5.990 ± 2.586	1.317 ± 0.600	**0.304 ± 0.113**	83.740 ± 5.762
obesity	0.287 ± 0.337	1.640 ± 0.992	2.601 ± 0.085	1.213 ± 1.935	2.692	0.349 ± 0.768	4.702 ± 5.667	0.368 ± 0.063	0.948 ± 0.500	0.193 ± 0.057	**0.036 ± 0.023**	10.496 ± 2.717
diabetes	1.776 ± 5.212	1.105 ± 0.353	0.953 ± 0.723	0.542 ± 0.953	–	0.263 ± 0.077	3.167 ± 1.559	0.796 ± 1.901	0.996 ± 1.917	**0.058 ± 0.023**	0.062 ± 0.015	27.853 ± 17.394
german_credit	1.858 ± 5.398	1.687 ± 1.583	8.982 ± 0.810	0.254 ± 0.129	–	0.435 ± 0.334	2.723 ± 1.713	0.581 ± 0.146	1.105 ± 0.711	0.103 ± 0.027	**0.071 ± 0.031**	16.535 ± 1.822
bank_marketing	0.856 ± 0.799	0.585 ± 0.580	2.557 ± 1.284	3.818 ± 12.599	–	1.044 ± 0.846	12.075 ± 30.720	0.902 ± 0.377	8.494 ± 29.701	**0.127 ± 0.061**	0.237 ± 0.236	1835.097 ± 5924.108
census_income	1.344 ± 1.265	0.799 ± 0.778	3.091 ± 1.218	21.677 ± 39.564	–	1.784 ± 1.003	7.884 ± 15.023	2.709 ± 1.327	1.540 ± 1.072	**0.188 ± 0.072**	0.891 ± 0.055	113.086 ± 33.070
surv_melanoma	0.975 ± 2.300	–	–	–	–	0.801 ± 1.452	4.625 ± 4.366	0.859 ± 0.984	0.386 ± 0.476	**0.073 ± 0.027**	0.271 ± 0.304	4.857 ± 2.184
surv_pbc	1.520 ± 2.591	–	–	–	–	0.475 ± 0.287	1.865 ± 0.658	2.770 ± 9.431	0.439 ± 0.332	0.057 ± 0.052	**0.057 ± 0.030**	4.204 ± 0.987
surv_pbc3	1.247 ± 0.927	–	–	–	–	1.710 ± 1.121	11.014 ± 5.263	2.785 ± 0.549	1.009 ± 0.265	0.287 ± 0.064	**0.222 ± 0.061**	28.665 ± 2.528
surv_retinopathy	0.982 ± 2.703	–	–	–	–	0.824 ± 0.867	2.196 ± 1.095	0.966 ± 0.197	1.526 ± 0.668	0.320 ± 0.102	**0.035 ± 0.050**	2.611 ± 0.508
surv_whas500	1.544 ± 1.357	–	–	–	–	0.955 ± 0.983	16.775 ± 10.154	1.925 ± 1.097	0.928 ± 1.290	0.244 ± 0.080	**0.205 ± 0.061**	67.247 ± 23.059
surv_gbsg2	11.501 ± 46.501	–	–	–	–	2.080 ± 3.255	8.209 ± 14.802	1.058 ± 0.462	0.470 ± 0.379	**0.134 ± 0.020**	0.835 ± 0.070	62.397 ± 16.244
surv_colon	0.417 ± 1.688	–	–	–	–	1.398 ± 2.134	5.128 ± 3.767	0.730 ± 0.497	0.493 ± 0.805	**0.054 ± 0.009**	0.180 ± 0.032	26.894 ± 8.398
surv_grace	0.219 ± 0.264	–	–	–	–	0.998 ± 0.921	9.721 ± 23.582	0.345 ± 0.082	0.194 ± 0.296	**0.014 ± 0.016**	0.086 ± 0.064	11.076 ± 5.627
surv_trace	0.308 ± 1.020	–	–	–	–	0.659 ± 0.398	3.817 ± 2.281	3.200 ± 8.118	0.410 ± 0.376	0.022 ± 0.017	**0.016 ± 0.008**	8.624 ± 0.478
surv_framingham	0.209 ± 0.105	–	–	–	–	0.623 ± 0.708	10.909 ± 10.687	4.702 ± 9.962	0.590 ± 0.403	**0.048 ± 0.019**	0.168 ± 0.038	29.902 ± 8.845
tab_blood_transfusion	1.560 ± 3.196	–	–	–	–	1.406 ± 0.086	3.423 ± 2.467	1.012 ± 0.429	0.921 ± 0.542	0.212 ± 0.073	**0.146 ± 0.084**	38.142 ± 4.669
tab_credit_g	2.865 ± 0.397	–	–	–	–	0.532 ± 0.671	3.872 ± 4.041	0.574 ± 0.243	0.919 ± 0.376	0.153 ± 0.105	**0.064 ± 0.013**	17.004 ± 2.819

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
tab_qsar_biodeg	14.141 ± 1.739	–	–	–	–	4.580 ± 1.868	59.061 ± 54.876	9.061 ± 5.276	10.922 ± 9.358	2.540 ± 1.324	**0.348 ± 0.255**	754.443 ± 102.561
tab_maternal_risk	0.275 ± 0.121	–	–	–	–	1.322 ± 3.054	5.393 ± 6.872	0.583 ± 0.543	0.632 ± 0.218	**0.082 ± 0.052**	0.710 ± 0.568	18.273 ± 4.293
tab_website_phishing	0.562 ± 1.887	–	–	–	–	0.119 ± 0.090	1.272 ± 0.917	0.099 ± 0.087	0.198 ± 0.118	**0.031 ± 0.011**	0.052 ± 0.014	0.612 ± 0.098

Table 25: Wasserstein (↓) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.319 ± 0.238	0.391 ± 0.066	0.597 ± 0.056	**0.111 ± 0.027**	0.958	0.247 ± 0.038	0.685 ± 0.228	0.308 ± 0.058	0.142 ± 0.063	0.125 ± 0.023	0.124 ± 0.096	1.085 ± 0.106
breast_cancer	0.309 ± 0.054	0.692 ± 0.122	0.393 ± 0.103	0.150 ± 0.026	1.851	0.309 ± 0.043	1.567 ± 0.561	0.510 ± 0.065	0.167 ± 0.172	0.107 ± 0.063	**0.090 ± 0.021**	2.098 ± 0.235
liver_disorders	0.378 ± 0.237	0.863 ± 0.148	0.423 ± 0.139	**0.109 ± 0.024**	0.113	0.356 ± 0.066	0.486 ± 0.236	0.425 ± 0.089	0.236 ± 0.149	0.112 ± 0.019	0.121 ± 0.019	1.872 ± 0.095
parkinsons	0.264 ± 0.139	0.622 ± 0.122	0.406 ± 0.061	0.120 ± 0.014	1.227	0.384 ± 0.126	0.471 ± 0.482	0.540 ± 0.058	0.164 ± 0.045	0.138 ± 0.076	**0.081 ± 0.029**	1.477 ± 0.048
obesity	0.119 ± 0.042	0.308 ± 0.151	0.358 ± 0.054	0.296 ± 0.123	0.300	0.172 ± 0.133	0.539 ± 0.618	0.175 ± 0.016	0.187 ± 0.091	0.146 ± 0.030	**0.037 ± 0.002**	0.727 ± 0.069
diabetes	0.235 ± 0.361	0.245 ± 0.016	0.164 ± 0.063	0.184 ± 0.096	–	0.144 ± 0.036	0.471 ± 0.165	0.185 ± 0.196	0.207 ± 0.098	0.067 ± 0.008	**0.026 ± 0.000**	1.483 ± 0.399
german_credit	0.225 ± 0.341	0.252 ± 0.099	0.663 ± 0.091	0.077 ± 0.028	–	0.157 ± 0.070	0.406 ± 0.157	0.185 ± 0.024	0.190 ± 0.115	0.069 ± 0.015	**0.056 ± 0.006**	1.113 ± 0.097
bank_marketing	0.162 ± 0.050	0.133 ± 0.066	0.245 ± 0.048	0.272 ± 0.184	–	0.184 ± 0.067	0.648 ± 0.882	0.201 ± 0.019	0.221 ± 0.179	0.087 ± 0.012	**0.072 ± 0.022**	7.511 ± 8.825
census_income	0.279 ± 0.245	0.141 ± 0.084	0.264 ± 0.113	0.762 ± 1.192	–	0.248 ± 0.047	0.623 ± 0.925	0.202 ± 0.050	0.194 ± 0.096	0.113 ± 0.007	**0.059 ± 0.001**	3.209 ± 0.490
surv_melanoma	0.327 ± 0.570	–	–	–	–	0.273 ± 0.278	0.726 ± 0.538	0.333 ± 0.108	0.183 ± 0.068	**0.108 ± 0.016**	0.138 ± 0.082	0.686 ± 0.139
surv_pbc	0.262 ± 0.135	–	–	–	–	0.226 ± 0.064	0.417 ± 0.154	0.354 ± 0.853	0.215 ± 0.095	0.058 ± 0.076	**0.054 ± 0.044**	0.510 ± 0.041
surv_pbc3	0.227 ± 0.144	–	–	–	–	0.290 ± 0.122	0.718 ± 0.237	0.385 ± 0.055	0.168 ± 0.022	0.113 ± 0.028	**0.086 ± 0.026**	1.194 ± 0.035
surv_retinopathy	0.164 ± 0.153	–	–	–	–	0.204 ± 0.179	0.305 ± 0.222	0.193 ± 0.075	0.215 ± 0.153	0.066 ± 0.017	**0.049 ± 0.017**	0.323 ± 0.024
surv_whas500	0.223 ± 0.155	–	–	–	–	0.176 ± 0.083	0.615 ± 0.359	0.298 ± 0.126	0.127 ± 0.031	0.070 ± 0.015	**0.063 ± 0.018**	1.128 ± 0.056
surv_gbsg2	0.270 ± 0.367	–	–	–	–	0.405 ± 0.369	0.801 ± 0.599	0.321 ± 0.052	0.124 ± 0.032	0.132 ± 0.017	**0.069 ± 0.034**	2.844 ± 0.265
surv_colon	0.137 ± 0.011	–	–	–	–	0.268 ± 0.234	0.501 ± 0.337	0.166 ± 0.051	0.117 ± 0.081	**0.043 ± 0.008**	0.071 ± 0.007	1.267 ± 0.162
surv_grace	0.159 ± 0.104	–	–	–	–	0.336 ± 0.199	0.967 ± 1.322	0.247 ± 0.036	0.153 ± 0.104	**0.048 ± 0.018**	0.080 ± 0.017	0.941 ± 0.126
surv_trace	0.099 ± 0.143	–	–	–	–	0.209 ± 0.034	0.547 ± 0.186	0.432 ± 0.568	0.175 ± 0.088	0.040 ± 0.016	**0.037 ± 0.010**	0.885 ± 0.017
surv_framingham	0.135 ± 0.036	–	–	–	–	0.232 ± 0.127	0.985 ± 0.375	0.581 ± 0.872	0.214 ± 0.130	0.053 ± 0.010	**0.052 ± 0.005**	1.341 ± 0.154
tab_blood_transfusion	0.397 ± 0.744	–	–	–	–	0.459 ± 0.094	0.658 ± 0.258	0.323 ± 0.090	0.167 ± 0.116	**0.084 ± 0.036**	0.126 ± 0.008	2.406 ± 0.154
tab_credit_g	0.178 ± 0.076	–	–	–	–	0.177 ± 0.186	0.464 ± 0.445	0.194 ± 0.033	0.173 ± 0.042	0.086 ± 0.035	**0.060 ± 0.019**	1.133 ± 0.086
tab_qsar_biodeg	0.281 ± 0.076	–	–	–	–	0.192 ± 0.058	0.787 ± 0.514	0.244 ± 0.025	0.184 ± 0.084	0.104 ± 0.020	**0.068 ± 0.013**	3.328 ± 0.200
tab_maternal_risk	0.146 ± 0.084	–	–	–	–	0.323 ± 0.394	0.694 ± 0.565	0.304 ± 0.062	0.174 ± 0.060	**0.124 ± 0.016**	0.255 ± 0.059	1.182 ± 0.096
tab_website_phishing	–	–	–	–	–	–	–	–	–	–	–	–

Table 26: Association error (↓) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.057 ± 0.005	0.059 ± 0.005	0.070 ± 0.006	0.055 ± 0.007	0.150	0.069 ± 0.012	0.094 ± 0.022	0.060 ± 0.012	0.058 ± 0.008	**0.050 ± 0.020**	0.071 ± 0.013	0.063 ± 0.009
breast_cancer	0.278 ± 0.019	0.347 ± 0.010	0.195 ± 0.007	**0.051 ± 0.017**	0.188	0.107 ± 0.028	0.290 ± 0.050	0.240 ± 0.022	0.298 ± 0.045	0.099 ± 0.008	0.053 ± 0.028	0.357 ± 0.012
liver_disorders	0.077 ± 0.095	0.222 ± 0.028	0.156 ± 0.099	0.052 ± 0.005	0.080	0.118 ± 0.122	0.216 ± 0.077	0.207 ± 0.025	0.171 ± 0.019	**0.051 ± 0.010**	0.095 ± 0.055	0.232 ± 0.048
parkinsons	0.242 ± 0.046	0.444 ± 0.026	0.335 ± 0.008	0.056 ± 0.018	0.224	0.175 ± 0.029	0.366 ± 0.025	0.369 ± 0.044	0.378 ± 0.054	0.133 ± 0.021	**0.047 ± 0.007**	0.450 ± 0.016
obesity	0.047 ± 0.002	0.103 ± 0.004	0.086 ± 0.008	0.069 ± 0.006	0.115	0.039 ± 0.058	0.091 ± 0.017	0.074 ± 0.008	0.087 ± 0.013	0.037 ± 0.010	**0.031 ± 0.001**	0.105 ± 0.002
diabetes	0.055 ± 0.056	0.090 ± 0.048	0.056 ± 0.028	0.071 ± 0.014	–	0.027 ± 0.003	0.079 ± 0.027	0.075 ± 0.018	0.065 ± 0.010	**0.016 ± 0.004**	0.037 ± 0.008	0.098 ± 0.006
german_credit	0.108 ± 0.509	0.085 ± 0.004	0.111 ± 0.014	0.072 ± 0.006	–	0.048 ± 0.010	0.082 ± 0.003	0.063 ± 0.013	0.080 ± 0.004	0.047 ± 0.009	**0.033 ± 0.005**	0.086 ± 0.003
bank_marketing	0.038 ± 0.004	0.075 ± 0.021	0.079 ± 0.023	0.067 ± 0.004	–	**0.034 ± 0.014**	0.084 ± 0.009	0.043 ± 0.023	0.072 ± 0.005	0.038 ± 0.005	0.048 ± 0.009	0.089 ± 0.003
census_income	0.059 ± 0.024	0.081 ± 0.007	0.075 ± 0.013	0.108 ± 0.017	–	0.044 ± 0.011	0.116 ± 0.015	0.070 ± 0.036	0.104 ± 0.007	0.056 ± 0.004	**0.025 ± 0.007**	0.127 ± 0.002
surv_melanoma	0.074 ± 0.023	–	–	–	–	0.086 ± 0.068	0.119 ± 0.076	0.113 ± 0.023	0.109 ± 0.086	0.059 ± 0.026	**0.055 ± 0.005**	0.130 ± 0.008
surv_pbc	0.109 ± 0.112	–	–	–	–	0.060 ± 0.045	0.087 ± 0.016	0.082 ± 0.046	0.073 ± 0.026	**0.051 ± 0.037**	0.056 ± 0.025	0.090 ± 0.037
surv_pbc3	0.081 ± 0.019	–	–	–	–	0.068 ± 0.034	0.099 ± 0.013	0.086 ± 0.009	0.096 ± 0.002	0.056 ± 0.019	**0.051 ± 0.007**	0.097 ± 0.010
surv_retinopathy	0.082 ± 0.086	–	–	–	–	0.088 ± 0.029	0.120 ± 0.010	0.093 ± 0.022	0.097 ± 0.038	0.056 ± 0.020	**0.040 ± 0.048**	0.120 ± 0.007
surv_whas500	0.069 ± 0.020	–	–	–	–	0.055 ± 0.015	0.084 ± 0.029	0.064 ± 0.005	0.066 ± 0.009	0.048 ± 0.003	**0.046 ± 0.008**	0.071 ± 0.004
surv_gbsg2	0.062 ± 0.042	–	–	–	–	0.098 ± 0.088	0.126 ± 0.008	0.090 ± 0.043	0.092 ± 0.036	**0.044 ± 0.021**	0.090 ± 0.002	0.121 ± 0.030
surv_colon	0.038 ± 0.017	–	–	–	–	0.082 ± 0.055	0.061 ± 0.013	0.043 ± 0.013	0.049 ± 0.015	**0.034 ± 0.013**	0.048 ± 0.007	0.057 ± 0.003
surv_grace	0.039 ± 0.020	–	–	–	–	0.044 ± 0.018	0.088 ± 0.060	0.060 ± 0.023	0.040 ± 0.039	**0.021 ± 0.012**	0.061 ± 0.033	0.055 ± 0.010
surv_trace	0.034 ± 0.019	–	–	–	–	0.062 ± 0.053	0.095 ± 0.030	0.104 ± 0.045	0.066 ± 0.040	0.021 ± 0.016	**0.019 ± 0.007**	0.109 ± 0.007
surv_framingham	0.023 ± 0.014	–	–	–	–	0.032 ± 0.005	0.144 ± 0.045	0.107 ± 0.003	0.113 ± 0.020	**0.018 ± 0.004**	0.073 ± 0.011	0.152 ± 0.008
tab_blood_transfusion	0.233 ± 0.628	–	–	–	–	0.120 ± 0.110	0.369 ± 0.086	0.315 ± 0.176	0.266 ± 0.145	0.066 ± 0.039	**0.064 ± 0.033**	0.402 ± 0.057
tab_credit_g	0.064 ± 0.003	–	–	–	–	0.047 ± 0.008	0.081 ± 0.012	0.061 ± 0.007	0.077 ± 0.006	0.047 ± 0.005	**0.034 ± 0.005**	0.086 ± 0.003
tab_qsar_biodeg	0.154 ± 0.022	–	–	–	–	0.083 ± 0.009	0.147 ± 0.019	0.117 ± 0.001	0.142 ± 0.008	0.063 ± 0.008	**0.032 ± 0.005**	0.147 ± 0.003
tab_maternal_risk	0.079 ± 0.068	–	–	–	–	0.081 ± 0.037	0.166 ± 0.044	0.114 ± 0.044	0.160 ± 0.034	**0.041 ± 0.010**	0.084 ± 0.041	0.205 ± 0.010
tab_website_phishing	0.081 ± 0.082	–	–	–	–	0.058 ± 0.013	0.105 ± 0.005	0.076 ± 0.012	0.067 ± 0.014	**0.047 ± 0.014**	0.051 ± 0.010	0.106 ± 0.008

SDMetrics quality report

Table 27: Overall quality (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.374 ± 0.062	0.369 ± 0.021	0.307 ± 0.016	**0.423 ± 0.010**	0.341	0.387 ± 0.010	0.297 ± 0.060	0.348 ± 0.017	0.415 ± 0.021	0.419 ± 0.004	0.420 ± 0.028	0.317 ± 0.015
breast_cancer	0.707 ± 0.025	0.600 ± 0.005	0.719 ± 0.033	0.849 ± 0.004	0.704	0.783 ± 0.017	0.404 ± 0.149	0.614 ± 0.031	0.747 ± 0.003	0.840 ± 0.022	**0.876 ± 0.008**	0.522 ± 0.009
liver_disorders	0.805 ± 0.080	0.597 ± 0.056	0.705 ± 0.040	0.870 ± 0.003	0.866	0.787 ± 0.024	0.635 ± 0.104	0.646 ± 0.040	0.760 ± 0.052	**0.871 ± 0.014**	0.851 ± 0.024	0.564 ± 0.021
parkinsons	0.766 ± 0.041	0.556 ± 0.036	0.669 ± 0.012	0.847 ± 0.012	0.726	0.731 ± 0.059	0.603 ± 0.128	0.564 ± 0.020	0.724 ± 0.019	0.813 ± 0.030	**0.861 ± 0.006**	0.520 ± 0.018
obesity	0.843 ± 0.054	0.775 ± 0.035	0.747 ± 0.050	0.816 ± 0.008	0.850	0.836 ± 0.053	0.718 ± 0.200	0.695 ± 0.033	0.792 ± 0.019	0.894 ± 0.032	**0.914 ± 0.004**	0.578 ± 0.006
diabetes	0.385 ± 0.023	0.391 ± 0.005	0.399 ± 0.009	0.404 ± 0.014	–	0.396 ± 0.016	0.336 ± 0.060	0.388 ± 0.022	0.388 ± 0.025	0.421 ± 0.003	**0.428 ± 0.000**	0.272 ± 0.011
german_credit	0.715 ± 0.129	0.761 ± 0.019	0.557 ± 0.073	0.821 ± 0.007	–	0.816 ± 0.015	0.732 ± 0.022	0.785 ± 0.023	0.750 ± 0.014	0.840 ± 0.017	**0.863 ± 0.013**	0.542 ± 0.010
bank_marketing	0.704 ± 0.005	0.709 ± 0.017	0.690 ± 0.024	0.702 ± 0.014	–	0.648 ± 0.100	0.511 ± 0.111	0.586 ± 0.116	0.614 ± 0.053	**0.718 ± 0.013**	0.692 ± 0.007	0.404 ± 0.004
census_income	0.518 ± 0.085	0.590 ± 0.026	0.607 ± 0.044	0.597 ± 0.060	–	0.541 ± 0.089	0.402 ± 0.289	0.531 ± 0.106	0.502 ± 0.049	**0.611 ± 0.010**	0.585 ± 0.033	0.305 ± 0.002

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
surv_melanoma	0.341 ± 0.124	–	–	–	–	0.352 ± 0.049	0.285 ± 0.119	0.317 ± 0.033	0.376 ± 0.007	**0.388 ± 0.006**	0.387 ± 0.014	0.318 ± 0.006
surv_pbc	0.447 ± 0.127	–	–	–	–	0.472 ± 0.224	0.445 ± 0.228	0.390 ± 0.086	0.500 ± 0.280	0.520 ± 0.263	**0.528 ± 0.287**	0.404 ± 0.150
surv_pbc3	0.431 ± 0.184	–	–	–	–	0.434 ± 0.227	0.354 ± 0.198	0.378 ± 0.205	0.452 ± 0.198	0.475 ± 0.216	**0.488 ± 0.220**	0.344 ± 0.158
surv_retinopathy	0.768 ± 0.324	–	–	–	–	0.763 ± 0.106	0.606 ± 0.073	0.618 ± 0.016	0.720 ± 0.021	0.846 ± 0.013	**0.899 ± 0.016**	0.623 ± 0.030
surv_whas500	0.516 ± 0.068	–	–	–	–	0.534 ± 0.018	0.424 ± 0.078	0.467 ± 0.061	0.530 ± 0.044	0.546 ± 0.028	**0.560 ± 0.033**	0.419 ± 0.030
surv_gbsg2	0.671 ± 0.222	–	–	–	–	0.605 ± 0.208	0.562 ± 0.115	0.457 ± 0.120	0.720 ± 0.110	**0.742 ± 0.128**	0.605 ± 0.083	0.437 ± 0.135
surv_colon	0.876 ± 0.360	–	–	–	–	0.864 ± 0.065	0.669 ± 0.094	0.842 ± 0.007	0.827 ± 0.105	0.926 ± 0.022	**0.929 ± 0.003**	0.634 ± 0.037
surv_grace	0.381 ± 0.017	–	–	–	–	0.340 ± 0.048	0.265 ± 0.200	0.350 ± 0.012	0.379 ± 0.026	**0.404 ± 0.005**	0.396 ± 0.004	0.316 ± 0.010
surv_trace	0.403 ± 0.028	–	–	–	–	0.381 ± 0.018	0.330 ± 0.025	0.347 ± 0.099	0.392 ± 0.016	0.418 ± 0.003	**0.418 ± 0.001**	0.307 ± 0.006
surv_framingham	**0.898 ± 0.008**	–	–	–	–	0.867 ± 0.060	0.547 ± 0.021	0.692 ± 0.174	0.751 ± 0.053	0.898 ± 0.001	0.877 ± 0.031	0.600 ± 0.025
tab_blood_transfusion	0.610 ± 0.344	–	–	–	–	0.608 ± 0.043	0.474 ± 0.129	0.561 ± 0.056	0.643 ± 0.065	0.732 ± 0.024	**0.733 ± 0.005**	0.427 ± 0.026
tab_credit_g	0.667 ± 0.044	–	–	–	–	0.899 ± 0.057	0.748 ± 0.202	0.871 ± 0.056	0.856 ± 0.019	0.928 ± 0.014	**0.964 ± 0.017**	0.612 ± 0.003
tab_qsar_biodeg	0.707 ± 0.050	–	–	–	–	0.768 ± 0.010	0.550 ± 0.116	0.694 ± 0.024	0.712 ± 0.028	0.813 ± 0.028	**0.851 ± 0.023**	0.475 ± 0.013
tab_maternal_risk	0.746 ± 0.100	–	–	–	–	0.689 ± 0.112	0.539 ± 0.194	0.573 ± 0.005	0.782 ± 0.021	**0.850 ± 0.017**	0.690 ± 0.036	0.537 ± 0.045
tab_website_phishing	0.852 ± 0.243	–	–	–	–	0.910 ± 0.025	0.692 ± 0.099	0.902 ± 0.031	0.886 ± 0.005	0.956 ± 0.011	**0.959 ± 0.011**	0.750 ± 0.009

Table 28: Column shapes (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.748 ± 0.124	0.738 ± 0.042	0.614 ± 0.032	**0.846 ± 0.020**	0.681	0.773 ± 0.019	0.595 ± 0.120	0.696 ± 0.033	0.830 ± 0.041	0.838 ± 0.007	0.841 ± 0.056	0.634 ± 0.029
breast_cancer	0.756 ± 0.052	0.649 ± 0.016	0.724 ± 0.082	0.868 ± 0.015	0.631	0.782 ± 0.028	0.255 ± 0.287	0.550 ± 0.060	0.882 ± 0.039	0.897 ± 0.034	**0.911 ± 0.016**	0.495 ± 0.027
liver_disorders	0.643 ± 0.117	0.487 ± 0.106	0.612 ± 0.083	0.771 ± 0.011	0.772	0.632 ± 0.018	0.601 ± 0.179	0.535 ± 0.082	0.731 ± 0.068	**0.774 ± 0.012**	0.767 ± 0.016	0.434 ± 0.052
parkinsons	0.773 ± 0.044	0.591 ± 0.048	0.700 ± 0.027	0.863 ± 0.015	0.676	0.714 ± 0.066	0.669 ± 0.236	0.522 ± 0.048	0.849 ± 0.016	0.856 ± 0.057	**0.886 ± 0.013**	0.522 ± 0.035
obesity	0.851 ± 0.016	0.856 ± 0.045	0.821 ± 0.027	0.893 ± 0.011	0.889	0.885 ± 0.003	0.794 ± 0.185	0.848 ± 0.024	0.879 ± 0.007	0.931 ± 0.016	**0.937 ± 0.003**	0.664 ± 0.013
diabetes	0.770 ± 0.046	0.782 ± 0.009	0.798 ± 0.017	0.808 ± 0.028	–	0.791 ± 0.032	0.672 ± 0.119	0.776 ± 0.044	0.776 ± 0.050	0.843 ± 0.007	**0.856 ± 0.001**	0.543 ± 0.022
german_credit	0.799 ± 0.052	0.872 ± 0.017	0.639 ± 0.011	0.925 ± 0.007	–	0.885 ± 0.019	0.817 ± 0.061	0.870 ± 0.016	0.854 ± 0.020	**0.927 ± 0.002**	0.925 ± 0.002	0.622 ± 0.009
bank_marketing	0.856 ± 0.020	0.852 ± 0.019	0.812 ± 0.023	0.852 ± 0.021	–	0.809 ± 0.085	0.667 ± 0.139	0.767 ± 0.041	0.783 ± 0.052	**0.881 ± 0.005**	0.849 ± 0.015	0.515 ± 0.007
census_income	0.663 ± 0.088	0.796 ± 0.057	0.782 ± 0.031	0.818 ± 0.109	–	0.726 ± 0.076	0.572 ± 0.403	0.730 ± 0.102	0.710 ± 0.102	**0.822 ± 0.010**	0.771 ± 0.021	0.426 ± 0.008
surv_melanoma	0.682 ± 0.247	–	–	–	–	0.703 ± 0.097	0.569 ± 0.237	0.634 ± 0.066	0.752 ± 0.013	**0.777 ± 0.013**	0.774 ± 0.028	0.637 ± 0.013
surv_pbc	0.767 ± 0.042	–	–	–	–	0.767 ± 0.038	0.712 ± 0.053	0.744 ± 0.213	0.777 ± 0.025	**0.832 ± 0.024**	**0.832 ± 0.016**	0.679 ± 0.016
surv_pbc3	0.781 ± 0.060	–	–	–	–	0.766 ± 0.031	0.619 ± 0.190	0.661 ± 0.043	0.819 ± 0.028	0.850 ± 0.012	**0.871 ± 0.016**	0.613 ± 0.008
surv_retinopathy	0.786 ± 0.090	–	–	–	–	0.794 ± 0.063	0.747 ± 0.078	0.795 ± 0.017	0.794 ± 0.016	0.846 ± 0.007	**0.851 ± 0.005**	0.768 ± 0.001
surv_whas500	0.826 ± 0.083	–	–	–	–	0.855 ± 0.039	0.725 ± 0.145	0.767 ± 0.057	0.880 ± 0.016	0.903 ± 0.011	**0.906 ± 0.004**	0.689 ± 0.016
surv_gbsg2	0.771 ± 0.154	–	–	–	–	0.687 ± 0.205	0.635 ± 0.242	0.701 ± 0.024	0.824 ± 0.030	**0.830 ± 0.016**	0.801 ± 0.007	0.512 ± 0.023
surv_colon	0.799 ± 0.582	–	–	–	–	0.754 ± 0.118	0.699 ± 0.154	0.815 ± 0.022	0.838 ± 0.021	**0.874 ± 0.005**	0.860 ± 0.005	0.630 ± 0.017
surv_grace	0.761 ± 0.033	–	–	–	–	0.679 ± 0.097	0.531 ± 0.400	0.700 ± 0.024	0.757 ± 0.053	**0.808 ± 0.011**	0.791 ± 0.007	0.632 ± 0.020
surv_trace	0.805 ± 0.056	–	–	–	–	0.763 ± 0.035	0.660 ± 0.049	0.695 ± 0.198	0.783 ± 0.031	0.836 ± 0.005	**0.836 ± 0.003**	0.613 ± 0.011
surv_framingham	0.806 ± 0.034	–	–	–	–	0.759 ± 0.058	0.494 ± 0.018	0.615 ± 0.313	0.768 ± 0.049	0.839 ± 0.008	**0.847 ± 0.001**	0.599 ± 0.016
tab_blood_transfusion	0.579 ± 0.307	–	–	–	–	0.531 ± 0.006	0.434 ± 0.207	0.576 ± 0.070	0.694 ± 0.052	**0.746 ± 0.030**	0.721 ± 0.007	0.370 ± 0.023
tab_credit_g	0.792 ± 0.018	–	–	–	–	0.918 ± 0.056	0.823 ± 0.180	0.919 ± 0.006	0.911 ± 0.022	0.967 ± 0.005	**0.971 ± 0.013**	0.654 ± 0.008
tab_qsar_biodeg	0.789 ± 0.064	–	–	–	–	0.849 ± 0.029	0.574 ± 0.167	0.775 ± 0.019	0.873 ± 0.005	0.910 ± 0.012	**0.916 ± 0.003**	0.417 ± 0.015
tab_maternal_risk	0.779 ± 0.023	–	–	–	–	0.732 ± 0.233	0.588 ± 0.268	0.724 ± 0.031	0.867 ± 0.044	**0.899 ± 0.021**	0.788 ± 0.014	0.585 ± 0.011
tab_website_phishing	0.908 ± 0.097	–	–	–	–	0.936 ± 0.032	0.769 ± 0.094	0.953 ± 0.033	0.920 ± 0.012	**0.982 ± 0.007**	0.972 ± 0.011	0.841 ± 0.014

Table 29: Column-pair trends (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**-0.000 ± 0.000**	**0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**-0.000 ± 0.000**
breast_cancer	0.658 ± 0.015	0.550 ± 0.009	0.714 ± 0.017	0.830 ± 0.015	0.777	0.784 ± 0.021	0.553 ± 0.016	0.678 ± 0.008	0.611 ± 0.039	0.782 ± 0.010	**0.841 ± 0.001**	0.549 ± 0.010
liver_disorders	0.967 ± 0.044	0.707 ± 0.008	0.798 ± 0.098	**0.969 ± 0.006**	0.959	0.942 ± 0.060	0.668 ± 0.035	0.757 ± 0.004	0.790 ± 0.036	0.968 ± 0.017	0.934 ± 0.062	0.694 ± 0.018
parkinsons	0.759 ± 0.038	0.520 ± 0.025	0.637 ± 0.017	0.831 ± 0.009	0.777	0.749 ± 0.052	0.537 ± 0.028	0.606 ± 0.030	0.599 ± 0.042	0.771 ± 0.008	**0.836 ± 0.006**	0.518 ± 0.008
obesity	0.834 ± 0.099	0.695 ± 0.026	0.673 ± 0.074	0.738 ± 0.008	0.812	0.788 ± 0.104	0.642 ± 0.217	0.541 ± 0.070	0.705 ± 0.037	0.856 ± 0.048	**0.890 ± 0.005**	0.492 ± 0.010
diabetes	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	–	**-0.000 ± 0.000**	**-0.000 ± 0.000**	**-0.000 ± 0.000**	**0.000 ± 0.000**	**-0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**
german_credit	0.630 ± 0.310	0.650 ± 0.033	0.475 ± 0.151	0.716 ± 0.020	–	0.746 ± 0.011	0.647 ± 0.045	0.701 ± 0.059	0.646 ± 0.039	0.753 ± 0.033	**0.802 ± 0.024**	0.462 ± 0.015
bank_marketing	0.552 ± 0.017	0.566 ± 0.016	**0.568 ± 0.027**	0.553 ± 0.014	–	0.488 ± 0.116	0.356 ± 0.117	0.406 ± 0.192	0.445 ± 0.054	0.556 ± 0.021	0.535 ± 0.009	0.293 ± 0.007
census_income	0.373 ± 0.083	0.383 ± 0.025	**0.431 ± 0.062**	0.375 ± 0.027	–	0.355 ± 0.103	0.233 ± 0.177	0.333 ± 0.124	0.293 ± 0.011	0.400 ± 0.009	0.399 ± 0.044	0.185 ± 0.003
surv_melanoma	**0.000 ± 0.000**	–	–	–	–	**0.000 ± 0.000**	**0.000 ± 0.000**	**-0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**-0.000 ± 0.000**
surv_pbc	0.127 ± 0.289	–	–	–	–	0.178 ± 0.449	0.178 ± 0.505	0.035 ± 0.076	0.222 ± 0.559	0.207 ± 0.516	**0.223 ± 0.559**	0.129 ± 0.306
surv_pbc3	0.082 ± 0.353	–	–	–	–	0.102 ± 0.439	0.090 ± 0.386	0.096 ± 0.411	0.086 ± 0.368	0.099 ± 0.427	**0.106 ± 0.454**	0.075 ± 0.324
surv_retinopathy	0.751 ± 0.558	–	–	–	–	0.733 ± 0.153	0.465 ± 0.103	0.440 ± 0.024	0.646 ± 0.036	0.846 ± 0.026	**0.947 ± 0.028**	0.479 ± 0.062
surv_whas500	0.206 ± 0.067	–	–	–	–	**0.214 ± 0.069**	0.123 ± 0.023	0.166 ± 0.067	0.181 ± 0.074	0.189 ± 0.067	0.213 ± 0.070	0.148 ± 0.072
surv_gbsg2	0.572 ± 0.326	–	–	–	–	0.522 ± 0.234	0.489 ± 0.109	0.213 ± 0.264	0.616 ± 0.192	**0.655 ± 0.261**	0.408 ± 0.169	0.362 ± 0.256
surv_colon	0.954 ± 0.138	–	–	–	–	0.974 ± 0.064	0.639 ± 0.044	0.869 ± 0.017	0.816 ± 0.201	0.978 ± 0.049	**0.998 ± 0.001**	0.638 ± 0.083
surv_grace	**0.000 ± 0.000**	–	–	–	–	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**
surv_trace	**-0.000 ± 0.000**	–	–	–	–	**0.000 ± 0.000**	**-0.000 ± 0.000**	**0.000 ± 0.000**	**-0.000 ± 0.000**	**-0.000 ± 0.000**	**-0.000 ± 0.000**	**0.000 ± 0.000**
surv_framingham	**0.991 ± 0.018**	–	–	–	–	0.975 ± 0.071	0.600 ± 0.061	0.769 ± 0.040	0.734 ± 0.093	0.957 ± 0.007	0.908 ± 0.061	0.602 ± 0.040
tab_blood_transfusion	0.642 ± 0.408	–	–	–	–	0.685 ± 0.085	0.514 ± 0.076	0.547 ± 0.101	0.593 ± 0.129	0.718 ± 0.019	**0.746 ± 0.007**	0.484 ± 0.040
tab_credit_g	0.541 ± 0.069	–	–	–	–	0.879 ± 0.061	0.673 ± 0.228	0.823 ± 0.119	0.801 ± 0.022	0.890 ± 0.028	**0.957 ± 0.022**	0.569 ± 0.006
tab_qsar_biodeg	0.625 ± 0.036	–	–	–	–	0.688 ± 0.042	0.526 ± 0.088	0.613 ± 0.039	0.551 ± 0.051	0.716 ± 0.066	**0.787 ± 0.049**	0.532 ± 0.035
tab_maternal_risk	0.712 ± 0.178	–	–	–	–	0.645 ± 0.014	0.491 ± 0.137	0.422 ± 0.023	0.698 ± 0.068	**0.801 ± 0.044**	0.593 ± 0.082	0.490 ± 0.079
tab_website_phishing	0.796 ± 0.390	–	–	–	–	0.883 ± 0.021	0.616 ± 0.106	0.851 ± 0.031	0.851 ± 0.005	0.929 ± 0.018	**0.947 ± 0.014**	0.660 ± 0.021

FAITH framework

Table 30: FAITH overall (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.813 ± 0.090	0.884 ± 0.084	0.739 ± 0.014	0.827 ± 0.035	0.870	0.808 ± 0.057	0.916 ± 0.054	0.758 ± 0.044	0.804 ± 0.075	0.834 ± 0.048	0.817 ± 0.059	**0.967 ± 0.002**
breast_cancer	0.388 ± 0.057	0.883 ± 0.001	0.804 ± 0.131	0.897 ± 0.084	0.856	**0.925 ± 0.036**	0.870 ± 0.019	0.770 ± 0.070	0.900 ± 0.032	0.899 ± 0.081	0.854 ± 0.046	0.883 ± 0.002
liver_disorders	0.870 ± 0.089	0.903 ± 0.010	0.710 ± 0.013	0.859 ± 0.035	0.835	0.800 ± 0.039	0.845 ± 0.046	0.804 ± 0.038	0.819 ± 0.071	0.852 ± 0.032	0.837 ± 0.062	**0.923 ± 0.008**
parkinsons	0.298 ± 0.133	0.852 ± 0.013	0.763 ± 0.091	0.868 ± 0.003	0.825	**0.877 ± 0.086**	0.847 ± 0.018	0.715 ± 0.063	0.860 ± 0.057	0.862 ± 0.056	0.818 ± 0.030	0.852 ± 0.008
obesity	0.561 ± 0.212	0.944 ± 0.014	0.790 ± 0.042	0.957 ± 0.012	0.854	0.902 ± 0.133	0.941 ± 0.047	0.907 ± 0.034	0.947 ± 0.004	0.877 ± 0.011	0.855 ± 0.015	**0.961 ± 0.003**
diabetes	0.745 ± 0.090	0.862 ± 0.037	0.769 ± 0.034	0.924 ± 0.015	–	0.853 ± 0.015	0.877 ± 0.168	0.841 ± 0.056	0.865 ± 0.024	0.886 ± 0.002	0.842 ± 0.007	**0.964 ± 0.003**
german_credit	0.374 ± 0.180	0.881 ± 0.025	0.724 ± 0.004	0.859 ± 0.014	–	0.847 ± 0.019	0.853 ± 0.022	0.814 ± 0.009	0.878 ± 0.027	0.852 ± 0.011	0.829 ± 0.016	**0.960 ± 0.003**
bank_marketing	0.810 ± 0.054	0.922 ± 0.021	0.749 ± 0.024	0.936 ± 0.026	–	0.932 ± 0.050	0.960 ± 0.018	0.903 ± 0.066	0.941 ± 0.028	0.927 ± 0.018	0.919 ± 0.016	**0.974 ± 0.002**
census_income	0.350 ± 0.158	0.949 ± 0.021	0.769 ± 0.010	0.982 ± 0.021	–	0.957 ± 0.072	0.974 ± 0.009	0.967 ± 0.059	0.976 ± 0.013	0.959 ± 0.012	0.935 ± 0.036	**0.984 ± 0.001**
surv_melanoma	0.791 ± 0.042	–	–	–	–	0.772 ± 0.065	0.880 ± 0.067	0.763 ± 0.012	0.804 ± 0.050	0.806 ± 0.014	0.808 ± 0.011	**0.902 ± 0.025**
surv_pbc	0.655 ± 0.432	–	–	–	–	0.797 ± 0.082	0.828 ± 0.084	0.836 ± 0.209	0.797 ± 0.074	0.812 ± 0.068	0.812 ± 0.072	**0.913 ± 0.041**
surv_pbc3	0.795 ± 0.064	–	–	–	–	0.827 ± 0.039	0.907 ± 0.025	0.742 ± 0.026	0.808 ± 0.024	0.839 ± 0.038	0.833 ± 0.024	**0.959 ± 0.007**
surv_retinopathy	0.657 ± 0.382	–	–	–	–	0.807 ± 0.031	0.830 ± 0.068	0.818 ± 0.033	0.817 ± 0.052	0.809 ± 0.014	0.804 ± 0.028	**0.859 ± 0.016**
surv_whas500	0.295 ± 0.046	–	–	–	–	0.822 ± 0.053	0.942 ± 0.032	0.761 ± 0.044	0.830 ± 0.040	0.838 ± 0.024	0.839 ± 0.028	**0.965 ± 0.003**
surv_gbsg2	0.664 ± 0.492	–	–	–	–	0.840 ± 0.046	0.887 ± 0.076	0.802 ± 0.076	0.843 ± 0.056	0.853 ± 0.044	0.869 ± 0.060	**0.954 ± 0.014**
surv_colon	0.824 ± 0.074	–	–	–	–	0.845 ± 0.093	0.887 ± 0.038	0.805 ± 0.056	0.822 ± 0.064	0.841 ± 0.060	0.830 ± 0.052	**0.956 ± 0.004**
surv_grace	0.803 ± 0.016	–	–	–	–	0.792 ± 0.059	0.915 ± 0.123	0.767 ± 0.031	0.795 ± 0.018	0.817 ± 0.005	0.805 ± 0.007	**0.953 ± 0.017**
surv_trace	0.804 ± 0.052	–	–	–	–	0.801 ± 0.026	0.878 ± 0.063	0.867 ± 0.137	0.820 ± 0.046	0.827 ± 0.003	0.826 ± 0.012	**0.944 ± 0.001**
surv_framingham	0.816 ± 0.014	–	–	–	–	0.839 ± 0.050	0.916 ± 0.026	0.873 ± 0.223	0.823 ± 0.022	0.839 ± 0.005	0.826 ± 0.005	**0.939 ± 0.003**
tab_blood_transfusion	0.474 ± 0.440	–	–	–	–	0.798 ± 0.084	0.836 ± 0.111	0.811 ± 0.025	0.811 ± 0.007	0.834 ± 0.029	0.787 ± 0.036	**0.881 ± 0.020**
tab_credit_g	0.379 ± 0.070	–	–	–	–	0.840 ± 0.079	0.888 ± 0.144	0.826 ± 0.023	0.872 ± 0.016	0.866 ± 0.020	0.841 ± 0.030	**0.958 ± 0.001**
tab_qsar_biodeg	0.363 ± 0.113	–	–	–	–	0.921 ± 0.042	0.938 ± 0.014	0.865 ± 0.033	0.938 ± 0.008	0.883 ± 0.046	0.847 ± 0.029	**0.938 ± 0.001**
tab_maternal_risk	0.527 ± 0.550	–	–	–	–	0.834 ± 0.007	0.872 ± 0.072	0.789 ± 0.053	0.847 ± 0.038	0.853 ± 0.031	0.844 ± 0.042	**0.925 ± 0.007**
tab_website_phishing	0.351 ± 0.645	–	–	–	–	0.509 ± 0.009	0.530 ± 0.024	0.509 ± 0.005	0.514 ± 0.012	0.507 ± 0.004	0.509 ± 0.003	**0.581 ± 0.014**

Table 31: Factuality (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.978 ± 0.043	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
breast_cancer	0.000 ± 0.000	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
liver_disorders	0.999 ± 0.005	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
parkinsons	0.000 ± 0.000	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
obesity	0.416 ± 0.407	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
diabetes	0.859 ± 0.302	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
german_credit	0.000 ± 0.000	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
bank_marketing	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
census_income	0.091 ± 0.248	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
surv_melanoma	0.998 ± 0.009	–	–	–	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
surv_pbc	0.621 ± 1.316	–	–	–	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
surv_pbc3	0.986 ± 0.024	–	–	–	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
surv_retinopathy	0.544 ± 1.251	–	–	–	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
surv_whas500	0.000 ± 0.000	–	–	–	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
surv_gbsg2	0.631 ± 1.122	–	–	–	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
surv_colon	0.998 ± 0.010	–	–	–	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
surv_grace	**1.000 ± 0.000**	–	–	–	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
surv_trace	1.000 ± 0.001	–	–	–	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
surv_framingham	1.000 ± 0.000	–	–	–	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
tab_blood_transfusion	0.313 ± 1.016	–	–	–	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
tab_credit_g	0.000 ± 0.000	–	–	–	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
tab_qsar_biodeg	0.029 ± 0.120	–	–	–	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
tab_maternal_risk	0.453 ± 1.138	–	–	–	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
tab_website_phishing	0.667 ± 1.434	–	–	–	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**

Table 32: Alignment (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.910 ± 0.018	0.877 ± 0.029	**0.944 ± 0.019**	0.916 ± 0.011	0.790	0.893 ± 0.025	0.808 ± 0.034	0.904 ± 0.014	0.887 ± 0.008	0.927 ± 0.024	0.887 ± 0.047	0.879 ± 0.010
breast_cancer	0.882 ± 0.021	0.532 ± 0.002	0.747 ± 0.010	0.933 ± 0.022	0.722	0.861 ± 0.037	0.479 ± 0.077	0.713 ± 0.032	0.627 ± 0.055	0.873 ± 0.005	**0.934 ± 0.029**	0.532 ± 0.007
liver_disorders	0.917 ± 0.089	0.707 ± 0.020	0.789 ± 0.149	**0.943 ± 0.020**	0.901	0.858 ± 0.101	0.600 ± 0.138	0.725 ± 0.035	0.782 ± 0.052	0.941 ± 0.004	0.896 ± 0.071	0.693 ± 0.028
parkinsons	0.832 ± 0.080	0.411 ± 0.052	0.598 ± 0.001	0.919 ± 0.009	0.678	0.767 ± 0.060	0.393 ± 0.072	0.545 ± 0.051	0.547 ± 0.077	0.844 ± 0.022	**0.933 ± 0.007**	0.409 ± 0.033
obesity	0.936 ± 0.030	0.845 ± 0.012	0.855 ± 0.011	0.929 ± 0.020	0.824	0.946 ± 0.098	0.819 ± 0.098	0.903 ± 0.007	0.866 ± 0.012	0.958 ± 0.016	**0.971 ± 0.013**	0.845 ± 0.011
diabetes	0.912 ± 0.116	0.888 ± 0.031	0.915 ± 0.046	0.910 ± 0.008	–	0.965 ± 0.008	0.868 ± 0.057	0.903 ± 0.016	0.906 ± 0.031	**0.979 ± 0.003**	0.946 ± 0.004	0.859 ± 0.008
german_credit	0.920 ± 0.094	0.841 ± 0.016	0.892 ± 0.010	0.914 ± 0.027	–	0.919 ± 0.014	0.858 ± 0.013	0.901 ± 0.021	0.863 ± 0.023	0.943 ± 0.012	**0.956 ± 0.021**	0.843 ± 0.010
bank_marketing	0.932 ± 0.017	0.936 ± 0.027	0.877 ± 0.074	0.919 ± 0.069	–	0.960 ± 0.035	0.891 ± 0.003	0.959 ± 0.062	0.912 ± 0.012	**0.970 ± 0.027**	0.959 ± 0.020	0.897 ± 0.007
census_income	0.938 ± 0.035	0.941 ± 0.027	0.929 ± 0.012	0.974 ± 0.005	–	0.964 ± 0.015	0.905 ± 0.020	0.956 ± 0.014	0.939 ± 0.014	0.978 ± 0.010	**0.979 ± 0.004**	0.935 ± 0.005
surv_melanoma	0.903 ± 0.042	–	–	–	–	0.841 ± 0.132	0.772 ± 0.032	0.831 ± 0.024	0.853 ± 0.100	**0.931 ± 0.010**	0.911 ± 0.035	0.811 ± 0.084
surv_pbc	0.879 ± 0.124	–	–	–	–	0.913 ± 0.031	0.853 ± 0.067	0.877 ± 0.008	0.891 ± 0.050	0.935 ± 0.049	**0.940 ± 0.044**	0.856 ± 0.044
surv_pbc3	0.892 ± 0.042	–	–	–	–	0.898 ± 0.028	0.816 ± 0.042	0.863 ± 0.011	0.859 ± 0.020	0.925 ± 0.012	**0.929 ± 0.013**	0.836 ± 0.032
surv_retinopathy	0.953 ± 0.037	–	–	–	–	0.830 ± 0.057	0.642 ± 0.040	0.775 ± 0.068	0.693 ± 0.038	0.890 ± 0.028	**0.957 ± 0.105**	0.630 ± 0.052
surv_whas500	0.897 ± 0.050	–	–	–	–	0.917 ± 0.006	0.824 ± 0.038	0.900 ± 0.011	0.892 ± 0.014	0.928 ± 0.006	**0.933 ± 0.010**	0.865 ± 0.012
surv_gbsg2	0.898 ± 0.088	–	–	–	–	0.865 ± 0.118	0.777 ± 0.101	0.862 ± 0.089	0.865 ± 0.050	**0.944 ± 0.063**	0.903 ± 0.063	0.818 ± 0.054
surv_colon	0.949 ± 0.030	–	–	–	–	0.901 ± 0.085	0.840 ± 0.012	0.927 ± 0.005	0.900 ± 0.045	**0.960 ± 0.027**	0.932 ± 0.007	0.842 ± 0.027
surv_grace	0.946 ± 0.037	–	–	–	–	0.937 ± 0.032	0.855 ± 0.044	0.914 ± 0.028	0.947 ± 0.045	**0.975 ± 0.015**	0.913 ± 0.043	0.903 ± 0.056
surv_trace	0.958 ± 0.030	–	–	–	–	0.922 ± 0.069	0.824 ± 0.009	0.859 ± 0.042	0.913 ± 0.062	0.974 ± 0.017	**0.975 ± 0.012**	0.852 ± 0.004
surv_framingham	0.964 ± 0.014	–	–	–	–	0.957 ± 0.001	0.724 ± 0.011	0.849 ± 0.021	0.842 ± 0.042	**0.974 ± 0.005**	0.898 ± 0.014	0.766 ± 0.009
tab_blood_transfusion	0.718 ± 0.340	–	–	–	–	0.867 ± 0.136	0.564 ± 0.110	0.649 ± 0.173	0.718 ± 0.196	**0.926 ± 0.010**	0.831 ± 0.094	0.530 ± 0.077
tab_credit_g	0.953 ± 0.031	–	–	–	–	0.920 ± 0.037	0.853 ± 0.078	0.903 ± 0.001	0.865 ± 0.020	0.944 ± 0.025	**0.963 ± 0.003**	0.831 ± 0.003

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
tab_qsar_biodeg	0.732 ± 0.077	–	–	–	–	0.864 ± 0.011	0.751 ± 0.057	0.820 ± 0.003	0.768 ± 0.018	0.903 ± 0.020	**0.952 ± 0.003**	0.754 ± 0.005
tab_maternal_risk	0.908 ± 0.071	–	–	–	–	0.896 ± 0.076	0.699 ± 0.032	0.853 ± 0.063	0.789 ± 0.009	**0.948 ± 0.023**	0.897 ± 0.054	0.713 ± 0.041
tab_website_phishing	–	–	–	–	–	–	–	–	–	–	–	–

Table 33: Integrity (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.386 ± 0.310	0.660 ± 0.307	0.013 ± 0.045	0.393 ± 0.144	0.690	0.338 ± 0.251	0.856 ± 0.248	0.126 ± 0.181	0.329 ± 0.304	0.407 ± 0.169	0.381 ± 0.210	**0.990 ± 0.016**
breast_cancer	0.670 ± 0.217	**1.000 ± 0.000**	0.470 ± 0.530	0.656 ± 0.360	0.703	0.838 ± 0.176	**1.000 ± 0.000**	0.367 ± 0.291	0.971 ± 0.079	0.724 ± 0.328	0.482 ± 0.178	**1.000 ± 0.000**
liver_disorders	0.565 ± 0.398	0.906 ± 0.059	0.051 ± 0.101	0.494 ± 0.153	0.438	0.342 ± 0.245	0.781 ± 0.058	0.493 ± 0.136	0.494 ± 0.293	0.465 ± 0.131	0.450 ± 0.320	**0.998 ± 0.005**
parkinsons	0.361 ± 0.468	0.998 ± 0.009	0.453 ± 0.365	0.553 ± 0.009	0.622	0.739 ± 0.330	0.994 ± 0.016	0.314 ± 0.207	0.895 ± 0.189	0.605 ± 0.223	0.338 ± 0.113	**1.000 ± 0.000**
obesity	0.478 ± 0.071	0.931 ± 0.065	0.304 ± 0.179	0.900 ± 0.048	0.590	0.663 ± 0.433	0.943 ± 0.092	0.725 ± 0.128	0.921 ± 0.004	0.549 ± 0.060	0.451 ± 0.071	**1.000 ± 0.000**
diabetes	0.350 ± 0.368	0.558 ± 0.125	0.160 ± 0.091	0.784 ± 0.061	–	0.446 ± 0.053	0.638 ± 0.725	0.459 ± 0.239	0.555 ± 0.122	0.566 ± 0.007	0.420 ± 0.025	**0.996 ± 0.004**
german_credit	0.574 ± 0.816	0.685 ± 0.097	0.006 ± 0.007	0.521 ± 0.045	–	0.470 ± 0.064	0.553 ± 0.083	0.356 ± 0.049	0.647 ± 0.109	0.464 ± 0.042	0.361 ± 0.070	**0.998 ± 0.004**
bank_marketing	0.309 ± 0.231	0.754 ± 0.079	0.118 ± 0.025	0.823 ± 0.040	–	0.769 ± 0.171	0.951 ± 0.069	0.654 ± 0.306	0.853 ± 0.107	0.737 ± 0.050	0.716 ± 0.066	**1.000 ± 0.000**
census_income	0.280 ± 0.256	0.855 ± 0.110	0.147 ± 0.030	0.955 ± 0.080	–	0.863 ± 0.273	0.991 ± 0.031	0.913 ± 0.235	0.964 ± 0.042	0.858 ± 0.039	0.762 ± 0.145	**1.000 ± 0.000**
surv_melanoma	0.264 ± 0.114	–	–	–	–	0.246 ± 0.189	0.750 ± 0.244	0.220 ± 0.055	0.364 ± 0.101	0.295 ± 0.061	0.321 ± 0.078	**0.799 ± 0.026**
surv_pbc	0.498 ± 1.079	–	–	–	–	0.276 ± 0.325	0.458 ± 0.404	0.469 ± 0.834	0.299 ± 0.311	0.312 ± 0.241	0.309 ± 0.285	**0.795 ± 0.174**
surv_pbc3	0.315 ± 0.309	–	–	–	–	0.410 ± 0.182	0.811 ± 0.095	0.103 ± 0.095	0.372 ± 0.107	0.429 ± 0.145	0.401 ± 0.103	**0.999 ± 0.005**
surv_retinopathy	0.585 ± 0.981	–	–	–	–	0.398 ± 0.169	0.677 ± 0.260	0.495 ± 0.077	0.576 ± 0.228	0.347 ± 0.039	0.261 ± 0.102	**0.805 ± 0.074**
surv_whas500	0.283 ± 0.182	–	–	–	–	0.371 ± 0.209	0.942 ± 0.100	0.145 ± 0.164	0.426 ± 0.152	0.426 ± 0.095	0.424 ± 0.113	**0.995 ± 0.007**
surv_gbsg2	0.495 ± 0.350	–	–	–	–	0.495 ± 0.274	0.769 ± 0.312	0.345 ± 0.286	0.505 ± 0.263	0.467 ± 0.231	0.574 ± 0.211	**0.998 ± 0.005**
surv_colon	0.351 ± 0.325	–	–	–	–	0.477 ± 0.457	0.706 ± 0.160	0.293 ± 0.226	0.387 ± 0.231	0.405 ± 0.244	0.388 ± 0.208	**0.983 ± 0.012**
surv_grace	0.265 ± 0.096	–	–	–	–	0.233 ± 0.261	0.805 ± 0.451	0.154 ± 0.116	0.233 ± 0.069	0.294 ± 0.031	0.308 ± 0.035	**0.909 ± 0.022**
surv_trace	0.259 ± 0.214	–	–	–	–	0.280 ± 0.165	0.687 ± 0.241	0.610 ± 0.505	0.366 ± 0.124	0.336 ± 0.007	0.329 ± 0.045	**0.924 ± 0.006**
surv_framingham	0.301 ± 0.069	–	–	–	–	0.399 ± 0.201	0.941 ± 0.103	0.643 ± 0.905	0.450 ± 0.130	0.383 ± 0.022	0.405 ± 0.024	**0.991 ± 0.004**
tab_blood_transfusion	0.551 ± 0.523	–	–	–	–	0.326 ± 0.411	0.782 ± 0.554	0.596 ± 0.129	0.528 ± 0.170	0.410 ± 0.104	0.316 ± 0.066	**0.995 ± 0.003**
tab_credit_g	0.561 ± 0.279	–	–	–	–	0.438 ± 0.336	0.700 ± 0.573	0.399 ± 0.093	0.624 ± 0.081	0.520 ± 0.095	0.400 ± 0.117	**0.999 ± 0.002**
tab_qsar_biodeg	0.663 ± 0.446	–	–	–	–	0.821 ± 0.181	**1.000 ± 0.000**	0.638 ± 0.133	0.984 ± 0.030	0.629 ± 0.182	0.437 ± 0.115	**1.000 ± 0.000**
tab_maternal_risk	0.295 ± 0.262	–	–	–	–	0.442 ± 0.090	0.789 ± 0.298	0.302 ± 0.151	0.599 ± 0.161	0.463 ± 0.111	0.481 ± 0.220	**0.989 ± 0.012**
tab_website_phishing	0.069 ± 0.290	–	–	–	–	0.035 ± 0.034	0.121 ± 0.094	0.038 ± 0.020	0.058 ± 0.049	0.029 ± 0.015	0.035 ± 0.011	**0.325 ± 0.058**

Table 34: Tracking (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	0.978 ± 0.043	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
breast_cancer	0.000 ± 0.000	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
liver_disorders	0.999 ± 0.005	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
parkinsons	0.000 ± 0.000	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
obesity	0.416 ± 0.407	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
diabetes	0.859 ± 0.302	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
german_credit	0.000 ± 0.000	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
bank_marketing	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
census_income	0.091 ± 0.248	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
surv_melanoma	0.998 ± 0.009	–	–	–	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
surv_pbc	0.621 ± 1.316	–	–	–	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
surv_pbc3	0.986 ± 0.024	–	–	–	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
surv_retinopathy	0.544 ± 1.251	–	–	–	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
surv_whas500	0.000 ± 0.000	–	–	–	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
surv_gbsg2	0.631 ± 1.122	–	–	–	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
surv_colon	0.998 ± 0.010	–	–	–	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
surv_grace	**1.000 ± 0.000**	–	–	–	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
surv_trace	1.000 ± 0.001	–	–	–	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
surv_framingham	1.000 ± 0.000	–	–	–	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
tab_blood_transfusion	0.313 ± 1.016	–	–	–	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
tab_credit_g	0.000 ± 0.000	–	–	–	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
tab_qsar_biodeg	0.029 ± 0.120	–	–	–	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
tab_maternal_risk	0.453 ± 1.138	–	–	–	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**
tab_website_phishing	0.667 ± 1.434	–	–	–	–	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**	**1.000 ± 0.000**

Other

Table 35: Parse valid-rate (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
heart_failure	**0.675 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–
breast_cancer	**0.693 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–
liver_disorders	**0.680 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–
parkinsons	**0.655 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–
obesity	**0.708 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–
diabetes	**0.711 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–
german_credit	**0.537 ± 0.708**	–	–	–	–	–	–	–	–	–	–	–
bank_marketing	**0.711 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–
census_income	**0.711 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	ARF	BayesNet	Marginals
surv_melanoma	**0.659 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–
surv_pbc	**0.474 ± 0.873**	–	–	–	–	–	–	–	–	–	–	–
surv_pbc3	**0.680 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–
surv_retinopathy	**0.682 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–
surv_whas500	**0.688 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–
surv_gbsg2	**0.696 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–
surv_colon	**0.700 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–
surv_grace	**0.702 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–
surv_trace	**0.708 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–
surv_framingham	**0.711 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–
tab_blood_transfusion	**0.529 ± 0.701**	–	–	–	–	–	–	–	–	–	–	–
tab_credit_g	**0.702 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–
tab_qsar_biodeg	**0.703 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–
tab_maternal_risk	**0.688 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–
tab_website_phishing	**0.698 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–